TRACK P (PLANETARY): BLUEPRINT

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Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun n-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\mathtt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) z_1 and momenta (actions) z_2. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of n planets are $q \in \mathbb{R}^{2n}$; planet i has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \mathtt{Fin}\, n \times \mathtt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).
- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* 2π (CP10 footnote 1, p. 546), so the parameterization winds by 2π in the fast directions. Consequently the frequency vector w of a torus is the physical angular frequency divided by 2π (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.
- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds r_*, C, c of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and μ_0, d, D (resp. ε_0, d, D) are chosen before μ (resp. ε).

1.2. Track-K definitions used by track P.

The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, ω is (γ, τ)-*Diophantine* if $\gamma/|k|_1^\tau \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.

Frozen Lean declaration: `KAM.IsDiophantine`

Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

Definition 1.2 (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\, -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with D the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With z_1 the position and z_2 the momentum, $\dot z_1 = \partial_{z_2} H$, $\dot z_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of Ω.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

1.3. **Elementary planar and index definitions.**

Definition 1.3 (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

Definition 1.4 (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

Definition 1.5 (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

Definition 1.6 (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first a and last b coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

Definition 1.7 (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p, q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 (A_2), pp. 547 and 51.

Definition 1.8 (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

Definition 1.9 (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where H_0 is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

1.4. **Definitions for the properly degenerate theorem.**

Definition 1.10 (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V, r) = \Big\{ z \in \mathrm{Ph}(a+b) :\ z_2^{\mathrm{f}} \in V,\ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2\ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every j: a *product of discs*, not a box. Its fundamental cell for the fast angles of period 2π is

$$\mathcal{F}(V, r) = \big\{ z \in \mathcal{C}(V, r) :\ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi\ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period 2π).

Definition 1.11 (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

Definition 1.12 (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

Definition 1.13 (Fast diffeomorphism)**.** For H_0 and $W \subset \mathbb{R}^a$, HasFastDiffeo(H_0, W) holds if (i) H_0 is real analytic at every point of W (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class C^1 on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus ω_{H_0} is a bijection of W onto its open image with C^1 inverse g (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

Definition 1.14 (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all j, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;

(ii) for every $\epsilon > 0$ there is δ with $0 < \delta \le r_0$ such that for all p, q with $p_j^2 + q_j^2 < \delta^2$ for all j,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big(P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon\, |(p, q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p, q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that δ may depend on (μ, I, ϵ) in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 (A_2), p. 51), with the oscillator-disc guards of audit P1.

Definition 1.15 (Cylinder torus)**.** Let $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a+b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and H is differentiable at $K(\theta)$, for all θ;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all θ;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all θ, u, v;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

Definition 1.16 (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a+b)$, $\mathrm{IsCylKol}(H, V, r, \mu, C, S)$ holds if S is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$ and θ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); $\mathrm{IsCylTorus}(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a+b$ with w (γ, τ)-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

Definition 1.17 (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big(\big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i)\big)_i,\ \big(m_i\sqrt{Gm_0/a_i}\,(-\sin 2\pi\phi_i, \cos 2\pi\phi_i)\big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open η-neighbourhood in the phase max norm, $U_\eta = \{z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n,\ |z - \mathrm{circ}(\phi)|_\infty < \eta\}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

Definition 1.18 (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: $\mathrm{NoColl}(q)$ means $|q^{(i)}|_2 > 0$ for all i and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \ne j$ (heliocentric configurations, the Sun at the origin). For N points $Q_0, \dots, Q_{N-1} \in \mathbb{R}^2$, $\mathrm{PairBounds}(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \ne j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

Definition 1.19 (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu\sum_i m_i}\sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0}\sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i}p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet i at index i in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

Definition 1.20 (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. $\mathrm{IsNewton}(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \ne j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative

$$\sum_{j \ne i} \frac{GM_j}{|Q_j(t) - Q_i(t)|_2^3}\,(Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with G restored.

Definition 1.21 (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate $\mathrm{IsFixedSol}(G, m_0, m, \varepsilon, Q, V)$ holds

if for all t and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, Q_i has derivative $V_i(t)$ and V_i has derivative

$$-\frac{Gm_0}{|Q_i(t)|_2^3}\, Q_i(t) + \varepsilon \sum_{j \neq i} \frac{Gm_j}{|Q_j(t) - Q_i(t)|_2^3}(Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.

Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`

Source: Ours (fixed-center law obtained by differentiating H_{fix}), after Fej04r (28).

Definition 1.22 (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[\big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{Gm_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[\frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{Gm_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by μ, with G restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu|p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu|\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).

Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`

Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and G restored.

Definition 1.23 (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[\frac{|p^{(i)}|_2^2}{2m_i} - \frac{Gm_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{Gm_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient ε.

Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`

Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

Definition 1.24 (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, $\mathrm{IsPhysTorus}(H, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^d$;
(ii) H is differentiable at $K(\theta)$ for all θ;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with w (γ, τ)-Diophantine;
(viii) for all θ, t, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such K the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d - 1$ here against $\tau > a + b$ in Definition 1.16.)

Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`

Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

Definition 1.25 (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;
(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi + tw)_1),\ t \mapsto v(K(\phi + tw)_2)\big)$, where x, v are the barycentric reconstructions of Definition 1.19;

(c) $\mathrm{PairBounds}\big(d, D;\, x(K(\phi)_1)\big)$ (all pairs including the Sun; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`

Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

Definition 1.26 (Fixed-Sun Kolmogorov set)**.** For ε, η, d, D and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ and for every ϕ: $\mathrm{NoColl}(K(\phi)_1)$; $\mathrm{IsFixedSol}\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; $\mathrm{PairBounds}\big(d, D;\, (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.

Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`

Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

Remark 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* (ω_{H_0}, X_H, DK) are 0 at points of non-differentiability. ω_{H_0} is used only through `HasFastDiffeomorphism`, which asserts analyticity of H_0 on $W \supset \overline{V}$; X_H appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of H on the range of K (for the cylinder tori at points of $\mathcal{C}(V, r)$, where H_μ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on P and H_0); DK is guarded by analyticity of K. Injectivity of DK and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.
- *Bochner integrals* (P_{av}) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2 + q_j^2 < r_0^2$, and every point used in (ii) has $\delta \le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G, m_0, a_i > 0$; in x_0 and v_0 the denominators $m_0 + \mu \sum m_i$ and m_0 are positive since $\mu > 0$; the factor $1/m_i$ needs $m_i > 0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0 < r < r_* < 1$ has positive base $\log(1/r) > 0$; $\sqrt{r}$ has $r > 0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1 \ge 1$ for $k \ne 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1 - C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality $\mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F} < \mathrm{meas}(S \cap \mathcal{F})$ is then nontrivial only when $1 - C\sqrt{r} > 0$, but must be proved for every $r < r_*$ as stated (it follows from $0 < \mathrm{meas}(S \cap \mathcal{F})$ when $1 - C\sqrt{r} \le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V, r)$ are Borel and S is measurable by the conclusion; $\mathrm{meas}\,\mathcal{F}(V, r) < \infty$ since V is bounded.

1.6. **The frozen theorems.**

Theorem 1.28 (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a, b \ge 1$. A set $V \subset \mathbb{R}^a$ that is open, connected and bounded; an open set $W \subset \mathbb{R}^a$ with $\overline{V} \subset W$; a boundary collar condition: there are $A, d > 0$ such that for all $0 < t < d$*

$$\mathrm{meas}\{I \in \mathbb{R}^a :\ \exists J \in \partial V,\ |I - J|_\infty < t\} \le A\, t.$$

Reals $r_0, \mu_0, B > 0$. Functions $H_0 : \mathbb{R}^a \to \mathbb{R}$, $P : \mathbb{R} \times \mathrm{Ph}(a + b) \to \mathbb{R}$, $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$.

Hypotheses.

(H1) $(\mu, z) \mapsto P(\mu, z)$ is real analytic at every point of $[0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$ (jointly, as a function on $\mathbb{R} \times \mathrm{Ph}(a + b)$);

(H2) *for* $\mu \in [0,\mu_0]$, $z \in \mathcal{C}(V,r_0)$ *and* $k \in \mathbb{Z}^a$: $P(\mu, z+\mathrm{shift}(k)) = P(\mu,z)$;
(H3) $\mathrm{HasFastDiffeo}(H_0, W)$ *(Definition 1.13);*
(H4) $\mathrm{HasBirkhoff}(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *(Definition 1.14);*
(H5) $|\det \beta(\mu, I)| \geq B$ *for all* $\mu \in [0,\mu_0]$, $I \in V$;
(H6) $\beta(\mu, I)$ *is symmetric for all* $\mu \in [0,\mu_0]$, $I \in V$.

Conclusion. *There exist* r_*, C, c *with* $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$ *such that for all* r, μ *with*

$$0 < r < r_*, \qquad 0 < \mu < \mu_0, \qquad \mu < \frac{r^6}{(\log(1/r))^c},$$

there is $S \subset \mathrm{Ph}(a+b)$ *with* $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C, S)$ *(with* $H_\mu = H_0 + \mu P(\mu,\cdot)$ *of Definition 1.11) and*

$$0 < \mathrm{meas}(S \cap \mathcal{F}(V,r)), \qquad \mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F}(V,r) < \mathrm{meas}(S \cap \mathcal{F}(V,r)) < \mathrm{meas}\,\mathcal{F}(V,r).$$

Here r_*, C, c *depend on all the data and hypotheses above and not on* r *or* μ; C *is simultaneously the frequency bound of Definition 1.16 and the measure constant.*

Frozen Lean declaration: `KAM.Planetary.properlyDegenerateKAM`

Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with $c_* = 2b$, $\rho = r$*). It is* not *the printed theorem: the neighbourhood* W *(CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only* C^1 *uniform dependence) and the* 2π *period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

Source gap: CP11 Thm. 11.1 states $\mathrm{meas}\,P_\epsilon > \mathrm{meas}\,K$ *without proof; the strict inequality* $\mathrm{meas}(S \cap \mathcal{F}) < \mathrm{meas}\,\mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is possible because the lower bound has slack. CP10 Thm. 1.3 also needs* $P_{\mathrm{av}} - N = O(\epsilon^5)$ *uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.*

Theorem 1.29 (P-PLANAR: two planar planets with a moving Sun)**.** *Let* $G, m_0 > 0$*; let* $m, a : \mathtt{Fin}\,2 \to \mathbb{R}$ *(planets* $0, 1$*;* m *the planet weights,* a *the axes) with* $m_i > 0$ *and* $a_i > 0$ *for* $i = 0, 1$ *and* $a_0 < a_1$ *(strictly ordered axes, no numerical separation); let* $\eta > 0$*. Then there exist* μ_0, d, D *with* $0 < \mu_0$ *and* $0 < d < D$ *such that for every* μ *with* $0 < \mu < \mu_0$ *there is* $S \subset \mathrm{Ph}(2 \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

The order is: data (G, m_0, m, a, η)*, then* (μ_0, d, D)*, then* μ*, then* S*. The tori are* 4*-dimensional in the* 8*-dimensional* $\mathrm{Ph}(4)$.

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (`statement.md`*, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).*

Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see `PLAN.md`*.*

Theorem 1.30 (P-FIXED: n planar planets around a fixed Sun)**.** *Let* $n \geq 2$, $G, m_0 > 0$, $m, a \in \mathbb{R}^n$ *with all* $m_i > 0$ *and* $a_i > 0$, $a_i < a_j$ *whenever* $i < j$*, and* $\eta > 0$*. Then there exist* ε_0, d, D *with* $0 < \varepsilon_0$ *and* $0 < d < D$ *such that for every* ε *with* $0 < \varepsilon < \varepsilon_0$ *there is* $S \subset \mathrm{Ph}(n \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

The order is data (n, G, m_0, m, a, η)*, then* (ε_0, d, D)*, then* ε*, then* S.

Frozen Lean declaration: `KAM.Planetary.fixedSunPlanetaryKAM`

Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.

Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.

Remark 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.

2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville

Track P produces tori in coordinates that are not the frozen Cartesian ones: CP's action–angle variables with angles of period 2π, oscillator action–angle discs, the Poincaré chart. This part supplies, once, the four tools that carry a torus found in such a chart to a frozen predicate ([P:`bpP:p0:cyltorus`], [P:`bpP:p0:phystorus`]):

- `torus-transport` (Theorem 2.9): an analytic embedding Ψ with $\Psi^*\Omega = c\,\Omega$ that is compatible with the angle lattice sends a lattice torus of $(H \circ \Psi)/c$ to a torus of H, clause by clause of the two frozen predicates;
- `period-convert` (Lemma 2.11 and Corollaries 2.12, 2.13): the passage between 2π-angles and the frozen period-one parameter is the chart $S_{2\pi}$, so the frozen frequency is $w = \tilde{\omega}/2\pi$ and the dictionary of [P:`bpP:p3:period-dictionary`] is a special case;
- `diophantine-convert` (Lemmas 2.15–2.16): CP10 (19) and CP09 (6) imply the frozen `KAM.IsDiophantine` with the exponents the two torus predicates require;
- `measure-symplectic` (Lemmas 2.18–2.20): conformally symplectic charts scale Lebesgue measure by c^d, and a union of whole tori that is a Lipschitz image is measurable with controlled measure.

The `ode` item is cited, not repeated (§2.7). Everything here is standard; the point of writing it is to fix signs, lattices and Lean totality exactly as frozen.

2.1. **Conventions and external facts.** Phase space is $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$, points $z = (z_1, z_2)$ (position, momentum), max norms, Lebesgue measure ([P:`bpP:p0:conventions`]). Let $M_0 = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $J_0 = -M_0$. The frozen form ([P:`bpP:p0:hamsymp`]) is $\Omega(v, w) = \sum_i (v_{2,i} w_{1,i} - v_{1,i} w_{2,i}) = v^\top M_0 w$. For a linear form $\ell : \mathrm{Ph}(d) \to \mathbb{R}$ put $X_\ell := \big((\ell(0, e_i))_i, (-\ell(e_i, 0))_i\big)$; thus $X_H(z) = X_{DH(z)}$, and $X_H(z) = 0$ if H is not differentiable at z (Lean totality, [P:`bpP:p0:totality`]). Write $\iota_k := \big((k_i)_{i<d}, 0\big) \in \mathrm{Ph}(d)$ for $k \in \mathbb{Z}^d$ (integers coerced to reals: an integer translation of the positions; the nested form $((k_f, 0_b), 0)$ is reserved for the split $d = a + b$ and for `fastShift`) and, for $c > 0$, $S_c(z_1, z_2) := (c z_1, z_2)$.

(X1) Chain rule for `fderiv` and `HasDerivAt`; compositions of real analytic maps are real analytic (`AnalyticAt.comp` and `AnalyticOnNhd.comp`, `Analysis/Analytic/Composition.lean`; the latter needs the range of the inner map inside the set where the outer map is analytic, here $K(\mathbb{R}^d) \subset O$); sums and products of analytic maps are analytic.

(X2) Inverse function theorem: a C^1 map Ψ with $D\Psi(z_0)$ invertible is a homeomorphism from a neighbourhood of z_0 onto an open set, with C^1 inverse (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph` and `HasStrictFDerivAt.to_localInverse` (`Analysis/Calculus/InverseFunctionTheorem/FDeriv.lean`); for C^1 regularity of the inverse `ContDiffAt.to_localInverse` (`.../ContDiff.lean`). A C^1 map has a strict derivative, and a bijective linear map of $\mathbb{R}^{2d}$ is a continuous linear equivalence).

(X3) Area formula (`MeasureTheory/Function/Jacobian.lean`; the derivative field is a family of continuous linear maps with `HasFDerivWithinAt` on the set, supplied by analyticity of Ψ on O): `MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv` (*A Borel*, no

injectivity) gives $\operatorname{meas}\Psi(A) \le \int_A |\det D\Psi|$; `MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image` ($A$ Borel, Ψ injective on A) and its subscript-0 version (name with the suffix `image`$_0$, i.e. `addHaar_image` followed by the subscript-zero character; A `NullMeasurableSet`) give equality. Also `addHaar_image_eq_zero_of_differentiableOn_of_addHaar_eq_zero`: differentiable images of null sets are null.

(X4) A Lebesgue measurable subset of $\mathbb{R}^N$ is the union of an F_σ set and a null set (inner regularity, `MeasureTheory/Measure/Regular.lean`); a closed set is σ-compact; continuous images of compact sets are compact. *Null sets need not be Borel*: "measurable" below means `NullMeasurableSet` (Lebesgue) unless the word Borel is used, and the frozen predicates need `MeasurableSet`, i.e. Borel. Injective continuous images of Borel sets are Borel: `MeasurableSet.image_of_continuousOn_injOn` (`MeasureTheory/Constructions/Polish/Basic.lean`).

(X5) $\det(AB) = \det A \det B$, $\det A^\top = \det A$, and $\det M_0 = 1$.

Track K: [K:`bp:p1:holomorphic-analytic`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:lip-image`], [K:`bp:p1:mcshane`].

Lemma 2.1 (Algebra of Ω and X_H)**.** *(a) $M_0^\top = -M_0$, $M_0^2 = -\mathbf{1}$, Ω is nondegenerate, and $\Omega(w, X_\ell) = \ell(w)$ for all w and all linear ℓ. Hence X_ℓ is the unique vector with this property, and $\Omega(w, X_H(z)) = DH(z)w$ for every z (both sides vanish if H is not differentiable at z). (b) If A is a real $2d \times 2d$ matrix with $\Omega(Au, Av) = c\,\Omega(u, v)$ for all u, v ($c \neq 0$), then A is invertible, $|\det A| = |c|^d$, and $X_{\ell\circ A} = c\,A^{-1}X_\ell$ for every linear form ℓ. (c) The relation $\Omega(Au, Av) = c\Omega(u, v)$ is equivalent to $A^\top(-M_0)A = c(-M_0)$: it does not depend on the sign convention for the symplectic form (CP10's $dI \wedge d\varphi$ is Ω; CP09 Thm. 3 prints $dx \wedge dy$ with x the angles, which is $-\Omega$; both define the same class of maps, and the vector field is the frozen X_H, not a function of the printed sign).*

Proof. **Step S2.1: Matrices.** Direct from the block form; $X_\ell = J_0\nabla\ell$ so $M_0X_\ell = -M_0^2\nabla\ell = \nabla\ell$, i.e. $\Omega(w, X_\ell) = w^\top M_0 X_\ell = \ell(w)$. Uniqueness: M_0 invertible.

Step S2.2: (b). $A^\top M_0 A = cM_0$ and (X5) give $(\det A)^2 = c^{2d}$. For all w: $\Omega(w, cA^{-1}X_\ell) = \Omega(Aw, AA^{-1}X_\ell) = \Omega(Aw, X_\ell) = \ell(Aw) = (\ell \circ A)(w)$, so $cA^{-1}X_\ell = X_{\ell\circ A}$ by uniqueness. (c) is the displayed matrix identity. □

Remark 2.2 (Reordering of coordinates). Part 3 writes points as (J, θ) (action, angle). The relabelling $\pi(J, \theta) = (\theta, J)$ to the frozen order (position, momentum) = (angle, action) turns $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ into X_H of [P:`bpP:p0:hamsymp`] and the form $dJ \wedge d\theta$ into Ω; no sign changes. All statements of Part 3 are read in the frozen order after applying π.

2.2. Conformally symplectic charts.

Definition 2.3 (Chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $c > 0$. A *c-chart on O* is a map $\Psi : \mathrm{Ph}(d) \to \mathrm{Ph}(d)$ (total) that is real analytic at every point of O and satisfies $\Omega(D\Psi(z)u, D\Psi(z)v) = c\,\Omega(u, v)$ for $z \in O$, $u, v \in \mathrm{Ph}(d)$. A 1-chart is a *symplectic* chart. For $H : \mathrm{Ph}(d) \to \mathbb{R}$ the *pulled-back Hamiltonian* is $H^\Psi := c^{-1}\,(H \circ \Psi)$.

By Lemma 2.1(b), $D\Psi(z)$ is invertible on O, so Ψ is a local analytic diffeomorphism there ((X2); analyticity of the local inverse is not needed below).

Lemma 2.4 (Charts and Hamiltonian fields)**.** *Let Ψ be a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $z \in O$. Then H^Ψ is differentiable at z iff H is differentiable at $\Psi(z)$, and then*

$$D\Psi(z)\, X_{H^\Psi}(z) = X_H(\Psi(z)).$$

If H is not differentiable at $\Psi(z)$ then both fields are 0 (totality) and the identity still holds.

Proof. **Step S2.3: Differentiability.** ($\Leftarrow$) chain rule (X1). ($\Rightarrow$) by (X2) there is a C^1 local inverse Ψ_{loc}^{-1} near $\Psi(z)$ and $H = c\,H^{\Psi} \circ \Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$, a composition of differentiable maps.

Step S2.4: The field. With $\ell = DH(\Psi z)$ and $A = D\Psi(z)$, $D(H^{\Psi})(z) = c^{-1}\ell \circ A$, so $X_{H^{\Psi}}(z) = c^{-1}X_{\ell\circ A} = A^{-1}X_{\ell}$ by Lemma 2.1(b). Hence $AX_{H^{\Psi}}(z) = X_{\ell} = X_H(\Psi z)$. In the non-differentiable case both $D(H^{\Psi})(z)$ and $DH(\Psi z)$ are 0 by the Lean convention. □

Corollary 2.5 (Orbits transfer)**.** *Let $y : J \to O$ ($J \subset \mathbb{R}$ an interval) satisfy: H^{Ψ} is differentiable at $y(t)$ and y has derivative $X_{H^{\Psi}}(y(t))$ at t, for all $t \in J$. Then $x = \Psi \circ y$ has derivative $X_H(x(t))$ at every $t \in J$ and H is differentiable at $x(t)$.*

Proof. Chain rule (X1) and Lemma 2.4. □

Remark 2.6 (What is new compared with track N). [N:bpN:p5:symplectic] and [N:bpN:p5:canonical](b),(c) give $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ and the orbit transfer for the time-one map $\Phi = X_{\phi}^1$ of a *holomorphic* generating function on a complex domain, with symplecticity *derived* from the flow. Here the chart is an arbitrary real analytic map (Poincaré chart, polar chart of oscillators, scalings) with symplecticity *assumed*, in the frozen real setting, with the factor c, with the frozen sign of Ω (Lemma 2.1) and with the Lean-totality guard (differentiability of H is transported, not assumed). The algebra is that of [N:bpN:p5:canonical](b); it is repeated only because the hypotheses differ. Time-one maps of Part N are charts in the sense of Definition 2.3 with $c = 1$ (Lemma 2.7).

Lemma 2.7 (Holomorphic charts, real restriction)**.** *Let $c > 0$, $V \subset \mathbb{C}^{2d}$ be open, $\Psi : V \to \mathbb{C}^{2d}$ holomorphic, real at real points ($\Psi(V \cap \mathbb{R}^{2d}) \subset \mathbb{R}^{2d}$), and assume $D\Psi(z)^{\top} M_0 D\Psi(z) = cM_0$ (complex matrices) for $z \in V \cap \mathbb{R}^{2d}$. Identify $\mathbb{R}^{2d} \subset \mathbb{C}^{2d}$ through the real embedding; $O = V \cap \mathbb{R}^{2d}$ is open in* $\mathrm{Ph}(d)$*. Then any total map $\mathrm{Ph}(d) \to \mathrm{Ph}(d)$ that agrees with Ψ on O (choose any extension off O; no property off O is ever used) is a c-chart on O, with real derivative $D\Psi(z)|_{\mathbb{R}^{2d}}$.*

Proof. [K:bp:p1:real-restriction] (via [K:bp:p1:holomorphic-analytic]) gives analyticity and that the real derivative is the restriction of the complex one; the identity of matrices is the defining condition of Definition 2.3 (Lemma 2.1(c)). □

2.3. **Transport of tori.**

Definition 2.8 (Lattice torus in a chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $G : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$, $K : \mathbb{R}^d \to \mathrm{Ph}(d)$. $\mathrm{LT}(G, O, w, K)$ holds if

- (L1) K is real analytic on $\mathbb{R}^d$, $K(\theta) \in O$ and G is differentiable at $K(\theta)$ for all θ;
- (L2) $K(\theta + k) = K(\theta) + \iota_k$ for $\theta \in \mathbb{R}^d$, $k \in \mathbb{Z}^d$;
- (L3) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
- (L4) $DK(\theta)$ is injective;
- (L5) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all u, v;
- (L6) $s \mapsto K(\theta + sw)$ has derivative $X_G(K(\theta + tw))$ at $s = t$, for all θ, t.

Note that (L2) is the period-one form of "angles are shifted by integers" in the chart; the frozen cylinder predicate differs from it only by the image shift in clause (iii), and the frozen physical predicate by periodicity.

Theorem 2.9 (Torus transport)**.** *Let $c > 0$, $O \subset \mathrm{Ph}(d)$ open, Ψ a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $G = H^{\Psi} = c^{-1}H \circ \Psi$, $w \in \mathbb{R}^d$, and K with $\mathrm{LT}(G, O, w, K)$. Assume the lattice compatibility*

- *(Q1) $O + \iota_k = O$ and $\Psi(z + \iota_k) = \Psi(z) + \sigma_k$ for $z \in O$, $k \in \mathbb{Z}^d$, for vectors $\sigma_k \in \mathrm{Ph}(d)$;*
- *(Q2) (injectivity modulo the lattice) if $z, z' \in O$, $k \in \mathbb{Z}^d$ and $\Psi(z') = \Psi(z) + \sigma_k$, then $z' = z + \iota_m$ for some $m \in \mathbb{Z}^d$.*

Put $\mathcal{K} := \Psi \circ K$. Then:

(a) (cylinder) *Let* $d = a + b$ *and* $\sigma_k = \mathrm{shift}((k_i)_{i<a})$ *(Definition 1.8). If* $U \supset \Psi(K(\mathbb{R}^d))$*, then* $\mathrm{IsCylTorus}(H, U, w, \mathcal{K})$ *(Definition 1.15).*

(b) (physical) *Let* $\sigma_k = 0$ *and assume* w *is* (γ, τ)*-Diophantine for some* $\gamma > 0$*,* $\tau > d - 1$*. Then* $\mathrm{IsPhysTorus}(H, w, \mathcal{K})$ *(Definition 1.24).*

In both cases $\mathcal{K}(\theta + tw)$ *is the transport of the chart orbit,* $\mathcal{K}(\mathbb{R}^d) = \Psi(K(\mathbb{R}^d))$*, and* $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$*.*

Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus, KAM.Planetary.IsPhysicalTorus`

Source: Standard (change of canonical coordinates). Ours as a statement for the frozen predicates; no source page.

Proof. Write $\mathcal{K} = \Psi \circ K$.

Step S2.5: (i) Analyticity, range, differentiability. $K(\mathbb{R}^d) \subset O$ and Ψ is analytic on O, so $\mathcal{K}$ is analytic on $\mathbb{R}^d$ by (X1). $G = H^\Psi$ is differentiable at $K\theta$ by (L1), hence H is differentiable at $\mathcal{K}(\theta)$ by Lemma 2.4. Range: $\mathcal{K}(\theta) \in \Psi(K(\mathbb{R}^d)) \subset U$ in (a).

Step S2.6: (ii) Lattice behaviour. By (L2) and (Q1): $\mathcal{K}(\theta + k) = \Psi(K\theta + \iota_k) = \mathcal{K}(\theta) + \sigma_k$. In (a) this is clause (iii) of Def. 1.15 (with $k \in \mathbb{Z}^{a+b}$, shift by $k_f = (k_i)_{i<a}$); in (b) it is periodicity, clause (iii) of Def. 1.24.

Step S2.7: (iii) Injectivity modulo the lattice. (a): let $k \in \mathbb{Z}^a$ and $\mathcal{K}(\theta') = \mathcal{K}(\theta) + \mathrm{shift}(k)$. Then $\Psi(K\theta') = \Psi(K\theta) + \sigma_{(k,0)}$, so by (Q2) $K\theta' = K\theta + \iota_m = K(\theta + m)$ for some $m \in \mathbb{Z}^d$ (by (L2)), and (L3) gives $\theta' - (\theta + m) \in \mathbb{Z}^d$, i.e. $\theta' - \theta \in \mathbb{Z}^d$. (b): $\mathcal{K}(\theta') = \mathcal{K}(\theta)$ is the case $k = 0$ ($\sigma_0 = 0$) of the same argument.

Step S2.8: (iv) Derivative and Lagrangian property. $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$; $D\Psi(K\theta)$ is invertible (Lemma 2.1(b)), so $D\mathcal{K}(\theta)$ is injective by (L4). $\Omega(D\mathcal{K}u, D\mathcal{K}v) = c\,\Omega(DKu, DKv) = 0$ by Definition 2.3 and (L5).

Step S2.9: (v) The flow clause. Fix θ, t and let $y(s) = K(\theta + sw) \in O$. By (L6) y has derivative $X_G(y(t))$ at t and G is differentiable at $y(t)$; Corollary 2.5 (pointwise) gives that $s \mapsto \Psi(y(s)) = \mathcal{K}(\theta + sw)$ has derivative $X_H(\mathcal{K}(\theta + tw))$ at t.

Step S2.10: (vi) Diophantine clause. In (b) it is the hypothesis; in (a) none is part of IsCylTorus (see Def. 1.16 for where it is imposed; Lemmas 2.15–2.16). □

Remark 2.10 (Typical charts). (1) The polar chart of one oscillator $(\psi, R) \mapsto (q, p) = \sqrt{2R}(\sin\psi, \cos\psi)$ on $\{R > 0\}$ is a 1-chart ($dp \wedge dq = dR \wedge d\psi$, which is Ω in the frozen order), invariant under $\psi \mapsto \psi + 2\pi$; Part 5 composes it with $S_{2\pi}$ below. (2) S_c is a c-chart (Lemma 2.11). (3) Composition: if Ψ_1 is a c_1-chart on O_1, Ψ_2 a c_2-chart on $O_2 \supset \Psi_1(O_1)$, then $\Psi_2 \circ \Psi_1$ is a c_1c_2-chart on O_1 (chain rule), and $(H \circ \Psi_2)^{\Psi_1}$ with the factor c_1^{-1} followed by c_2^{-1} equals $H^{\Psi_2 \circ \Psi_1}$.

2.4. **Period conversion.** CP10 (footnote 1, p. 546: "$\mathbb{T}^n$ denotes the standard n-dimensional flat torus $\mathbb{R}^n/(2\pi\mathbb{Z}^n)$") and CP09 (Thm. 3, p. 6: "$\mathbb{T}^n$ the usual n-dimensional torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$") use angles of period 2π. The frozen predicates use a parameter $\theta \in \mathbb{R}^d$ of period one. The relation is one explicit chart.

Lemma 2.11 (Period conversion)**.** *Let* $c > 0$ *and* $S_c(z_1, z_2) = (cz_1, z_2)$*.*

(a) S_c *is a* c*-chart on all of* $\mathrm{Ph}(d)$ *(globally analytic,* $\Omega(S_cu, S_cv) = c\,\Omega(u, v)$*),* $S_c(z + \iota_k) = S_c(z) + (ck, 0)$*, and* $|\det DS_c| = c^d$*.*

(b) (Hamiltonian dictionary) *For* $\tilde{H} : \mathrm{Ph}(d) \to \mathbb{R}$ *the pulled-back Hamiltonian is* $\tilde{H}^{S_c}(z_1, z_2) = c^{-1}\tilde{H}(cz_1, z_2)$*, and for every* z*:* $S_c\, X_{\tilde{H}^{S_c}}(z) = X_{\tilde{H}}(S_cz)$*. With* $c = 2\pi$ *and* $(I, \phi) = (z_2, 2\pi z_1)$ *this is* $H(I, \theta) = \tilde{H}(I, 2\pi\theta)/2\pi$ *of Lemma 4.5: orbits of* H *(angle* θ*) and of* $\tilde{H}$ *(angle* $\phi = 2\pi\theta$*) correspond with the* same *time;* $\omega = \tilde{\omega}/2\pi$*; Lagrangian for* Ω *iff after* S_c *(factor* c*).*

(c) (Parameter conversion) *Let* $\tilde\kappa : \mathbb{R}^d \to \mathrm{Ph}(d)$ *(parameter of period* 2π*),* $\tilde\omega \in \mathbb{R}^d$*,* $K(\theta) := \tilde\kappa(2\pi\theta)$*,* $w := \tilde\omega/2\pi$*. Then* K *analytic iff* $\tilde\kappa$ *is;* $DK(\theta) = 2\pi D\tilde\kappa(2\pi\theta)$*, so injectivity of* DK *and the Lagrangian property (factor* $4\pi^2$*) hold for* K *iff for* $\tilde\kappa$*;* $\tilde\kappa(\vartheta+2\pi k) = \tilde\kappa(\vartheta)+\tau_k$ *for all* ϑ, k *iff* $K(\theta+k) = K(\theta)+\tau_k$*;* $\tilde\kappa(\vartheta) = \tilde\kappa(\vartheta') \Rightarrow \vartheta'-\vartheta \in 2\pi\mathbb{Z}^d$ *iff* $K(\theta) = K(\theta') \Rightarrow \theta'-\theta \in \mathbb{Z}^d$*; and* $s \mapsto K(\theta+sw)$ *has derivative* v *at* t *iff* $s \mapsto \tilde\kappa(2\pi\theta + s\tilde\omega)$ *has derivative* v *at* t*. In particular, for every Hamiltonian* F*, the frozen flow clause for* (K, w) *is CP's flow clause for* $(\tilde\kappa, \tilde\omega)$ *and the frozen frequency is* $w = \tilde\omega/2\pi$*.*

(d) (Frozen shift) *The frozen fast shift* $\mathrm{shift}(k_f) = ((2\pi k_f, 0_b), 0)$ *(Def. 1.8) equals* $S_{2\pi}$ *applied to* $((k_f, 0_b), 0)$*.*

Proof. **Step S2.11: (a).** $DS_c = \mathrm{diag}(c\mathbf{1}, \mathbf{1})$, so $\Omega(S_cu, S_cv) = \sum_i(u_{2,i}\,cv_{1,i} - cu_{1,i}v_{2,i}) = c\Omega(u, v)$; $\det = c^d$; the lattice formula is linearity.

Step S2.12: (b). Lemma 2.4 gives $DS_c\, X_{\tilde H^{S_c}} = X_{\tilde H} \circ S_c$. Explicitly $\partial_{z_2}\tilde H^{S_c} = c^{-1}\partial_2\tilde H$, $\partial_{z_1}\tilde H^{S_c} = \partial_1\tilde H$, which are Part 3's equations $\dot\theta = \partial_I H = \dot\phi/2\pi$, $\dot I = -\partial_\theta H = -\partial_\phi\tilde H$.

Step S2.13: (c). All claims are composition with the linear bijection $\theta \mapsto 2\pi\theta$ of $\mathbb{R}^d$ (the chain rule is not even needed for the flow clause: $K(\theta + sw) = \tilde\kappa(2\pi\theta + s\tilde\omega)$ exactly). (d) is (a). □

Corollary 2.12 (Cylinder pipeline (consumed by Part 5))**.** *Let* $d = a + b$*,* $H : \mathrm{Ph}(d) \to \mathbb{R}$*,* Φ *a* 1*-chart on an open* $O_0 \subset \mathrm{Ph}(d)$ *with* $O_0 + (2\pi k, 0) = O_0$ *and* $\Phi(z + (2\pi k, 0)) = \Phi(z) + \mathrm{shift}(k_f)$ *for* $z \in O_0$*,* $k \in \mathbb{Z}^d$*, and (*$\Phi(z') = \Phi(z) + \mathrm{shift}(k_f) \Rightarrow z' = z + (2\pi m, 0)$ *for some* $m \in \mathbb{Z}^d$*); put* $O = S_{2\pi}^{-1}(O_0)$*,* $\Psi = \Phi \circ S_{2\pi}$*,* $G_3 = H^\Psi = (2\pi)^{-1}(H \circ \Phi)(2\pi z_1, z_2)$*. If* $\mathrm{LT}(G_3, O, w, K_3)$ *(for instance,* K_3 *the torus of Prop. 4.3(a)–(d) with* π *of Remark 2.2 applied, and the injectivity modulo translations of that proposition for (L3)) and* $U \supset \Psi(K_3(\mathbb{R}^d))$*, then* $\mathrm{IsCylTorus}(H, U, w, \Psi \circ K_3)$*,* with the same w*: the frozen frequency equals Part 3's frequency* $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$*, and CP's own angular frequency is* $2\pi w$*.*

Proof. Ψ is a 2π-chart on O (Remark 2.10(3)); (Q1) holds because $\Psi(z+\iota_k) = \Phi(S_{2\pi}z+(2\pi k, 0)) = \Psi(z)+\mathrm{shift}(k_f)$; (Q2) because $\Psi(z') = \Psi(z)+\mathrm{shift}(k_f)$ gives $S_{2\pi}z' = S_{2\pi}z+(2\pi m, 0)$, i.e. $z' = z+\iota_m$. Apply Theorem 2.9(a). □

Corollary 2.13 (Physical pipeline (consumed by Parts 12, 13))**.** *Let* $H : \mathrm{Ph}(d) \to \mathbb{R}$*,* Φ *a* 1*-chart on an open* O_0 *with* $O_0+(2\pi k, 0) = O_0$*,* $\Phi(z+(2\pi k, 0)) = \Phi(z)$ *and (*$\Phi(z') = \Phi(z) \Rightarrow z' = z+(2\pi m, 0)$*, some* $m \in \mathbb{Z}^d$*). Let* $\tilde\kappa : \mathbb{R}^d \to O_0$ *be a torus in the* 2π *convention of CP09 Thm. 3 for* $\tilde G = H \circ \Phi$*: i.e. the clauses (L1)–(L5) hold for* $\tilde\kappa$ *with period* 2π *as in Lemma 2.11(c) (*$\tau_k = (2\pi k, 0)$*) and* $s \mapsto \tilde\kappa(\vartheta+s\tilde\omega)$ *has derivative* $X_{\tilde G}(\tilde\kappa(\vartheta+t\tilde\omega))$ *at* t*, and* $\tilde\omega$ *is* $(\tilde\gamma, \tau)$*-Diophantine with* $\tilde\gamma > 0$ *and* $\tau > d-1$ *(the positivity of* $\tilde\gamma$ *is a hypothesis:* `KAM.IsDiophantine` *itself has no positivity guard, and* $\tilde\omega = 0$ *is* $(0, \tau)$*-Diophantine). Then* $\mathrm{IsPhysTorus}(H, \tilde\omega/2\pi,\ \theta \mapsto \Phi(\tilde\kappa(2\pi\theta)))$*, and* $K(\mathbb{R}^d) = \Phi(\tilde\kappa(\mathbb{R}^d))$*.*

Proof. By Lemma 2.11(c), $K_0(\theta) = \tilde\kappa(2\pi\theta)$ satisfies (L1)–(L6) for $G = \tilde G$ and $w = \tilde\omega/2\pi$ with ι_k replaced by τ_k; the argument of Theorem 2.9(b) with $\Psi = \Phi$ (a 1-chart, $\sigma = 0$, lattice τ_k in place of ι_k) applies verbatim. The frequency w is $(\tilde\gamma/2\pi, \tau)$-Diophantine with $\tilde\gamma/2\pi > 0$ (Lemma 2.15(c)). □

2.5. **Diophantine conversion.**

Definition 2.14 (Two-scale Diophantine set)**.** For $n = n_1 + n_2$, $\gamma_1, \gamma_2, \tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is the set of $\omega = (\omega_1, \omega_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $|k \cdot \omega| \geq \gamma_1|k|_1^{-\tau}$ whenever $k_1 \neq 0$ and $|k \cdot \omega| \geq \gamma_2|k|_1^{-\tau}$ whenever $k_1 = 0$, $k \neq 0$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$). This is the set of Part 3 (§4.1); for $k_1 = 0$, $|k|_1 = |k_2|_1$.
Source: CP10 (19), p. 551 (with $\gamma_2 = \mu\bar\gamma_2$; norm $|k| = \sum|k_j|$, CP10 footnote 5, p. 547), and the definition of $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, p. 551.

Lemma 2.15 (Elementary properties of `KAM.IsDiophantine`)**.** *Let* $\omega \in \mathbb{R}^d$*,* $\gamma, \tau > 0$ *(the frozen definition is* $\gamma/|k|_1^\tau \leq |k \cdot \omega|$ *for* $k \neq 0$*, Def. 1.1).*

(a) *(monotone)* (γ,τ)*-Diophantine implies* (γ',τ')*-Diophantine for* $0<\gamma'\le\gamma$, $\tau'\ge\tau$. *In particular* $\tau>d-1$ *and* $\tau>a+b$ *can be arranged by raising* τ.
(b) *(norms) If* $|k\cdot\omega|\ge\gamma\,\nu(k)^{-\tau}$ *for all* $k\neq0$ *with* $0<\nu(k)\le|k|_1$ *(e.g.* $\nu=|\cdot|_2$ *or* $|\cdot|_\infty$*, which are positive on* $k\neq0$*), then* ω *is* (γ,τ)*-Diophantine.*
(c) *(scaling) For* $s>0$*:* ω *is* (γ,τ)*-Diophantine iff* ω/s *is* $(\gamma/s,\tau)$*-Diophantine.*
(d) *(two scales)* $\mathcal{D}_{\gamma_1,\gamma_2,\tau}\subset\{\omega:\ \omega$ *is* $(\min\{\gamma_1,\gamma_2\},\tau)$*-Diophantine*$\}$*; in particular, if* $\gamma_1\ge\gamma_2$*, the set of* (γ_2,τ)*-Diophantine vectors.*

Proof. (a) For $k\neq0$, $|k|_1\ge1$, so $|k|_1^{\tau}\le|k|_1^{\tau'}$ (real powers of a base ≥1 are nondecreasing in the exponent; the Lean base is ≥1, so no totality issue). (b) On the positive bases $0<\nu(k)\le|k|_1$ one has $\nu(k)^{-\tau}\ge|k|_1^{-\tau}$ ($\tau>0$). The positivity guard is necessary: Lean has $0^{-\tau}=0$, and $\nu\equiv0$ would satisfy the hypothesis for $\omega=0$. (c) $|k\cdot\omega/s|=|k\cdot\omega|/s\ge(\gamma/s)|k|_1^{-\tau}$. (d) For each $k\neq0$ the bound in the definition is one of the two lines, each $\ge\min\{\gamma_1,\gamma_2\}|k|_1^{-\tau}$. □

Lemma 2.16 (CP10 and CP09 give the frozen Diophantine clauses)**.** *(a)* (CP10, cylinder tori) *Let* $\omega\in\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ *with* $\gamma_1,\gamma_2>0$ *and* $\tau>n=a+b$ *(CP10:* $\tau_*>n$*, footnote 12, p. 550; Prop. 4.3 assumes* $\tau>n$*). Then* ω *is* $(\min\{\gamma_1,\gamma_2\},\tau)$*-Diophantine, so the clause "*$\exists\gamma>0,\tau>a+b$*,* w (γ,τ)*-Diophantine" of Def. 1.16 holds for* $w=\omega$*. If* $w=\tilde\omega/2\pi$ *with* $\tilde\omega\in\mathcal{D}_{\tilde\gamma_1,\tilde\gamma_2,\tau}$ *then* $w\in\mathcal{D}_{\tilde\gamma_1/2\pi,\tilde\gamma_2/2\pi,\tau}$.
(b) (CP09, physical tori) *Let* $\tilde\omega=\omega^\star(b)$ *satisfy* $|\langle k,\tilde\omega\rangle|\ge\gamma|k|_2^{-\tau}$ *for all* $k\in\mathbb{Z}^n\setminus\{0\}$ *(CP09 Thm. 3(iv), (6), p. 7:* $|k|_2$ *the Euclidean norm), with* $\gamma>0$ *and* $\tau>n\mu\ge n$ *(*$\mu\in\mathbb{Z}_{\ge1}$ *by CP09 Def. 2, p. 6). Then* $\tilde\omega$ *is* (γ,τ)*-Diophantine,* $w=\tilde\omega/2\pi$ *is* $(\gamma/2\pi,\tau)$*-Diophantine, and* $\tau>n-1$*: the clause (vii) of Def. 1.24 holds with* $d=n$.

Proof. (a) Lemma 2.15(d) and (c). (b) Lemma 2.15(b) with $\nu=|\cdot|_2\le|\cdot|_1$, then (c); $\tau>n\mu\ge n>n-1$. □

Remark 2.17 (Dimension). For the planar theorems the torus has $d=2n$ frequencies; CP09 Thm. 3 is applied with its n equal to this d, so its $\tau>n\mu\ge d$ exceeds $d-1$ as the frozen clause requires.

2.6. **Liouville measure and unions of tori.**

Lemma 2.18 (Charts scale Lebesgue measure)**.** *Let* Ψ *be a* c*-chart on* O *and* $A\subset O$ *Lebesgue measurable. Then* $\Psi(A)$ *is Lebesgue measurable (null-measurable) and* $\operatorname{meas}\Psi(A)\le c^d\operatorname{meas}A$*, with equality if* Ψ *is injective on* A*. An* injective *symplectic chart (*$c=1$*) preserves the measure of* A*; without injectivity only the upper bound is asserted (the polar chart of Remark 2.10(1) maps* $[0,4\pi)\times[1,2]$*, of area* 4π*, onto an annulus of area* 2π*). If* A *is Borel then* $\Psi(A)$ *is Borel.*

Proof. **Step S2.14: Determinant.** $|\det D\Psi|=c^d$ on O (Lemma 2.1(b)).

Step S2.15: Measurability. By (X2) cover O by countably many open O_j on which Ψ is a homeomorphism onto an open set with continuous inverse Ψ_j^{-1} (Lindelöf). Write $A\cap O_j=B\cup N$, B F_σ, N null ((X4)); $\Psi(B)$ is σ-compact, $\Psi(N)$ is null because Ψ is locally Lipschitz ([K:`bp:p1:lip-image`] on small cubes, countable union). So $\Psi(A)=\bigcup_j\Psi(A\cap O_j)$ is Lebesgue measurable.

Step S2.16: Borel sets. If A is Borel, $\Psi(A\cap O_j)=(\Psi_j^{-1})^{-1}(A\cap O_j)$ is the preimage of a Borel set under the continuous map Ψ_j^{-1} on the open set $\Psi(O_j)$, hence Borel; the countable union is Borel.

Step S2.17: Measure. For Borel A, (X3) gives $\operatorname{meas}\Psi(A)\le\int_A|\det D\Psi|=c^d\operatorname{meas}A$, with equality if Ψ is injective on A. For Lebesgue measurable A write $A=B\cup N$ with B Borel, N null: $\operatorname{meas}\Psi(A)\le\operatorname{meas}\Psi(B)+0$, and for equality under injectivity use the subscript-0 version of (X3). □

Lemma 2.19 (Quotient version for whole tori)**.** *Assume the hypotheses of Theorem 2.9 with* $\sigma_k=0$*. Let* $A\subset O$ *be measurable with* $A+\iota_k=A$ *for all* $k\in\mathbb{Z}^d$*, and* $Q=\{z:z_{1,i}\in[0,1)\ \forall i\}$*. Then*

$\Psi(A)$ is Lebesgue measurable, $\Psi(A) = \Psi(A \cap Q)$ and $\operatorname{meas}\Psi(A) = c^d \operatorname{meas}(A \cap Q)$. *Consequently* $0 < \operatorname{meas}\Psi(A)$ *iff* $0 < \operatorname{meas} A$, *and* $\operatorname{meas}\Psi(A) < \infty$ *if $A \cap Q$ is bounded (e.g. the momenta of A are bounded).*

Proof. $A = \bigcup_m (A \cap Q) + \iota_m$, and $\Psi((A \cap Q) + \iota_m) = \Psi(A \cap Q)$ by (Q1) with $\sigma = 0$. If $z, z' \in A \cap Q$ and $\Psi(z) = \Psi(z')$, (Q2) gives $z' = z + \iota_m$ and $z, z' \in Q$ forces $m = 0$: Ψ is injective on $A \cap Q$; Lemma 2.18 gives $\operatorname{meas}\Psi(A \cap Q) = c^d \operatorname{meas}(A \cap Q)$. *Borel version:* if A is Borel, so is $A \cap Q$, and Ψ is continuous and injective on it, so $\Psi(A) = \Psi(A \cap Q)$ is Borel by (X4) (`MeasurableSet.image_of_continuousOn_injOn`); this is the form needed for the frozen `MeasurableSet` clauses. Finally $\operatorname{meas} A = \sum_m \operatorname{meas}((A \cap Q) + \iota_m)$ is 0 iff $\operatorname{meas}(A \cap Q) = 0$. □

Lemma 2.20 (Unions of whole tori as Lipschitz images)**.** *Let $E \subset \mathbb{R}^d$ be Lebesgue measurable, $A = E \times [0,1)^d \subset \mathbb{R}^{2d}$, $\mathcal{F} : A \to \mathbb{R}^{2d}$ with $|\mathcal{F}(x) - \mathcal{F}(y)| \le \Lambda |x - y|$ (max norms). Then $\mathcal{F}(A)$ is Lebesgue measurable (`NullMeasurableSet`;* not *asserted Borel) and* $\operatorname{meas}\mathcal{F}(A) \le \Lambda^{2d} \operatorname{meas} E$. ***Borel consumers (the frozen predicates need `MeasurableSet`):*** *(B1) if E is σ-compact (e.g. compact, as $\mathcal{K}_\star$ in CP09 Thm. 3), then $\mathcal{F}(A)$ is σ-compact, hence Borel, with no injectivity needed; (B2) if E is Borel and $\mathcal{F}$ is continuous and injective on A, then $\mathcal{F}(A)$ is Borel ((X4)). In both cases the saturated union $\bigcup_k (\mathcal{F}(A) + \iota_k)$ below is a countable union of Borel sets. If moreover $|\mathcal{F}(x) - \mathcal{F}(y)| \ge \lambda |x - y|$ with $\lambda > 0$, then* $\operatorname{meas}\mathcal{F}(A) \ge \lambda^{2d} \operatorname{meas} E$; *in particular* $0 < \operatorname{meas}\mathcal{F}(A)$ *iff* $0 < \operatorname{meas} E$, *and* $\operatorname{meas}\mathcal{F}(A) < \infty$ *iff* $\operatorname{meas} E < \infty$. *If $\mathcal{F}(\omega, \vartheta) = \varphi_\omega(\vartheta)$ with $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + \iota_k$, then $\bigcup_{\omega \in E} \varphi_\omega(\mathbb{R}^d) = \bigcup_{k \in \mathbb{Z}^d} (\mathcal{F}(A) + \iota_k)$ is measurable.*

Proof. **Step S2.18: Extension.** McShane ([K:`bp:p1:mcshane`], componentwise in the max norm) gives a Λ-Lipschitz $\hat{\mathcal{F}} : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ extending $\mathcal{F}$.

Step S2.19: Measurability. A is measurable; $A = B \cup N$ with B F_σ, N null ((X4)); $\hat{\mathcal{F}}(B)$ is σ-compact and $\operatorname{meas}\hat{\mathcal{F}}(N) = 0$ by [K:`bp:p1:lip-image`]; so $\mathcal{F}(A) = \hat{\mathcal{F}}(B) \cup \hat{\mathcal{F}}(N)$ is Lebesgue measurable (completeness of Lebesgue measure; this does not give a Borel set).

Step S2.20: Borel cases. (B1): $E \times [0,1)^d$ is a countable union of compact sets and continuous images of compact sets are compact. (B2): (X4).

Step S2.21: Bounds. [K:`bp:p1:lip-image`]: $\operatorname{meas}\mathcal{F}(A) \le \Lambda^{2d} \operatorname{meas} A = \Lambda^{2d} \operatorname{meas} E$. Under the lower bound $\mathcal{F}$ is injective with λ^{-1}-Lipschitz inverse on $\mathcal{F}(A)$, so $\operatorname{meas} E = \operatorname{meas}\mathcal{F}^{-1}(\mathcal{F}(A)) \le \lambda^{-2d} \operatorname{meas}\mathcal{F}(A)$ (same lemma). The last claim is the periodicity and countability of $\mathbb{Z}^d$. □

Remark 2.21 (Use). *P-PLANAR/P-FIXED.* The chart-level family is $X_0(b, \xi)$ of CP09 Thm. 3(ii), bi-Lipschitz on $\mathcal{K}_\star \times \mathbb{T}^d$ with $\mathcal{K}_\star$ compact, so the chart-level union is compact (B1), hence Borel, and of positive finite measure (Lemma 2.20, applied *at the chart level*). The Cartesian union is its image under the symplectic chart Φ of Corollary 2.13; Lemma 2.19 gives meas = $\operatorname{meas}(A \cap Q_{2\pi})$ with $c = 1$ and the cell $Q_{2\pi} = \{z_{1,i} \in [0, 2\pi)\}$ (the lattice of the chart is $2\pi\mathbb{Z}^d$ there). In the period-one formulation $\Psi = \Phi \circ S_{2\pi}$ one has $c = 2\pi$ and the cell Q_1, and the two formulas agree because $S_{2\pi}$ multiplies cell volume by $(2\pi)^d$. The lower Lipschitz bound of Lemma 2.20 must *not* be applied to a wrapped (periodic) Cartesian embedding, which cannot satisfy it on a whole cell. Borelness of the Cartesian union follows from Lemma 2.19 (Borel version). *P-G.* The family is that of Prop. 4.3; its parameter set $E = \omega_0(D) \cap \mathcal{D}$ must be Borel (or σ-compact) for (B2)/(B1), and injectivity of the family modulo translations is part of that proposition; Part 5 must retain whole tori and Borelness when deleting parameters; the measure bounds of the cell are Part 4.

2.7. **The ODE layer (cited).** Track N's Part 3 supplies, for the three tracks, existence, uniqueness, continuation, energy and the identification of the frozen solution predicates. Track P uses only the following, always by these labels.

[N:bpN:p3:gen-uniqueness]	uniqueness on every interval for C^1 fields (orbits of a torus are *the* solutions; Lemma 2.22)
[N:bpN:p3:ham-smooth], [N:bpN:p3:ham-energy], [N:bpN:p3:ham-periodic]	X_H is C^1 where H is C^2; energy conservation; angle periodicity
[N:bpN:p3:newton-reduction], [N:bpN:p3:newton-continuation]	`IsNewtonSolution`, `IsFixedSolution` as `IsSolIn` (Part 6)
[N:bpN:p3:gen-continuation], [N:bpN:p3:gen-confinement]	only if a later part needs continuation (not used in Part 1)
[N:bpN:p5:symplectic], [N:bpN:p5:canonical]	time-one maps are symplectic; $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ (Remark 2.6)

Lemma 2.22 (Orbits on a torus)**.** *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$ be C^2 on an open $W \supset \mathcal{K}(\mathbb{R}^d)$ and let $\mathcal{K}, w$ satisfy the flow clause of Defs. 1.15 (vii) or 1.24 (viii). Then $t \mapsto \mathcal{K}(\theta + tw)$ is a solution of $\dot z = X_H(z)$ in W on $\mathbb{R}$, and every solution $y : \mathbb{R} \to W$ with $y(0) = \mathcal{K}(\theta)$ equals it. $H \circ \mathcal{K}(\cdot + tw)$ is constant in t.*

Proof. The flow clause is the solution property (with values in W by hypothesis). X_H is C^1 on W ([N:bpN:p3:ham-smooth]); uniqueness is [N:bpN:p3:gen-uniqueness] on $J = \mathbb{R}$; energy is [N:bpN:p3:ham-energy]. □

Remark 2.23 (Orbit transfer to Newton). That the frozen clause (b) of MovKol and the fixed-Sun clause hold along torus orbits is Part 6: the barycentric reconstruction is a linear map composed with the orbit, the chain rule is Corollary 2.5 or elementary, and the translation of the Newton predicates is [N:bpN:p3:newton-reduction]; nothing of the existence theory is used.

2.8. **Sources, deviations, forward references.**
Sources. CP10 footnote 1 (p. 546), (19) (p. 551), footnote 5 (p. 547) and footnote 12 (p. 550), all read; CP09 Def. 2 (p. 6), Thm. 3 and (6) (pp. 6–7), read. The rest is standard (the algebra of symplectic matrices, the inverse function theorem, the area formula).
Remarks on the sources. (1) CP09 Thm. 3 prints the symplectic form as $dx \wedge dy$ with $x \in \mathbb{T}^n$ the angles; with the frozen $\Omega = \sum dz_2 \wedge dz_1$ this is $-\Omega$; the class of symplectic maps and the Lagrangian property are unaffected (Lemma 2.1(c)), and the printed Hamiltonian field in CP09 Remark 3(i) ($\xi \mapsto \xi + \omega^\star t$) is the frozen one. (2) The exponent in CP09 (6) is typeset as $|k|_2^\tau$ with a Euclidean norm; we only use $|k|_2 \le |k|_1$. (3) CP09 Thm. 3 gives no statement that $X(b, \cdot, \cdot)$ is defined on all of $\mathbb{R}^{2n}$ or injective modulo 2π translations of ξ; (L3) for $X_0(b, \cdot)$ follows from (ii) (a homeomorphism of $\mathcal{K}_\star \times \mathbb{T}^n$ onto its image).
Forward references. [P:bpP:p5:kam-application] (cylinder pipeline), Parts 6, 12, 13 (physical pipeline, orbit transfer) cite Theorem 2.9 and Corollaries 2.12, 2.13. Part 3 cites [P:bpP:p1:period-convert] (Lemma 2.11; its Lemma 4.5 is the case (b) with $c = 2\pi$). Every other cited label exists in tracks K and N.
Label index. `bpP:p1:omega-algebra`, `chart`, `chart-field`, `chart-orbit`, `hol-chart`, `lattice-torus`, `torus-transport`, `period-convert`, `cyl-pipeline`, `phys-pipeline`, `two-scale`, `dio-basic`, `dio-sources`, `liouville`, `liouville-lattice`, `torus-union`, `torus-orbit`.

3. Averaging with two frequency scales

This part proves the averaging statement that Part 3 consumes, Lemma 4.8 (CP10 Proposition 1 with $m = 0$ and two non-resonance scales $\alpha_1 \ge \alpha_2$), and the averaging step of CP10 §4, Step 1 (p. 563) that Part 5 consumes. CP10 proves its Lemma A.1 only by reference to the unpinned [4, Lemma A.5]; here everything is derived from track N's Parts 4–6.
What track N gives, and what it does not. The one-scale Normal Form Lemma [N:bpN:p6:normal-form] cannot simply be applied with $\alpha = \alpha_2$, for four reasons, each of which is repaired below. (1) *Exponential averaging needs an iteration*: one homological solve and one flow

leave an angle-dependent quadratic remainder; the exponential bound $e^{-Ks/6}E$ comes from the finite iteration of [N:bpN:p6:iterative] (as in [N:bpN:p6:normal-form], CP10 p. 565), not from one step. (2) *The strip.* [N:bpN:p6:normal-form] shrinks the strip from S to $S/6$; the interface needs $\bar{s}+s \to \bar{s}+s/6$, that is, a loss budget s that is unrelated to the initial width. The iteration is therefore re-run with N's step lemma as a black box (Proposition 3.4). (3) *Two scales.* The first-action displacement with α_1 is not a consequence of the scalar statement (since α_1/α_2 is unbounded); it comes from the splitting of the generating function (Lemma 3.3, CP10 (70)–(71)). (4) *Hypotheses.* [N:bpN:p6:iterative] asks for non-resonance on the base set plus a Lipschitz bound L with $KLr \le \alpha/p$; Part 3 provides non-resonance on the whole complex neighbourhood, and no Lipschitz constant enters the statement. Lemma 3.2 records that N's proof only uses the divisor bound on the neighbourhood.

3.1. Conventions: norm, widths, dictionary.

Norm ruling. Track P uses track N's norm ([N:bpN:p4:norm], ruling of the coordinator after review P3-03): for u holomorphic on $\mathcal{D}_{r,s} = T_s \times V_rD$ and 1-periodic in θ, $|u|_{r,s} = \sup_{I\in V_rD}\sum_k |\hat{u}_k(I)|e^{2\pi|k|_1 s}$, where s is the *geometric* half-width of the strip $T_s = \{|\mathrm{Im}\,\theta|_\infty < s\}$ and $V_rD = \{I \in \mathbb{C}^n : \mathrm{dist}_\infty(I, D) < r\}$ ([N:bpN:p4:domains]). This is not the norm $\sum_k \sup_I$ used in Part 3 as printed; only $|\cdot|, \le \sum_k \sup_I$ holds.

Dictionary to Part 3. (Lemma 4.5 and [N:bpN:p4:dictionary]). Part 3 writes widths as *coefficient radii* σ (weight $e^{\sigma|k|_1}$, strip $\{|\mathrm{Im}\,\theta| < \sigma/2\pi\}$, notation $\mathbb{T}_\sigma$). Hence $\mathbb{T}^{\mathrm{P3}}_\sigma = T_{\sigma/2\pi}$, and Part 3's $(\bar{s}, s)$ correspond to $(\bar{s}/2\pi, s/2\pi)$ here. Frequencies, divisors, the Hamilton equations $\dot{\theta} = \partial_I H$, $\dot{I} = -\partial_\theta H$ and the period-one angles are the same in both; N's $\bar{\alpha} = 2\pi\alpha$ and $\kappa = 2\pi Ks$ are the quantities $2\pi\alpha$ and "Ks of Part 3". Throughout this part

$$s \text{ (geometric)}, \qquad \kappa := 2\pi Ks \quad (\text{so } \kappa = Ks_{\mathrm{P3}}), \qquad \bar{\alpha}_j := 2\pi\alpha_j, \qquad Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}.$$

The statement for Part 3 in its own notation is Corollary 3.6.

Splitting of indices. $n = n_1 + n_2$, $k = (k_1, k_2)$, $\theta = (\theta_1, \theta_2)$, $I = (I_1, I_2)$. A function is *independent of* θ_1 if all its Fourier modes have $k_1 = 0$.

External facts. (all inherited): those of [N:bpN:p4:s-ext], [N:bpN:p6:s-ext]; $12\ln 2 > 8$ and $\sum_{i\ge1} 4^{1-i} = \frac{4}{3}$ ([N:bpN:p6:ext-numerics]); $|k\cdot v| \le |k|_1|v|_\infty$.

3.2. N's step lemma under a direct divisor hypothesis.

Definition 3.1 (Direct divisor setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $q \ge 1$, $\Lambda \subset \mathbb{Z}^n$, and h holomorphic on V_rD, $\omega = \partial_I h$. The data satisfy (Div_q) if $|k\cdot\omega(I)| \ge \alpha/q$ for all $I \in V_rD$ and $k \in Z_K \setminus \Lambda$.

Lemma 3.2 (Iterative Lemma, direct form)**.** [N:bpN:p6:homological] *and* [N:bpN:p6:iterative] *(with its Remark* [N:bpN:p6:iterative-special]*) hold verbatim, with the same* $\phi = \sum_{k\in Z_K\setminus\Lambda} \frac{\hat{f}_k}{2\pi i k\cdot\omega}\chi_k$*, the same constants and the same estimates, if the hypotheses* [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L]*,* $KLr \le \alpha/p$ *are replaced by* (Div_q)*,* $q \ge 1$ *arbitrary. In particular* $q = 1$ *is allowed.*

Source: ours; N Part 6 as written.

Source gap: This is a variant of the written N statement. N's own proof is unchanged (check below); N may adopt (Div_q) *as its primary hypothesis.*

Proof. **Step S3.1: Where the hypotheses are used.** In [N:bpN:p6:homological], [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L] and $KLr \le \alpha/p$ enter only through [N:bpN:p6:divisor], whose conclusion is exactly (Div_q); and [N:bpN:p6:divisor] is used only in Steps [N:stepN:p6:hom-def] (holomorphy, nonvanishing denominators) and [N:stepN:p6:hom-norm] (the bound $|\hat{\phi}_k| \le q|\hat{f}_k|/\bar{\alpha}$). The proof of [N:bpN:p6:iterative] uses the hypotheses only through [N:bpN:p6:homological]; the identity $p^{-1} + q^{-1} = 1$ occurs only in [N:bpN:p6:divisor].

The rest (Lemmas [N:bpN:p5:flow], [N:bpN:p5:composition], [N:bpN:p5:canonical-map], [N:bpN:p5:transform], [N:bpN:p4:bracket]) never mentions ω. □

Lemma 3.3 (Two-scale step: splitting of ϕ and componentwise displacement)**.** *Let $D \subset \mathbb{C}^n$, $r, s > 0$, $0 < \alpha_2 \le \alpha_1$, $q \ge 1$, $\Lambda = \{0\}$, $K \ge 0$, and assume*

$$(\mathrm{Div}_q^{(2)}) \qquad |k\cdot\omega(I)| \ge \alpha_1/q \ \ (k_1 \neq 0), \qquad |k\cdot\omega(I)| \ge \alpha_2/q \ \ (k_1 = 0,\ k \neq 0), \qquad I \in V_rD,\ k \in Z_K.$$

Let $f \in \mathcal{O}(r,s)$ with $\delta = |f|_{r,s}$, $0 < \rho < r/2$, $0 < \sigma < s/2$, $u = q\delta/(\bar\alpha_2\rho\sigma) \le \frac14$ (the hypothesis of Lemma 3.2 with $\alpha = \alpha_2$), ϕ and $\Phi = X^1_\phi$ as there. Put $\phi^{(1)} = \sum_{k\in Z_K, k_1\neq 0} \hat\phi_k \chi_k$, $\phi^{(2)} = \phi - \phi^{(1)}$. Then

(a) $\phi^{(2)}$ is independent of θ_1; $|\phi^{(1)}|_{r,s} \le q\delta/\bar\alpha_1$ and $|\phi|_{r,s} \le q\delta/\bar\alpha_2$;
(b) on $\mathcal{D}_{r-\frac32\rho, s-\frac32\sigma}$:

$$|\Phi_{I_1} - I_1| \le \frac{q\delta}{e\,\bar\alpha_1\sigma}, \qquad |\Phi_{I_2} - I_2| \le \frac{q\delta}{e\,\bar\alpha_2\sigma}, \qquad |\Phi_\theta - \theta| \le \frac{q\delta}{\bar\alpha_2\rho}.$$

Source: CP10 (70)–(71), p. 565 (splitting of H_ϕ, bound for $|I_1 - I_1'|$); the displacement bookkeeping is ours.

Proof. **Step S3.2: Coefficients.** By the formula for ϕ ([N:bpN:p6:homological]), $\hat\phi_k = \hat f_k/(2\pi i k\cdot\omega)$, a finite family of holomorphic functions. For $k_1 \neq 0$, $(\mathrm{Div}_q^{(2)})$ gives $|\hat\phi_k| \le q|\hat f_k|/\bar\alpha_1$; for every $k \in Z_K \setminus \{0\}$, $\alpha_1 \ge \alpha_2$ gives $|\hat\phi_k| \le q|\hat f_k|/\bar\alpha_2$. Multiply by $e^{2\pi|k|_1 s}$, sum, take the supremum: (a) except independence, which holds because $\phi^{(2)}$ has only modes with $k_1 = 0$. Both $\phi^{(i)}$ lie in $\mathcal{O}(r,s)$ with finite norm ([N:bpN:p4:coefficients](c)).

Step S3.3: Field. Put $\varepsilon = |\phi|_{r,s}/(\rho\sigma) \le u \le \frac14$, $\Omega = \mathcal{D}_{r-\rho,s-\sigma}$, $\nu = \nu_{\rho,\sigma}$ ([N:bpN:p5:setting]). By [N:bpN:p5:flow](a) and Step [N:stepN:p5:cm-ab] of [N:bpN:p5:canonical-map], for $z \in \mathcal{D}_{r-\frac32\rho, s-\frac32\sigma}$ the trajectory $\gamma(t) = X^t_\phi(z)$, $0 \le t \le 1$, stays in $\bar B_\nu(z,\varepsilon) \subset \Omega$ and solves $\dot\gamma = X_\phi(\gamma) = (\partial_I\phi, -\partial_\theta\phi)$. The first n_1 components of $\dot\gamma_I$ are $-\partial_{\theta_1}\phi = -\partial_{\theta_1}\phi^{(1)}$ because $\partial_{\theta_1}\phi^{(2)} = 0$. By [N:bpN:p5:vf-bound] applied to $\phi^{(1)}$, $|\partial_{\theta_1}\phi^{(1)}| \le |\phi^{(1)}|_{r,s}/(e\sigma) \le q\delta/(e\bar\alpha_1\sigma)$ on Ω. Likewise [N:bpN:p5:vf-bound] for ϕ bounds $|\partial_{\theta_2}\phi| \le q\delta/(e\bar\alpha_2\sigma)$ and $|\partial_I\phi| \le q\delta/(\bar\alpha_2\rho)$ on Ω.

Step S3.4: Integrate. The mean value inequality along γ ([K:bp:p1:ext-mvt]) gives (b) with $t = 1$. □

3.3. The averaging proposition in N's conventions.

Proposition 3.4 (Averaging with a loss budget and two scales)**.** *Let $n = n_1 + n_2$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar s \ge 0$, $s > 0$, $K > 0$ with $\kappa = 2\pi K s \ge 6$, $0 < \alpha_2 \le \alpha_1$. Let h be holomorphic on V_rD and real on $V_rD \cap \mathbb{R}^n$, $\omega = \partial_I h$, and assume*

(A1) $|k\cdot\omega(I)| \ge \alpha_1$ if $k_1 \neq 0$, $\ge \alpha_2$ if $k_1 = 0$, for all $I \in V_rD$ and $k \in Z_K \setminus \{0\}$;
(A2) $f \in \mathcal{O}(r, \bar s + s)$, real on real points, with $E := |f|_{r,\bar s+s} \le \dfrac{\alpha_2 r}{128K}$.

Then there is a map $\Psi : \mathcal{D}_{r/2,\bar s+s/6} \to \mathcal{D}_{r,\bar s+s}$, $\Psi(\theta', I') = (\theta, I)$, and functions g, f_* on $\mathcal{D}_{r/2,\bar s+s/6}$ such that

(a) Ψ is holomorphic, injective, real on real points, $\Psi(\theta' + k, I') = \Psi(\theta', I') + (k, 0)$ ($k \in \mathbb{Z}^n$), and $(D\Psi)^\top J\, D\Psi = J$ (Ψ preserves $d\theta \wedge dI$, hence $dI \wedge d\theta$);
(b) $H \circ \Psi = h + g + f_*$, $H = h + f$, with $g = g(I')$ independent of θ', real on real points, and $\sup_{V_{r/2}D} |g - \hat f_0| \le |g - \hat f_0|_{r/2,\bar s+s/6} \le E/4$ (indeed $\le E/\kappa$);
(c) f_* is holomorphic, periodic, real on real points and $|f_*|_{r/2,\bar s+s/6} \le e^{-\kappa/6}E$;
(d) $\bar\alpha_1 s\,|I_1 - I_1'| \le 3E$, $\bar\alpha_2 s\,|I_2 - I_2'| \le 3E$, $\bar\alpha_2 r\,|\theta - \theta'| \le 9E$ on $\mathcal{D}_{r/2,\bar s+s/6}$.

Constants depend on nothing: no n, τ, q or L enter.
Source: CP10 Prop. 1 (p. 552) with $m = 0$, Lemma A.1 (p. 564), proof pp. 564–565; Pöschel 1993 pp. 193–196 via [N:bpN:p6:normal-form]. *Constants recomputed (see Remark 3.5).*

Proof. Throughout $q = 1$: by (A1), ($\mathrm{Div}_1^{(2)}$) holds on V_rD with $\Lambda = \{0\}$, and for every $r' \le r$ and $D' \subset D$-neighbourhoods inside V_rD. Write $\bar\alpha = \bar\alpha_2$, $\varepsilon := E$, $S_i := \bar s + s_i$ for the strip widths, where s_i (the *budget* still to be spent) is chosen below. From (A2): $\varepsilon \le \alpha_2 r/(128K) = \bar\alpha rs/(128\kappa)$. Every application of Lemma 3.2 below is on $\mathcal{D}_{r_i,S_i}$ with $r_i \le r$ and $S_i \ge s$, so its non-resonance hypothesis is (A1); its requirements $2\rho < r_i$, $2\sigma < S_i$ hold since the strip is at least the budget. Because the weights $|g|_*$ of [N:bpN:p6:setting] contain the widths only through the *differences* $s_j - s_i$, the strip offset $\bar s$ plays no role in any estimate.

Step S3.5: Step 0. Take $\rho_0 = r/8$, $\sigma_0 = s/6$ and $g = 0$ in Lemma 3.2. Then $y_0 := \varepsilon/(\bar\alpha\rho_0\sigma_0) = 48\varepsilon/(\bar\alpha rs) \le \frac{3}{8\kappa} \le \frac{1}{16}$, so $u = y_0 \le \frac14$. Lemma 3.2(d) with $|g|_* = 0$ gives

$$|f_1|_{\frac34 r,\bar s+\frac23 s} \le (1-2y_0)^{-1}\big(\tfrac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon, \qquad (1-2y_0)^{-1} \le \tfrac87$$

($2\pi K\sigma_0 = \kappa/6 \ge 1$). *Case A: $e^{-\kappa/6} \le y_0$.* Then

$$\varepsilon_1 := |f_1|_* \le \tfrac87\big(1+\tfrac2e\big)y_0\varepsilon \le 1.984\cdot\tfrac{3}{8\kappa}\,\varepsilon \le \frac{3\varepsilon}{4\kappa} \le \frac{\varepsilon}{8}. \tag{2.1}$$

Case B: $e^{-\kappa/6} > y_0$. Repeat Step 0 with the doubled losses $\rho' = r/4$, $\sigma' = s/3$ ($u' = y_0/4$): $|f_1|_{r/2,\bar s+s/3} \le (1-y_0/2)^{-1}(\frac{2}{e}\frac{y_0}{4} + e^{-\kappa/3})\varepsilon \le \frac{32}{31}(\frac{1}{2e}+\frac1e)e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$, using $e^{-\kappa/3} \le e^{-1}e^{-\kappa/6}$. We stop with $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,\bar s+s/3} \supset \mathcal{D}_{r/2,\bar s+s/6}$, $g = \hat f_0$, $f_* = f_1$. Clauses (a)–(c) follow from Lemma 3.2; (d): Lemma 3.3(b) gives $|\Delta I_j| \le \varepsilon/(e\bar\alpha_j\sigma') = 3\varepsilon/(e\bar\alpha_j s) \le 1.1\,\varepsilon/(\bar\alpha_j s)$ and $|\Delta\theta| \le \varepsilon/(\bar\alpha\rho') = 4\varepsilon/(\bar\alpha r)$. From now on Case A.

Step S3.6: Number of steps. $N := \lfloor \kappa/(12\ln 2)\rfloor$, so $N < \kappa/8$. If $N = 0$: $\kappa < 12\ln 2$, hence $e^{-\kappa/6} > \frac14$, and $\varepsilon_1 \le \varepsilon/8 \le e^{-\kappa/6}\varepsilon$ by (2.1); set $\Psi = \Phi_0$, $g = \hat f_0$, $f_* = f_1$ on $\mathcal{D}_{3r/4,\bar s+2s/3}$. Displacements: $|\Delta I_j| \le \varepsilon/(e\bar\alpha_j\sigma_0) = 6\varepsilon/(e\bar\alpha_j s) \le 2.21\,\varepsilon/(\bar\alpha_j s)$, $|\Delta\theta| \le 8\varepsilon/(\bar\alpha r)$. (a)–(d) follow. Let $N \ge 1$ and for $1 \le i \le N$

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac34 r - 2(i-1)\rho, \quad s_i = \tfrac23 s - 2(i-1)\sigma, \quad S_i = \bar s + s_i;$$

then $r_{N+1} = r/2$, $s_{N+1} = s/6$. $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, so $e^{-2\pi K\sigma} \le \frac18$; $\rho\sigma = rs/(32N^2)$. From $\kappa > 8N$, (A2) and (2.1):

$$\varepsilon \le \frac{\bar\alpha rs}{2^{10}N}, \qquad \varepsilon_1 \le \frac{3}{4\kappa}\varepsilon < \frac{3\varepsilon}{32N} \le \frac{3\,\bar\alpha rs}{2^{15}N^2} < \frac{\bar\alpha\rho\sigma}{2^8}. \tag{2.2}$$

Step S3.7: Induction. *Claim:* for $1 \le i \le N$ step i (Lemma 3.2 on $\mathcal{D}_{r_i,S_i}$, losses (ρ,σ)) is admissible, and $\varepsilon_{i+1} := |f_{i+1}|_{r_{i+1},S_{i+1}} \le \varepsilon_i/4$; hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ ($1 \le i \le N+1$). Here f_{i+1}, $g_i = g_{i-1} + (\hat f_i)_0$ are produced by the step, $g_0 := \hat f_0$, $|g_i - g_{i-1}| \le \varepsilon_i$ ([N:bpN:p4:projections]). *Admissibility:* $u_i = \varepsilon_i/(\bar\alpha\rho\sigma) \le 2^{-8}$ by (2.2). *Weights:* decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$; the term $j \ge 1$ lives on $\mathcal{D}_{r_j,S_j}$ with $r_j - r_i \ge 2\rho$, $s_j - s_i \ge 2\sigma$, weight $\le \frac12(\frac13+\frac13) \le \frac12$; the term g_0 lives on $\mathcal{D}_{r,\bar s+s}$ with $r - r_i \ge r/4$, $s - s_i \ge s/3$, weight $\le \frac12(\frac{1}{2N} + \frac{3}{4N}) \le \frac1N$. Thus $|g_{i-1}|_* \le \frac12\cdot\frac43\varepsilon_1 + \varepsilon/N \le \varepsilon_1 + \varepsilon/N \le \bar\alpha\rho\sigma\,(2^{-8}+2^{-5}) \le \bar\alpha\rho\sigma/2^4$ (by (2.2) and $\varepsilon/N \le \bar\alpha rs/(2^{10}N^2) = 2^{-5}\bar\alpha\rho\sigma$). *The step:* Lemma 3.2(d) with $\delta = \varepsilon_i$ gives $\varepsilon_{i+1} \le \frac{128}{127}\big(\frac2e(\frac{1}{16}+\frac{1}{2^8}) + \frac18\big)\varepsilon_i \le \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac14\varepsilon_i$. (This is the computation of [N:bpN:p6:normal-form], Step [N:stepN:p6:nf-induct], with $q = 1$.)

Step S3.8: Conclusion of the construction. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_N$ on $\mathcal{D}_{r/2,\bar s+s/6}$, $g = g_N$, $f_* = f_{N+1}$. Since $H_{i+1} = H_i\circ\Phi_i$, $H\circ\Psi = h + g + f_*$. g is a finite sum of θ-independent functions (each $(\hat f_i)_0$), so $g = g(I')$; $\hat f_0 = g_0$. [N:bpN:p6:telescope] with $U_i = \mathcal{D}_{r_i,S_i}$ gives (a) (holomorphic, canonical, injective; periodicity and reality are inherited from each Φ_i, [N:bpN:p6:iterative](a),(e)),

and reality of g, f_* is [N:bpN:p6:iterative](e). (c): $\varepsilon_{N+1} \le 4^{-N}\varepsilon_1 \le 4^{-N-1}\varepsilon \le e^{-\kappa/6}\varepsilon$, since $N+1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} < e^{-\kappa/6}$. (b): $|g-g_0| \le \sum_{i=1}^N \varepsilon_i \le \frac{4}{3}\varepsilon_1 \le \varepsilon/\kappa \le \varepsilon/6$ by (2.1); this is the bound of CP10 (25) in which no K appears in the numerator (review P3-04). The sup bound follows from [N:bpN:p4:sup-weighted](a).

Step S3.9: (d) Displacement. By [N:bpN:p6:telescope] (seminorms $|\Pi_{I_1}\cdot|$, $|\Pi_{I_2}\cdot|$, $|\Pi_\theta\cdot|$) and Lemma 3.3(b) for each step (valid on $V_{i+1} \subset \mathcal{D}_{r_i-\frac{3}{2}\rho_i, S_i-\frac{3}{2}\sigma_i}$, [N:bpN:p6:iterative](a)):

$$|\Delta I_j| \le \frac{\varepsilon}{e\bar\alpha_j\sigma_0} + \sum_{i=1}^N \frac{\varepsilon_i}{e\bar\alpha_j\sigma} \le \frac{1}{e\bar\alpha_j s}\Big(6\varepsilon + 4N\cdot\tfrac{4}{3}\varepsilon_1\Big), \qquad |\Delta\theta| \le \frac{\varepsilon}{\bar\alpha\rho_0} + \sum_{i=1}^N \frac{\varepsilon_i}{\bar\alpha\rho} \le \frac{1}{\bar\alpha r}\Big(8\varepsilon + 8N\cdot\tfrac{4}{3}\varepsilon_1\Big).$$

By (2.1) and $N < \kappa/8$: $N\varepsilon_1 \le \frac{3N}{4\kappa}\varepsilon < \frac{3}{32}\varepsilon$. Hence $\frac{1}{e}(6 + \frac{16}{3}\cdot\frac{3}{32}) = \frac{6.5}{e} < 2.4 \le 3$ and $8 + \frac{32}{3}\cdot\frac{3}{32} = 9$ (strict inequality, since $N < \kappa/8$ strictly). Both bounds are $\le$ the claimed ones. □

Remark 3.5 (Where the constants come from). Smallness: (A2) is the one-scale hypothesis (i) of [N:bpN:p6:normal-form] with $q = 1$ ($2^7 qK\varepsilon \le \alpha r$). The weaker requirement of CP10, $64e\,KE < \alpha_2 r$ (CP10 (23) with $m = 0$, $c_0 = e/2$, $d = rs$), implies it since $64e > 128$. The number of steps N, the losses, 4^{1-i} and $e^{-\kappa/6}$ are those of [N:bpN:p6:normal-form]; the improvements (E/κ instead of $16c\varepsilon^2$, $3E$ instead of $9E$ for the actions) come from keeping the exact expressions $6/e$ and $y_0 \le 3/(8\kappa)$ instead of the cruder $\kappa \ge 6 \Rightarrow 1/(\bar\alpha s) \le K/(6\alpha)$ of [N:bpN:p6:normal-form](c), which is a bound in K and is useless for the interface (the interface displacement is independent of K). The deviation of [N:bpN:p6:deviations](1) is respected: all steps satisfy $|\phi| \le \rho\sigma/4$.

3.4. The interface of Lemma 4.8.

Corollary 3.6 (Averaging with two scales, in Part 3's notation)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar s, s > 0$, $K > 0$ with $Ks \ge 6$. Let h be real analytic on D_r (holomorphic on $D_r = V_r D$, real on real points) with $\omega = \partial h$, and let f be holomorphic on $D_r \times \mathbb{T}_{\bar s+s}$, real analytic, 1-periodic in θ, with the* N-norm $E := \sup_{I\in D_r}\sum_k |\hat f_k(I)|e^{(\bar s+s)|k|_1} < \infty$. *Assume (A1) (that is, (N1) of Lemma 4.8) and*

$$64e\,KE < \alpha_2 r.$$

Then there is a holomorphic, injective, real, symplectic ($\Psi^*(dI\wedge d\theta) = dI'\wedge d\theta'$) map $\Psi : D_{r/2}\times\mathbb{T}_{\bar s+s/6} \to D_r\times\mathbb{T}_{\bar s+s}$ with $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', and $g = g(I')$, f_* with $H\circ\Psi = h + g + f_*$ on $D_{r/2}\times\mathbb{T}_{\bar s+s/6}$ and, in the N-norm with weight $e^{(\bar s+s/6)|k|_1}$ on $D_{r/2}$,

$$\sup_{D_{r/2}}|g - f_0| \le E/4, \qquad \|f_*\|_{r/2,\bar s+s/6} \le e^{-Ks/6}E,$$
$$\alpha_1 s|I_1 - I_1'| \le 3E\ (\le 9E), \quad \alpha_2 s|I_2 - I_2'| \le 3E\ (\le 9E), \quad 2\pi\alpha_2 r|\theta - \theta'| \le 9E.$$

Source: CP10 Prop. 1 (p. 552) with $m = 0$, $\Lambda = \{0\}$; proof: Proposition 3.4.

Proof. **Step S3.10: Dictionary.** Put $\bar s_N = \bar s/2\pi$, $s_N = s/2\pi$ (geometric). Then $\mathbb{T}_{\bar s+s} = T_{\bar s_N+s_N}$, the weight $e^{(\bar s+s)|k|_1} = e^{2\pi|k|_1(\bar s_N+s_N)}$, so $E = |f|_{r,\bar s_N+s_N}$ in the notation of Proposition 3.4; $\kappa = 2\pi K s_N = Ks \ge 6$; $\bar\alpha_j s_N = \alpha_j s$ and $\bar\alpha_2 r = 2\pi\alpha_2 r$.

Step S3.11: Hypotheses. $64e > 128$, so $64eKE < \alpha_2 r$ gives $E \le \alpha_2 r/(128K)$. The remaining hypotheses are identical. Proposition 3.4(a)–(d) are the claims; $e^{-\kappa/6} = e^{-Ks/6}$. □

Remark 3.7 (What Part 3 must change (the interface report)). Lemma 4.8 is *provable as stated, with the single change of norm*: replace the Part 3 norm $\|\cdot\|_{r,\sigma} = \sum_k \sup_I |f_k|e^{\sigma|k|}$ by N's $\sup_I\sum_k|\hat f_k(I)|e^{\sigma|k|_1}$ (this is Corollary 3.6). Precisely:

(1) Every occurrence of the norm (hypothesis $E \ge \|f\|_{\rho,\bar s+s}$ of [P:bpP:p3:two-scale-kam], E_j in the iteration, outputs $\|f_*\|$) is read in N's norm. Used properties: monotone under restriction of r, σ and of the base set; $\sup|f| \le \|f\|$ ([N:bpN:p4:sup-weighted](a)); Cauchy

estimates for g use only sup. Nothing in Part 3 uses the finer $\sum_k \sup_I$ structure, except the printed definition.

(2) The numerical constants: hypothesis (N2) $64eKE < \alpha_2 r$ and the conclusions $E/4$, $e^{-Ks/6}E$, $\alpha_2 r 2\pi|\Delta\theta| \le 9E$ are unchanged; $\alpha_i s|\Delta I_i| \le 9E$ may be replaced by $3E$ (so the constant $\frac{3}{4}$ of `st-avg` may become $\frac{1}{4}$); no other constant of Part 3 changes.

(3) Hypothesis (N1) is used on the full complex neighbourhood D_r, exactly as in Lemma 4.8; *no Lipschitz hypothesis is added* (Lemma 3.2). Reality of h and f on real points is used, as in Part 3's setting.

(4) The proof plan of Lemma 4.8 (one homological solve, one flow) must be replaced by Proposition 3.4: a finite iteration; the Fourier-tail remark of `avg-flow` is not the source of the exponential.

3.5. **Averaging with an oscillator pair (CP10 §4, Step 1).** CP10 §4, Step 1 (p. 563) averages over the n_1 fast angles while the oscillator variables $(p,q) \in \mathbb{C}^{n_2} \times \mathbb{C}^{n_2}$ are dynamical: the transformation moves them, symplectically for $dI \wedge d\theta + dp \wedge dq$. Track N's Parts 4–6 are written for (θ, I) only. The extension is mechanical but must be recorded; we do it with its own constants.
Setting. $n, m \ge 1$, $D \subset \mathbb{R}^n$, $v = (r, r_p, r_q)$, $s > 0$, $\bar{s} \ge 0$. $\mathcal{D}_{v,S} := T_S \times V_r D \times B_{r_p} \times B_{r_q} \ni (\theta, I, p, q)$ with $B_a = \{x \in \mathbb{C}^m : |x|_\infty < a\}$ (the complex neighbourhood of $\{0\}$ in the max norm; the Euclidean ball of CP10 contains it up to $\sqrt{m}$ in radius), $\mathcal{O}(v,S)$ and $|\cdot|_{v,S}$ as in [N:`bpN:p4:norm`] with the sup over (I,p,q) (I,p,q together play the rôle of N's parameter I; only θ is Fourier expanded). The field of u is $X_u = (\partial_I u, -\partial_\theta u;\ \partial_p u, -\partial_q u)$ in the order $(\dot\theta, \dot I; \dot q, \dot p)$ and $\{u,w\} = \sum_i (\partial_{\theta_i} u\, \partial_{I_i} w - \partial_{I_i} u\, \partial_{\theta_i} w) + \sum_j (\partial_{q_j} u\, \partial_{p_j} w - \partial_{p_j} u\, \partial_{q_j} w) = Du \cdot X_w$. Put $d := \min\{rs, r_p r_q\}$ and $b := 1+m$. For losses $\nu = (\rho, \rho_p, \rho_q, \sigma)$ put $\delta_\nu := \min\{\rho\sigma, \rho_p\rho_q\}$ and the scaled norm $\nu(a, b_I, c_q, c_p) = \max(|a|/\sigma, |b_I|/\rho, |c_q|/\rho_q, |c_p|/\rho_p)$, whose balls are the polydiscs of the four domain losses.

Lemma 3.8 (Extended bracket and field estimates)**.** *Let $u \in \mathcal{O}(v_0, S_0)$, $w \in \mathcal{O}(v, S)$ with finite norms, $0 < \rho < r$, $0 < \rho_p < r_p$, $0 < \rho_q < r_q$, $0 < \sigma < S$, $v - \nu < v_0$ componentwise, $S - \sigma < S_0$. Then $\{u,w\} \in \mathcal{O}(v - \nu, S - \sigma)$ and*

$$|\{u,w\}|_{v-\nu,S-\sigma} \le \frac{W}{\delta_\nu} |u|_{v_0,S_0} |w|_{v,S}, \quad W = \tfrac{1}{e}\Big(\tfrac{\rho}{r_0 - r + \rho} + \tfrac{\sigma}{S_0 - S + \sigma}\Big) + m\Big(\tfrac{\rho_q}{r_{q0} - r_q + \rho_q} + \tfrac{\rho_p}{r_{p0} - r_p + \rho_p}\Big) \le 2b\,R,$$

where R is the largest of the four ratios. Moreover for $\phi \in \mathcal{O}(v,S)$: on $\mathcal{D}_{v-\nu,S-\sigma}$, $\nu(X_\phi) \le |\phi|_{v,S}/\delta_\nu$.

Proof. **Step S3.12: Derivative bounds.** [N:`bpN:p4:cauchy`](b) is a statement about each coordinate of the parameter separately (its proof is a one-variable Cauchy integral in that coordinate; ∂_{p_j} costs the loss in the p-radius), and (a) concerns θ only. So, with the same proofs, $\sum_j |\partial_{p_j} u|_, \le m \max_j \le m|u|_{/,(r_{p0} - r_p + \rho_p)}$ after restriction, and likewise for q.

Step S3.13: Products. [N:`bpN:p4:algebra`] and the triangle inequality as in [N:`bpN:p4:bracket`]: the (θ, I) part gives $\frac{1}{e}(\frac{1}{(r_0 - r + \rho)\sigma} + \frac{1}{(S_0 - S + \sigma)\rho})$, the (q,p) part $m(\frac{1}{(r_{q0} - r_q + \rho_q)\rho_p} + \frac{1}{(r_{p0} - r_p + \rho_p)\rho_q})$. Since $\frac{1}{(r_0 - r + \rho)\sigma} = \frac{1}{\rho\sigma} \cdot \frac{\rho}{r_0 - r + \rho}$ etc. and $\rho\sigma, \rho_p\rho_q \ge \delta_\nu$, the bound follows; $\frac{1}{e} + \frac{1}{e} \le 2$ gives $W \le 2bR$.

Step S3.14: Field. As in [N:`bpN:p5:vf-bound`]: $|\partial_I \phi|/\sigma \le |\phi|_,/(\rho\sigma)$, $|\partial_\theta \phi|/\rho \le |\phi|_,/(e\sigma\rho)$, $|\partial_p \phi|/\rho_q \le |\phi|_,/(\rho_p\rho_q)$, $|\partial_q \phi|/\rho_p \le |\phi|_,/(\rho_q\rho_p)$. □

Lemma 3.9 (Iterative Lemma with an oscillator pair)**.** [N:`bpN:p6:iterative`] *(in direct form, Lemma 3.2, $\Lambda = \{0\}$, $q = 1$) holds on $\mathcal{D}_{v,S}$ with all losses ν, with $\rho\sigma$ replaced by δ_ν, the factor $\frac{2}{e}$ in the bracket estimate replaced by $2b$ (so $|g|_*$ is replaced by $|g|_R := \sum_j R_j |g_j|_,$), and $\Phi = X_\phi^1$ symplectic for $d\theta \wedge dI + dq \wedge dp$:*

(a) if $|f|_{v,S} = \delta \le \bar\alpha\delta_\nu/(16b)$ then $|\Phi_\theta - \theta| \le \delta/(\bar\alpha\rho)$, $|\Phi_I - I| \le \delta/(e\bar\alpha\sigma)$, $|\Phi_q - q| \le \delta/(\bar\alpha\rho_p)$, $|\Phi_p - p| \le \delta/(\bar\alpha\rho_q)$;

(b) $|f_+|_{v-2\nu,S-2\sigma} \le (1 - 2b\delta/(\bar\alpha\delta_\nu))^{-1}\big(\frac{2b}{\bar\alpha\delta_\nu}(|g|_R + \delta) + e^{-2\pi K\sigma}\big)\delta$.

Proof. **Step S3.15: Transfer.** [N:bpN:p6:homological] involves θ and the parameters only through coefficients and is unchanged ($|\phi|_, \le \delta/\bar\alpha$). [N:bpN:p5:flow] is for an arbitrary complex norm and open set; [N:bpN:p5:symplectic] uses only $DF = JS$ with S symmetric and $J^\top = -J$, true for the enlarged J; Corollary [N:bpN:p5:canonical] follows. Lemma [N:bpN:p5:adjoint], Theorem [N:bpN:p5:composition], Propositions [N:bpN:p5:transform], [N:bpN:p5:canonical-map] use the geometry only through the bracket bound and the field bound: replace [N:bpN:p4:bracket] and [N:bpN:p5:vf-bound] by Lemma 3.8 and $\rho\sigma$ by δ_ν; the ratio of the Lie series of [N:bpN:p5:composition] is multiplied by at most $e\,b$ (the bracket constant $2/e$ becomes $2b$), see the next step.

Step S3.16: Smallness range. With the new bracket constant the Lie series converges and equals the composition ([N:bpN:p5:composition]) as soon as $e\,b\,\eta \le \frac12$, $\eta = 2|\phi|_,/\delta_\nu$, that is $|\phi|_, \le \delta_\nu/(4eb)$. We therefore *assume* $\delta \le \bar\alpha\delta_\nu/(16b)$ (this replaces $u \le \frac14$); it holds at every step of Proposition 3.10 (there $\delta/(\bar\alpha\delta_\nu) \le y_0 \le 1/(64b\kappa)$ or is smaller by (2.3)). Then (a) and (b) follow as in [N:bpN:p6:iterative] with $\frac{2}{e} \mapsto 2b$ and $1 - \frac{2q\delta}{\bar\alpha\rho\sigma} \mapsto 1 - \frac{2b\delta}{\bar\alpha\delta_\nu}$.

Step S3.17: Displacement. As in Step S3.3: components of X_ϕ and Lemma 3.8: $|\partial_I\phi| \le |\phi|_,/\rho$, $|\partial_\theta\phi| \le |\phi|_,/(e\sigma)$, $|\partial_p\phi| \le |\phi|_,/\rho_p$, $|\partial_q\phi| \le |\phi|_,/\rho_q$. □

Source gap: Lemma 3.9 extends N Parts 5–6 by an oscillator pair; the extension is stated by its (short) list of changes, not rewritten in N. It is the oscillator part of CP10 Lemma A.1 / [4, Lemma A.5]. If track N prefers, it can be absorbed into N Part 5 ($\{\cdot,\cdot\}$ on $\mathbb{C}^n \times \mathbb{C}^n \times \mathbb{C}^m \times \mathbb{C}^m$). Constants differ from CP10's $c_m = e(1+em)/2$.

Proposition 3.10 (Averaging with oscillators, one scale)**.** *In the setting above, let h be holomorphic on V_rD, real on real points, $\omega = \partial h$ (h independent of p, q), $\alpha > 0$, $K > 0$, $\kappa = 2\pi Ks \ge 6$,*

(O1) $|k\cdot\omega(I)| \ge \alpha$ for $I \in V_rD$, $0 < |k|_1 \le K$;
(O2) $f \in \mathcal{O}(v, \bar s + s)$ real on real points, $E := |f|_{v,\bar s+s}$ with $2^{12}\,b\,KE\,s \le \alpha d$.

Then there is $\Psi : \mathcal{D}_{v/2,\bar s+s/6} \to \mathcal{D}_{v,\bar s+s}$ holomorphic, injective, real, $\Psi - \mathrm{id}$ periodic in θ, symplectic for $d\theta\wedge dI + dq\wedge dp$, with $H\circ\Psi = h + g + f_*$ on $\mathcal{D}_{v/2,\bar s+s/6}$, $H = h + f$, $g = g(I,p,q)$ independent of θ, real, and with $\bar\alpha = 2\pi\alpha$

$$|g - \hat f_0|_{v/2,\bar s+s/6} \le \frac{400\,b\,E^2}{\bar\alpha d}, \qquad |f_*|_{v/2,\bar s+s/6} \le e^{-\kappa/6}E,$$
$$\bar\alpha s|I - I'| \le 3E, \quad \bar\alpha r|\theta - \theta'| \le 9E, \quad \bar\alpha r_q|p - p'| \le 9E, \quad \bar\alpha r_p|q - q'| \le 9E.$$

Source: CP10 Prop. 1 (p. 552), Lemma A.1 (p. 564) with $\ell_2 = 0$, $\Lambda = \{0\}$; (25), (26). Constants ours.

Proof. Use the notation of Proposition 3.4 ($q = 1$, $\varepsilon = E$, $\bar\alpha = 2\pi\alpha$); (O2) reads $\varepsilon \le \bar\alpha d/(2^{12}b\kappa)$. Step 0 losses $\rho_0 = r/8$, $\rho_{p0} = r_p/8$, $\rho_{q0} = r_q/8$, $\sigma_0 = s/6$; then $\delta_{\nu_0} = \min\{rs/48, r_pr_q/64\} \ge d/64$ and $y_0 := \varepsilon/(\bar\alpha\delta_{\nu_0}) \le 64\varepsilon/(\bar\alpha d) \le \frac{1}{64b\kappa} \le \frac{1}{16}$.

Step S3.18: Step 0. Lemma 3.9(b) with $|g|_R = 0$: $\varepsilon_1 \le \frac87(2by_0 + e^{-\kappa/6})\varepsilon$. *Case A:* $e^{-\kappa/6} \le 2by_0$. $\varepsilon_1 \le \frac{32}{7}by_0\varepsilon \le \frac{32}{7}\cdot\frac{1}{64\kappa}\varepsilon \le 0.08\,\varepsilon/\kappa$, and $\varepsilon_1 \le \frac{32}{7}\cdot 64\,b\varepsilon^2/(\bar\alpha d) \le 294\,b\varepsilon^2/(\bar\alpha d)$. *Case B:* $e^{-\kappa/6} > 2by_0$. Doubled losses ($\delta_{\nu'} = 4\delta_{\nu_0}$): $|f_1|_, \le (1 - y_0/2)^{-1}(2b\frac{y_0}{4} + e^{-\kappa/3})\varepsilon \le \frac{32}{31}(\frac14 + \frac1e)e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$; stop, $g = \hat f_0$, displacements $\le \varepsilon/(e\bar\alpha\sigma') \le 1.1\varepsilon/(\bar\alpha s)$, $\le 4\varepsilon/(\bar\alpha r)$, $\le 4\varepsilon/(\bar\alpha r_q), 4\varepsilon/(\bar\alpha r_p)$.

Step S3.19: Iteration (Case A). $N = \lfloor\kappa/(12\ln 2)\rfloor < \kappa/8$; $N = 0$: stop, since then $e^{-\kappa/6} > \frac14 > \varepsilon_1/\varepsilon$. $N \ge 1$: losses $\rho = r/8N$, $\rho_p = r_p/8N$, $\rho_q = r_q/8N$, $\sigma = s/4N$, all four radii and the budget following the schedule of Step S3.6 (so every ratio R_j is $\le \frac13$ for $j \ge 1$ and $\le \frac{3}{4N}$ for g_0). Then $\delta_\nu \ge d/(64N^2)$ and by $\kappa > 8N$

$$\frac{\varepsilon}{N} \le \frac{\bar\alpha d}{2^{15}bN^2} \le \frac{\bar\alpha\delta_\nu}{2^9 b} \quad (\text{since } d \le 64N^2\delta_\nu), \qquad \varepsilon_1 \le \frac{0.08\,\varepsilon}{8N}. \tag{2.3}$$

Then $|g_{i-1}|_R + \varepsilon_i \le 2\varepsilon_1 + \varepsilon/N \le 1.02\,\varepsilon/N$ and $x := \frac{2b}{\bar\alpha\delta_\nu}(|g_{i-1}|_R + \varepsilon_i) \le 0.004$; Lemma 3.9(b) gives $\varepsilon_{i+1} \le \frac{128}{127}(0.004 + \frac18)\varepsilon_i < \frac14\varepsilon_i$ (using $e^{-2\pi K\sigma} \le \frac18$, $u_i \le 2^{-7}$). Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$.

Step S3.20: Conclusion. As Step S3.8: $|f_*|, \le 4^{-(N+1)}\varepsilon \le e^{-\kappa/6}\varepsilon$; $|g - \hat f_0|, \le \frac43\varepsilon_1 \le \frac43 \cdot 294\,b\varepsilon^2/(\bar\alpha d) \le 400\,b\varepsilon^2/(\bar\alpha d)$. Displacements by Lemma 3.9(a) and [N:bpN:p6:telescope]: I: $\frac{1}{\epsilon\bar\alpha s}(6\varepsilon + 4N\frac43\varepsilon_1) \le 2.3\,\varepsilon/(\bar\alpha s)$; θ: $\frac{1}{\bar\alpha r}(8\varepsilon + \frac{32}{3}N\varepsilon_1) \le 8.2\,\varepsilon/(\bar\alpha r)$ since $N\varepsilon_1 \le 0.01\varepsilon$; the same for p, q with r_q, r_p. □

Proposition 3.11 (Averaged remainder for [P:bpP:p0:pg]: CP10 §4, Step 1)**.** *Let $n = n_1 \ge 1$, $m = n_2 \ge 1$, $\bar D \subset \mathbb{R}^n$, $\bar\rho, \epsilon_0 > 0$, $s_0 > 0$ (geometric), $0 < \epsilon < e^{-1/5}$, $\mu > 0$, $v_0 = (\bar\rho, \epsilon_0, \epsilon_0)$. Let $H = H_0(I) + \mu P(\theta, I, p, q)$, H_0 holomorphic on $V_{\bar\rho}\bar D$ and real, $P \in \mathcal{O}(v_0, s_0)$ real with $|P|_{v_0,s_0} \le 1$, and $\langle P\rangle(I, p, q) := \int_{\mathbb{T}^n} P\,d\theta = \hat P_0$. Choose K by $e^{-2\pi K s_0/6} = \epsilon^5$, i.e. $K = \frac{15}{\pi s_0}\log\frac1\epsilon$ (so $\kappa = 30\log\frac1\epsilon \ge 6$; CP10 (36) with 2π-conversion $K_{\rm CP} = K$, $s_{\rm CP} = 2\pi s_0$). Assume, with $\bar\alpha = 2\pi\alpha$, $d = \min\{\bar\rho s_0, \epsilon_0^2\}$,*

(R1) $|k\cdot\partial H_0(I)| \ge \alpha$ for $I \in V_{\bar\rho}\bar D$, $0 < |k|_1 \le K$ (non-resonance on the complex neighbourhood, CP10 (39));

(R2) $2^{12}(1+m)\,K s_0\,\mu \le \alpha d$ (CP10 (65), implied by (61)).

Then there is a map $\bar\varphi : \mathcal{D}_{v_0/2,s_0/6} \to \mathcal{D}_{v_0,s_0}$ as in Proposition 3.10 (with $\bar s = 0$, $s = s_0$) such that on $\mathcal{D}_{v_0/2,s_0/6}$

$$H \circ \bar\varphi = H_0 + \mu\langle P\rangle + \widetilde P, \qquad |\widetilde P|_{v_0/2,s_0/6} \le \frac{400(1+m)\,\mu^2}{\bar\alpha d} + \mu\,\epsilon^5,$$

where $\widetilde P = (g - \mu\langle P\rangle) + f_*$, $g = g(I, p, q)$, $|f_*|, \le \mu\epsilon^5$ (so $f_* = \mu\epsilon^5\bar P$ with $|\bar P|, \le 1$, in CP10's (40) notation), and

$$\bar\alpha s_0|I - \bar I| \le 3\mu, \quad \bar\alpha\bar\rho|\theta - \bar\theta| \le 9\mu, \quad \bar\alpha\epsilon_0|p - \bar p|,\ \bar\alpha\epsilon_0|q - \bar q| \le 9\mu.$$

Source: CP10 §4, Step 1, p. 563 (36)–(40), (62)–(63); the estimates (41)–(42) of Step 1 of §3 (p. 557). Perturbation size: $E = \mu$ before averaging and $\mu\epsilon^5$ after (CP10 p. 560: $f = \mu\epsilon^5\check P$, $E = C\mu\epsilon^5$; not μ^5).

Proof. **Step S3.21: Apply.** $f = \mu P$, $E = \mu$, $\bar s = 0$, $s = s_0$, $r = \bar\rho$, $r_p = r_q = \epsilon_0$, $h = H_0$; (R1), (R2) are (O1), (O2). Proposition 3.10 gives $H \circ \bar\varphi = H_0 + g + f_*$, and $\hat f_0 = \mu\langle P\rangle$.

Step S3.22: Collect. $\widetilde P := (g - \mu\langle P\rangle) + f_*$; its norm is bounded by the sum of the two bounds of Proposition 3.10, and $e^{-\kappa/6}\mu = \epsilon^5\mu$ by the choice of K. □

Remark 3.12 (How Part 5 uses this). (1) The two terms of CP10 (63) correspond to $\mu^2/(\bar\alpha d)$ (the average-correction) and $\mu\epsilon^5$ (the tail). Part 5 must verify $\mu/(\bar\alpha d) \le C\epsilon^5$ from its choice of $\bar\gamma$ (CP10 (61), with $\mu < \epsilon^6/(\log\frac1\epsilon)^c$ of [P:bpP:p0:pg]); this is a different inequality from CP10's (since the proof here has no $K^{2\tau+1}/\bar\gamma^2$), and it is implied by (61) together with $\mu \le \epsilon^5$. (2) Distinguish three radii: the physical oscillator radius ϵ (only through K and the later domain), the ambient radius ϵ_0 of P, and Part 3's enlargement radius $\hat\rho$. (3) Period conversion: $\alpha = \tilde\alpha/2\pi$, μ unchanged, $s_0 = s_{\rm CP}/2\pi$ (Lemma 4.5). The max-norm polydisc B_{ϵ_0} replaces CP10's Euclidean ball.

3.6. Suspected errata in CP10 and open points.

(1) CP10 p. 564, Lemma A.1, (66): the target is printed as $W_{v,\sigma}$; it must be $W_{v,s}$ (the map lands in the original domain).

(2) CP10 p. 563, Step 1: "$r_p = r_q = 0$" together with $\rho_p = \rho_q = \epsilon_0/4$ and $B = B' = \{0\}$ is inconsistent with the hypotheses $\rho_p < r_p/2$ of Lemma A.1; the intended value is $r_p = r_q = \epsilon_0$ (the resulting domain $\tilde v = (\bar\rho/2, \epsilon_0/2)$ on the same page confirms it). Also p. 556 (Step 1 of §3) prints $r_p = r_q = \epsilon_0$ for the same set-up.

(3) CP10 p. 565 proves the exponential bound by "applying A.1 once and then proceeding as in (A21)–(A26) of [4]": the iteration is not in the pinned sources; it is Proposition 3.4, Steps S3.6–S3.8.

(4) Part 3's proof plan of Lemma 4.8 (one flow) does not give $e^{-Ks/6}E$ (review P3-01); the corrected proof is above.

Forward and cross-track references used: [N:bpN:p4:norm], [N:bpN:p4:domains], [N:bpN:p4:cauchy], [N:bpN:p4:bracket], [N:bpN:p4:projections], [N:bpN:p4:sup-weighted], [N:bpN:p4:coefficients], [N:bpN:p4:algebra], [N:bpN:p4:dictionary], [N:bpN:p5:setting], [N:bpN:p5:vf-bound], [N:bpN:p5:flow], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:adjoint], [N:bpN:p5:composition], [N:bpN:p5:transform], [N:bpN:p5:canonical-map], [N:bpN:p6:setting], [N:bpN:p6:homological], [N:bpN:p6:divisor], [N:bpN:p6:iterative], [N:bpN:p6:iterative-special], [N:bpN:p6:telescope], [N:bpN:p6:normal-form], [N:bpN:p6:deviations], [K:bp:p1:ext-mvt], [P:bpP:p0:pg].

4. Two-scale KAM theorem I: statement and the iterative lemma

This part (3a) states the two-scale KAM theorem of Chierchia–Pinzari (CP10 Proposition 3, pp. 554–555) in the frozen conventions and proves the iterative lemma (CP10 Lemma B.1 with Lemma B.2, pp. 565–570). Part 3b (limit, Lipschitz bounds, measure) completes the proof. *Every constant is recomputed here*: the numerical constants differ from those of CP10 (max norms, track N's weighted norm, no use of the unpinned reference [4]), the shape of $K, \hat{\rho}, L, \hat{E}$ and of the KAM condition is the same. Results of track K are cited as [K:label]; the averaging step is proved in Part 2 ([P:bpP:p2:avg], [P:bpP:p2:nfb]), which rests on track N, Parts 4–6.

4.1. Conventions, norms, constants.

- *Period one.* $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with coordinates θ; Fourier characters $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, $|k|_1 = \sum_i |k_i|$. The symplectic form is $dJ \wedge d\theta$, the Hamilton equations are $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ (this is the order (action, angle) used by CP10; the frozen order $(z_1, z_2) =$ (position, momentum) is restored in Part 5). A frequency $\omega \in \mathbb{R}^n$ is a rotation number per unit time. Part 0 keeps period 2π for the canonical fast angles of P-G; the passage is the one-line dictionary of Lemma 4.5, under which *all constants of this part are invariant.*
- *Norms.* $|\cdot|$ is the max norm on $\mathbb{R}^n, \mathbb{C}^n$; $\|A\| = \max_i \sum_j |A_{ij}|$ the induced operator norm (also for rectangular matrices). Hence for a bilinear form entries and operator norm are comparable ($|A_{ij}| \le \|A\|$).
- *Complex neighbourhoods.* For $D \subset \mathbb{R}^n$ and $r > 0$: $D_r = \{I \in \mathbb{C}^n : \exists I_0 \in D,\ |I - I_0| < r\}$ and $\mathrm{Re}(D_r) = D_r \cap \mathbb{R}^n = \{x \in \mathbb{R}^n : \mathrm{dist}_\infty(x, D) < r\}$; $(D_r)_{r'} \subset D_{r+r'}$ and $D_r \subset D_{r'}$ for $r \le r'$. For $\sigma > 0$, $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\mathrm{Im}\,\theta| < \sigma/2\pi\}$ (weight σ, strip width $\sigma/2\pi$).
- *Weighted norm (track N's norm, ruling P3-03).* For $f = \sum_k \hat f_k(I) e^{2\pi i k\cdot\theta}$ analytic on $D_r \times \mathbb{T}_\sigma$: $\|f\|_{r,\sigma} := \sup_{I\in D_r} \sum_{k\in\mathbb{Z}^n} |\hat f_k(I)|\, e^{\sigma|k|_1}$ ([N:bpN:p4:norm]; docs/fourier-conventions.md). *Width dictionary:* σ here is the coefficient radius, $\sigma = 2\pi\sigma_N$ with σ_N the geometric half-width ($\mathbb{T}_\sigma = T_{\sigma/2\pi}$ in the notation of Part 2). The printed CP10 norm is $\sum_k \sup_I$, which dominates this one; the two are not equivalent. Properties used in Part 3: monotone ($\|f\|_{r',\sigma'} \le \|f\|_{r,\sigma}$ for smaller base set, $r' \le r$, $\sigma' \le \sigma$), $\sup_{D_r\times\mathbb{T}_\sigma}|f| \le \|f\|_{r,\sigma}$, $\sup_{D_r}|f_0| \le \|f\|_{r,\sigma}$, and Cauchy estimates, applied to sup only; the averaging output (Corollary 3.6) is stated in this norm. Every constant below was rechecked for this change (none used $\sum_k \sup_I$; Part 2, Remark 3.7).
- *Two-scale Diophantine set.* For $n = n_1 + n_2$, $k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$, $\gamma_1 \ge \gamma_2 > 0$, $\tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau} = \{\omega \in \mathbb{R}^n :\ |k\cdot\omega| \ge \gamma_1 |k|_1^{-\tau}$ if $k_1 \ne 0$, $\ |k\cdot\omega| \ge \gamma_2 |k|_1^{-\tau}$ if $k_1 = 0, k \ne 0\}$. For $\gamma_1 = \gamma_2$ this is the set of Definition 1.1; $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \mathcal{D}_{\gamma_2,\gamma_2,\tau}$. (CP10 (19) with $\bar\gamma_2\mu = \gamma_2$; for $k_1 = 0$ CP10 writes $|k_2|^\tau$, the same as $|k|_1^\tau$.) For $D \subset \mathbb{R}^n$ and $\omega_0 : D_\rho \to \mathbb{C}^n$ put $D^{\mathrm{dio}} := \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau}) \cap D$ (this is $\mathcal{D}_{\gamma_1,\gamma_2,\tau_*}$ of CP10, "the ω_0-preimage in D").

- *Blocks.* $U = \partial^2 h$ is the $n \times n$ Hessian; $\hat{U}$ is the $n_2 \times n$ matrix of its last n_2 rows (the derivatives of $\omega_2 = \partial_{I_2} h$ with respect to all actions); T_1, T_2 are the first n_1 and last n_2 rows of U^{-1}.

Remark 4.1 (Misprint in the definition of $\hat{U}$). CP10 p. 554 defines $\hat{U}$ as "the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1+1 \le i \le n$, $1 \le j \le n$". The size and the index ranges disagree. Only the reading above ($n_2 \times n$, rows $n_1+1, \dots, n$, all columns) is used in the proof: $\hat{M}$ bounds the variation of ω_2 (CP10 (82), via the choice of $\hat{\rho}$). We adopt it.

Definition 4.2 (Constants of the two-scale theorem)**.** Let $n = n_1 + n_2$, $\tau > 0$, $\gamma_1 \ge \gamma_2 > 0$, $0 < s$, $\rho > 0$, and $E, M, \hat{M}, \bar{M} > 0$. Put

$$L := \max\{\bar{M},\ M^{-1},\ \hat{M}^{-1}\}, \qquad x := \frac{EM^2 L}{\gamma_1^2}, \qquad K := \frac{6}{s}\log_+\!\big(x^{-1}\big), \quad \log_+ a := \max\{1, \log a\},$$

$$\hat{\rho} := \min\Big\{\frac{\gamma_1}{3MK^{\tau+1}},\ \frac{\gamma_2}{3\hat{M}K^{\tau+1}},\ \rho\Big\}, \qquad \hat{E} := \frac{EL}{\hat{\rho}^{\,2}}, \qquad \hat{c} := \max\Big\{2^{13}n,\ \frac{24^{\tau+1}}{3}\Big\}.$$

The *KAM condition* is $\hat{c}\hat{E} < 1$. (CP10 (31): $\hat{c} = \max\{2^8 n, 24^{\tau+1}/6\}$; the larger constants come from the max-norm Cauchy estimates and the Lipschitz bookkeeping of Parts 3a–3b. Only $\hat{c} = \hat{c}(n,\tau)$, an absolute function of n and τ, matters downstream.) *Totality.* $E > 0$ guards $\log(1/x)$, and $\log_+ \ge 1$ gives $Ks \ge 6$; $K, \hat{\rho} > 0$.

Source: CP10 Prop. 3, p. 555, equations (31), (32), with the changes stated.

4.2. The theorem.

Proposition 4.3 (Two-scale KAM theorem)**.** ***Data.*** *Integers $n_1, n_2 \ge 1$, $n = n_1 + n_2$; $\tau > n$; $\gamma_1 \ge \gamma_2 > 0$; $s, \bar{s}$ with $0 < 4s \le \bar{s} < 1$; $\rho > 0$; an arbitrary set $D \subset \mathbb{R}^n$; $A := D_\rho$. A function $H(J,\theta) = h(J) + f(J,\theta)$,* real analytic *on $A \times \mathbb{T}_{\bar{s}+s}$, 1-periodic in θ (real on $\mathbb{R}^n \times \mathbb{R}^n$). Let $\omega_0 := \partial h$; assume ω_0 is an injective holomorphic map $A \to \mathbb{C}^n$ with $U = \partial^2 h$ nonsingular on A (hence a diffeomorphism of A onto an open set). Constants $M, \hat{M}, \bar{M}, \bar{M}_1, \bar{M}_2, E > 0$ with*

$$M \ge \sup_A\|U\|, \quad \hat{M} \ge \sup_A\|\hat{U}\|, \quad \bar{M} \ge \sup_A\|U^{-1}\|, \quad \bar{M}_i \ge \sup_A\|T_i\|\ (i=1,2), \quad E \ge \|f\|_{\rho,\bar{s}+s},$$

and (normalizations, always satisfiable by lowering the constants; see Remark 4.4) $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$. Let $L, K, \hat{\rho}, \hat{E}, \hat{c}$ be as in Definition 4.2 and assume the ***KAM condition*** *$\hat{c}\hat{E} < 1$.*

Conclusions. *Put $r := 32\,\hat{E}\hat{\rho}$, $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ and $D^{\mathrm{dio}} := \omega_0^{-1}(\Omega_*) \cap D$. There are maps φ_ω, $\omega \in \Omega_*$,*

$$\varphi_\omega : \mathbb{R}^n \to \mathrm{Re}(D_r) \times \mathbb{R}^n, \quad \varphi_\omega(\vartheta) = \big(v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)\big),$$

with v, u 1-periodic in ϑ, such that for every $\omega \in \Omega_*$:

- *(a) (analyticity) φ_ω is real analytic and extends holomorphically to $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\}$, with $|u| \le \frac{3}{4\pi}\hat{E}s$ there;*
- *(b) (conjugacy) $\frac{d}{dt}\varphi_\omega(\vartheta+t\omega) = X_H(\varphi_\omega(\vartheta+t\omega))$ for all $\vartheta \in \mathbb{R}^n, t \in \mathbb{R}$, where $X_H = (-\partial_\theta H, \partial_J H)$ (Hamilton equations for (J,θ));*
- *(c) (embedded Lagrangian torus) $D_\vartheta\varphi_\omega$ is injective (indeed $\|D_\vartheta u\| \le \frac{3}{4}n\hat{E}$) and $\sum_i \big(\partial_{\vartheta_a} v_i\, \partial_{\vartheta_b}(\vartheta_i + u_i) - \partial_{\vartheta_b} v_i\, \partial_{\vartheta_a}(\vartheta_i + u_i)\big) = 0$ for all a, b;*
- *(d) (estimates) with $I^0(\omega) := \omega_0^{-1}(\omega) \in D^{\mathrm{dio}}$ and $v = (v_1, v_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$,*

$$|v_1 - I_1^0(\omega)| \le 20\Big(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{M}\Big)\hat{E}\hat{\rho}, \quad |v_2 - I_2^0(\omega)| \le 20\Big(\frac{\bar{M}_2}{\bar{M}} + 1\Big)\hat{E}\hat{\rho}, \quad |u| \le \frac{3}{4\pi}\hat{E}s \quad (\text{on } \mathbb{R}^n).$$

Moreover the map $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$, $\mathbb{R}^n \times \Omega_* \to \mathbb{R}^{2n}$, is injective modulo the translations $\vartheta \mapsto \vartheta + k$, $k \in \mathbb{Z}^n$ (precisely: $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') \in \{0\} \times \mathbb{Z}^n$ implies $\omega = \omega'$ and $\vartheta - \vartheta' \in \mathbb{Z}^n$), and is Lipschitz in the following sense: it factors as $\Phi \circ (\check{\imath} \circ \omega_0^{-1}, \mathrm{id})$ with $\check{\imath}$ of Lipschitz constant $\le 1 + 2560n\hat{E}$ and

Φ *bi-Lipschitz in the rescaled norm of §5.1;* ω_0^{-1} *is* $\bar M$*-Lipschitz on* Ω_* *for pairs at distance* $< \rho/\bar M$ *(globally* $\bar M \max\{1, \operatorname{diam} D/\rho\}$*-Lipschitz if* D *is bounded). Finally, with* $K_\Omega := \bigcup_{\omega\in\Omega_*} \varphi_\omega(\mathbb{R}^n)$ *(a union of tori),* $c_n := 1 + (1+2^n)(1+(4/3)^n)$ *and* meas *the Lebesgue measure on* $\mathbb{R}^n \times \mathbb{T}^n$*,*

$$\operatorname{meas}\big((\operatorname{Re}(D_r)\times\mathbb{T}^n)\setminus K_\Omega\big) \le c_n\Big(\operatorname{meas}\big((D\setminus D^{\mathrm{dio}})\times\mathbb{T}^n\big) + \operatorname{meas}\big((\operatorname{Re}(D_r)\setminus D)\times\mathbb{T}^n\big)\Big). \tag{3.1}$$

Constant dependence. $\hat c, c_n$ *depend only on* n, τ*;* $K, \hat\rho, \hat E, L$ *depend only on* $(n_1, n_2, \tau, \gamma_1, \gamma_2, s, \rho, E, M, \hat M, \bar M)$*, in the order of Definition 4.2. The sets* D, h, f *enter only through those constants and the hypotheses;* $\bar s$ *enters only through* $4s \le \bar s < 1$ *and through the width of analyticity in (a).*
Not frozen itself: Part 5 consumes this statement (§5.6) to prove the frozen theorem [P:bpP:p0:pg].
Source: CP10 Prop. 3, pp. 554–555, proved in Appendix B, pp. 565–573. Constants recomputed; (35) of CP10 is corrected in (d).
Source gap: Prop. 3 has no other source ("not available in literature", CP10 p. 550); its printed proof relies on Prop. 1/Lemma A.1 through the unpinned [4, Lemma A.5]; that input is supplied by Part 2 (Lemma 4.8 is proved there). The scheme (Lemma B.1) is proved inductively on CP10 pp. 567–570; the pointer to Arnold [3, p. 144] in Remark 3 is historical, not a dependency. Misprints and repairs: §5.7.

Remark 4.4 (Normalizations $\hat M \le M$, $\bar M_i \le \bar M$). $\hat U$ and T_i are sub-blocks of U, U^{-1}, so $\|\hat U\| \le \|U\|$ and $\|T_i\| \le \|U^{-1}\|$ on A: the hypotheses may be stated with $\hat M \le M$ and $\bar M_i \le \bar M$ *if one replaces* $\hat M$*,* $\bar M_i$ *by* $\min\{\hat M, M\}$*,* $\min\{\bar M_i, \bar M\}$; this only decreases $\hat M$ (which can only increase $\hat\rho$, hence is not harmless for $\hat E$: one should lower after computing $\hat E$). We state the proposition with the normalization as a hypothesis, as CP10 uses it implicitly in (74), (77) and (35) ("$\le 5n\hat E\hat\rho$" needs $\bar M_i \le \bar M$). In P-G, $\hat M = C\mu \le M = C$ and $\bar M_i \le \bar M = C\mu^{-1}$.

Lemma 4.5 (Period dictionary)**.** *This is* [P:bpP:p1:period-convert] *with* $c = 2\pi$ *(chart* $S_{2\pi}(z_1, z_2) = (2\pi z_1, z_2)$*; in Part 3's order,* $Q(I,\theta) = (I, 2\pi\theta)$*). Let* $\tilde H(I,\phi) = \tilde h(I) + \tilde f(I,\phi)$ *be analytic on* $A \times \{|\operatorname{Im}\phi| < \sigma\}$*,* $\phi \in (\mathbb{R}/2\pi\mathbb{Z})^n$ *(CP10 convention: form* $dI \wedge d\phi$*, characters* $e^{ik\cdot\phi}$*, norm* $\sup_I \sum_k |\tilde f_k| e^{\sigma|k|_1}$*,* $\tilde\omega = \partial\tilde h$*; constants* $\tilde M, \tilde{\hat M}, \tilde{\bar M}, \tilde E, \tilde\gamma_i$ *and* $\tilde L, \tilde K, \tilde{\hat\rho}, \tilde{\hat E}$ *computed by Definition 4.2 with* $\tilde\gamma_i$ *the Diophantine constants of* $\tilde\omega$*). Put* $H(I,\theta) := \tilde H(I, 2\pi\theta)/2\pi$*. Then* $h = \tilde h/2\pi$*,* $f = \tilde f(\cdot, 2\pi\theta)/2\pi$*,* $\omega = \tilde\omega/2\pi$*,* $\|f\|_{\rho,\sigma} = \|\tilde f\|_{\rho,\sigma}/2\pi$ *and, with* $M = \tilde M/2\pi$*,* $\hat M = \tilde{\hat M}/2\pi$*,* $\bar M = 2\pi\tilde{\bar M}$*,* $\bar M_i = 2\pi\tilde{\bar M}_i$*,* $E = \tilde E/2\pi$*,* $\gamma_i = \tilde\gamma_i/2\pi$*:* $L = 2\pi\tilde L$ *and* $K = \tilde K$*,* $\hat\rho = \tilde{\hat\rho}$*,* $\hat E = \tilde{\hat E}$*,* $\hat c = \tilde{\hat c}$ identically. Tori. *If* $\varphi_\omega = (v, \mathrm{id} + u)$ *is a torus of Proposition 4.3 for* H *(coordinates* (I,θ)*, frequency* ω*), then* $\tilde\varphi(\phi) := Q\big(\varphi_\omega(\phi/2\pi)\big) = (v(\phi/2\pi),\ \phi + 2\pi u(\phi/2\pi))$ *is a torus for* $\tilde H$ *in CP's coordinates with flow* $\phi \mapsto \phi + t\tilde\omega$*,* $\tilde\omega = 2\pi\omega$*, and* $\tilde u := 2\pi u$ *satisfies* $|\tilde u| \le \frac{3}{2}\hat E s$ *(the* output angle is scaled*). The frozen parameterization is* $K(\theta) = Q(\varphi_\omega(\theta))$ *(up to the frozen order) with frozen frequency* $w = \omega = \tilde\omega/2\pi$*:* one *division by* 2π*, already contained in the passage* $\tilde H \mapsto H$*;* w *is not divided again (test:* $\tilde h(I) = 2\pi I$ *has* $\tilde\omega = 2\pi$ *and* $\omega = w = 1$*).*

Proof. **Step S4.1: Substitution.** $\tilde f(I,\phi) = \sum_k \tilde f_k(I) e^{ik\cdot\phi}$ and $\phi = 2\pi\theta$ give $f(I,\theta) = \frac{1}{2\pi}\sum_k \tilde f_k(I) e^{2\pi ik\cdot\theta}$: same weights, coefficients divided by 2π. $U = \tilde U/2\pi$, $U^{-1} = 2\pi\tilde U^{-1}$, sub-blocks likewise.

Step S4.2: Invariants. $EM^2L/\gamma_1^2 = \frac{\tilde E}{2\pi}\frac{\tilde M^2}{4\pi^2} 2\pi\tilde L \frac{4\pi^2}{\tilde\gamma_1^2} = \tilde E\tilde M^2\tilde L/\tilde\gamma_1^2$, hence $K = \tilde K$; $\gamma_1/(3MK^{\tau+1}) = \tilde\gamma_1/(3\tilde M K^{\tau+1})$ (same for $\gamma_2, \hat M$), so $\hat\rho = \tilde{\hat\rho}$; $EL/\hat\rho^2 = \tilde E\tilde L/\tilde{\hat\rho}^2$.

Step S4.3: Dynamics and tori. $\dot\phi = \partial_I\tilde H = 2\pi\partial_I H = 2\pi\dot\theta$, $\dot I = -\partial_\phi\tilde H = -\partial_\theta H$ ([P:bpP:p1:period-convert](b)); composing the conjugacy with Q gives $\tilde\varphi$, and [P:bpP:p1:period-convert](c) gives $w = \tilde\omega/2\pi = \omega$. □

4.3. **External facts and imported tools.**

(X1) *Holomorphic inverse function theorem.* An injective holomorphic $F : \Omega \to \mathbb{C}^n$ (Ω open) with $\det DF \neq 0$ is open, and $F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $D(F^{-1}) = (DF)^{-1} \circ F^{-1}$.

(X2) *Fourier coefficients of periodic continuous functions* ([K:`bp:p1:ext-fourier`]): $\hat{F}_k = \int_{[0,1]^n} F(\vartheta) e^{-2\pi i k\cdot\vartheta} d\vartheta$, used in Lemma 5.4 (which replaces a several-variable Vitali–Montel theorem; review P3-11).

(X3) *Weierstrass.* Locally uniform limits of holomorphic functions are holomorphic and derivatives converge locally uniformly ([K:`bp:p1:uniform-limits`]).

(X4) Banach fixed point ([K:`bp:p1:ext-banach`]); mean value inequality on convex sets ([K:`bp:p1:ext-mvt`]); Cauchy estimates on polydiscs ([K:`bp:p1:cauchy-open`]).

(X5) Neumann series: $\|B\| \le \frac{1}{2}$ implies $1 + B$ invertible with $\|(1 + B)^{-1}\| \le 2$.

(X6) Lipschitz measure lemmas [K:`bp:p1:lip-image`], McShane extension [K:`bp:p1:mcshane`] (max norm).

4.4. **Two elementary lemmas.**

Lemma 4.6 (Balls in the image of a diffeomorphism)**.** *Let $\Omega \subset \mathbb{C}^n$ be open, $F : \Omega \to \mathbb{C}^n$ injective holomorphic with $\|(DF)^{-1}\| \le m$ on Ω, and $B(x_0, \delta) \subset \Omega$ (max-norm ball). Then $F(B(x_0, \delta)) \supset B(F(x_0), \delta/m)$. Moreover, if $w_1, w_2 \in B(F(x_0), \delta/m)$ then $|F^{-1}(w_1) - F^{-1}(w_2)| \le m|w_1 - w_2|$, with F^{-1} evaluated in $B(x_0, \delta)$.*

Proof. **Step S4.4: Inverse.** By (X1), $G := F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $\|DG\| \le m$ and $F(B(x_0, \delta))$ is open.

Step S4.5: Path lifting. Let $w \in B(F(x_0), \delta/m)$, $\gamma(t) = F(x_0) + t(w - F(x_0))$, $\delta' = m|w - F(x_0)| < \delta$ and $T = \{t \in [0, 1] : \gamma([0, t]) \subset F(B(x_0, \delta))\}$. $0 \in T$ and T is relatively open. For $t \in T$ the path $G \circ \gamma$ on $[0, t]$ is C^1 with $|(G \circ \gamma)'| \le m|w - F(x_0)|$, so $|G(\gamma(t)) - x_0| \le t\delta'$ ((X4), mean value inequality along the path).

Step S4.6: Closedness. If $t_k \uparrow t^*$, $t_k \in T$, then $G(\gamma(t_k))$ is Cauchy (Lipschitz in t), with limit x^*, $|x^* - x_0| \le \delta' < \delta$, $F(x^*) = \gamma(t^*)$ by continuity of F, so $t^* \in T$. Hence $T = [0, 1]$ and $w = F(G(w)) \in F(B(x_0, \delta))$.

Step S4.7: Lipschitz bound. The segment $[w_1, w_2]$ lies in the convex ball $B(F(x_0), \delta/m) \subset F(B(x_0, \delta))$, so the mean value inequality for G gives the second claim. □

Lemma 4.7 (Lipschitz constants from sup bounds, non-convex sets)**.** *Let $S \subset \mathbb{R}^N$ be arbitrary, $r > 0$, $S_r \subset \mathbb{C}^N$ its complex r-neighbourhood in the max norm, and $\Delta : S_r \to \mathbb{C}^m$ holomorphic with $\sup_{S_r} |\Delta| \le A$. Then $|\Delta(x) - \Delta(y)| \le \frac{4NA}{r}|x - y|$ for all $x, y \in S$.*

Proof. **Step S4.8: Close pairs.** If $|x - y| < r/2$, every point z of the real segment $[x, y]$ has $|z - x| < r/2$, so the closed polydisc $\bar{P}(z, r/2) \subset S_r$ (z is within $r/2$ of $x \in S$, hence $\bar{P}(z, r/2) \subset S_r$ up to the open/closed boundary; use radius $r/2 - \epsilon$ and let $\epsilon \to 0$). By [K:`bp:p1:cauchy-open`], $|\partial_\ell \Delta_i(z)| \le 2A/r$, so $\|D\Delta(z)\| \le 2NA/r$, and the mean value inequality on the segment gives $|\Delta(x) - \Delta(y)| \le (2NA/r)|x - y|$.

Step S4.9: Distant pairs. If $|x - y| \ge r/2$, then $|\Delta(x) - \Delta(y)| \le 2A \le (4A/r)|x - y|$. Since $N \ge 1$, both are $\le 4NA/r$. □

4.5. **Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1).**

Lemma 4.8 (One averaging step with two scales)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s}, s > 0$, $K > 0$ with $Ks \ge 6$. Let h be analytic on D_r, real on real points, $\omega = \partial h$, f analytic on $D_r \times \mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in θ, and $E := \|f\|_{r,\bar{s}+s}$ (N-norm, §4.1). Assume*

(N1) (non-resonance on the complex neighbourhood) for all $I \in D_r$ and $k \in \mathbb{Z}^n$ with $0 < |k|_1 \le K$: $|k \cdot \omega(I)| \ge \alpha_1$ if $k_1 \neq 0$ and $\ge \alpha_2$ if $k_1 = 0$;

(N2) (smallness) $64e\,KE < \alpha_2 r$; equivalently $E < \alpha_2 d/(2^7 c_0 K s)$ with $d = rs$, $c_0 = e/2$ (CP10 (23), $m = 0$).

Then there is $\Psi : D_{r/2} \times \mathbb{T}_{\bar{s}+s/6} \to D_r \times \mathbb{T}_{\bar{s}+s}$ ($\Psi(I', \theta') = (I, \theta)$), holomorphic, injective, real on real points, $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', symplectic ($\Psi^*(dI \wedge d\theta) = dI' \wedge d\theta'$), with $H \circ \Psi = h + g + f_*$ on $D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}$, $g = g(I')$ independent of θ', and

$$\sup_{D_{r/2}} |g - f_0| \le E/4 \quad (\text{indeed} \le E/(Ks)), \qquad \|f_*\|_{r/2, \bar{s}+s/6} \le e^{-Ks/6} E,$$

$$\alpha_1 s |I_1 - I_1'| \le 9E, \quad \alpha_2 s |I_2 - I_2'| \le 9E, \quad 2\pi\alpha_2 r |\theta - \theta'| \le 9E \qquad \text{on } D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}.$$

Here $f_0(I) = \int_{\mathbb{T}^n} f(I, \theta) d\theta$ and $I = (I_1, I_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$.
Source: CP10 Prop. 1 (p. 552), $m = 0$, $\Lambda = \{0\}$, (23)–(26); Lemma A.1 (p. 564).

Proof. This is Corollary 3.6 of Part 2 verbatim (there the first two displacement constants are even $3E$; $9E$ is kept because Lemma 4.9 was computed with it). Its proof is Proposition 3.4: a *finite iteration* of track N's Iterative Lemma (the exponential $e^{-Ks/6}$ comes from the iteration, not from one flow), with the loss schedule $\rho_i = r/8N$, $\sigma_i = s/4N$, $N = \lfloor Ks/12 \ln 2 \rfloor$, and the generating-function splitting of Lemma 3.3 that gives the α_1-bound for I_1 at every stage. The bound on $g - f_0$ has denominator $\alpha_2 rs$ and no K in the numerator ($|g - f_0| \le E/\kappa$, $\kappa = Ks$; CP10 (25); review P3-04). Only (N1), (N2) are assumed: no Lipschitz hypothesis (Part 2, Remark 3.7). □

4.6. **One step of the scheme.**

Lemma 4.9 (One KAM step)**.** *Let the data of Proposition 4.3 be given with $H = h + f$ on $D_\rho \times \mathbb{T}_{\bar{s}+s}$, constants $M, \hat{M}, \bar{M}, \bar{M}_i, L, E, \ldots, K, \hat{\rho}, \hat{E}$ as in Definition 4.2, and in addition*

(S1) $\hat{c}\hat{E} < 1$ (so $\hat{E} < 2^{-13}/n$);

(S2) $L \ge \max\{\bar{M}, M^{-1}, \hat{M}^{-1}\}$ (this is the definition; it is stated to allow L larger, used in the iteration: replace L by any larger number, and then use the same formulas);

(S3) $\omega_0(D) \subset \mathcal{D}_{\gamma_1, \gamma_2, \tau}$.

Put $\rho_+ := \hat{\rho}/16$, $s_+ := s/12$, $M_+ := 2M$, $\hat{M}_+ := 2\hat{M}$, $\bar{M}_+ := 2\bar{M}$, $\bar{M}_{i+} := 2\bar{M}_i$, $L_+ := 2L$, $E_+ := E^2 L M^2/\gamma_1^2$ and let $K_+, \hat{\rho}_+, \hat{E}_+$ be given by Definition 4.2 with the + data. Then there are a map Ψ_+, a function h_+ and a set $D_+ \subset D_{\hat{\rho}/16}$, with

(i) $\Psi_+ : (D_+)_{\rho_+} \times \mathbb{T}_{\bar{s}+s_+} \to D_{\hat{\rho}} \times \mathbb{T}_{\bar{s}+s}$ holomorphic, real on reals, $\Psi_+ - \mathrm{id}$ periodic, symplectic, and $H_+ := H \circ \Psi_+ = h_+ + f_+$ with $h_+ = h + g$, $g = g(I)$ analytic on $D_{\hat{\rho}/2}$, $\sup_{D_{\hat{\rho}/2}} |g| \le \frac{5}{4} E$, and $\|f_+\|_{\rho_+, \bar{s}+s_+} \le E_+$;

(ii) $\omega_+ = \partial h_+$ is injective on $D_{\hat{\rho}/8}$ with $U_+ = \partial^2 h_+$ nonsingular there; $(D_+)_{\rho_+} \subset D_{\hat{\rho}/8}$, $\omega_+(D_+) = \omega_0(D)$ (so (S3) holds again);

(iii) $M_+ \ge \sup_{D_{\hat{\rho}/4}} \|U_+\|$, $\hat{M}_+ \ge \sup \|\hat{U}_+\|$, $\bar{M}_+ \ge \sup \|U_+^{-1}\|$, $\bar{M}_{i+} \ge \sup \|T_{i+}\|$ (sup over $D_{\hat{\rho}/4}$), and $L_+ \ge \max\{\bar{M}_+, M_+^{-1}, \hat{M}_+^{-1}\}$;

(iv) the map $\hat{\imath}_+ := (\omega_+|_{D_{\hat{\rho}/8}})^{-1} \circ \omega_0$ is defined and holomorphic on $D_{\hat{\rho}/16}$, $D_+ = \hat{\imath}_+(D)$, and

$$\sup_{D_{\hat{\rho}/16}} |\hat{\imath}_{+,i} - \mathrm{id}_i| \le 10 \frac{\bar{M}_i}{\bar{M}} \hat{E}\hat{\rho} \ (\le 10\hat{E}\hat{\rho}, \ i = 1, 2), \qquad L_D(\hat{\imath}_+ - \mathrm{id}) \le 640\, n \hat{E};$$

(where ω_0 here means the frequency map of the current h, and L_D is the Lipschitz constant on the set D);

(v) the displacement bounds, for $(I', \theta') \in D_{\hat{\rho}/2} \times \mathbb{T}_{\bar{s}+s/6}$ and $\Psi_+(I', \theta') = (I, \theta)$:

$$|I_1 - I_1'| \le \tfrac{3}{4} \tfrac{\hat{M}}{\bar{M}} \hat{E}\hat{\rho}, \quad |I_2 - I_2'| \le \tfrac{3}{4} \hat{E}\hat{\rho}, \quad 2\pi |\theta - \theta'| \le \tfrac{3}{4} \hat{E} s;$$

(vi) $\hat\rho_+ \ge \hat\rho/(2\cdot 24^{\tau+1})$, $K_+ \le 24K$, $\hat E_+ \le \frac{8}{9}(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$, *and* $\hat c\hat E_+ < 1$.

Proof. **Step S4.10: Non-resonance on $D_{\hat\rho}$.** Let $I \in D_{\hat\rho}$, pick $I_0 \in D$ with $|I - I_0| < \hat\rho \le \rho$; the segment $[I_0, I]$ lies in D_ρ, where $\|U\| \le M$, hence $|\omega_i(I) - \omega_i(I_0)| \le M\hat\rho$ for all i, and for $i > n_1$ (rows of $\hat U$) $\le \hat M\hat\rho$ ([K:bp:p1:ext-mvt]). For $0 < |k|_1 \le K$ and $k_1 \ne 0$: $|k\cdot\omega(I)| \ge \gamma_1|k|_1^{-\tau} - |k|_1 M\hat\rho \ge \gamma_1 K^{-\tau} - KM\hat\rho \ge \frac{2}{3}\gamma_1 K^{-\tau} =: \alpha_1$, since $\hat\rho \le \gamma_1/(3MK^{\tau+1})$ and $\omega(I_0) \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by (S3). For $k_1 = 0$ only ω_2 enters: $|k\cdot\omega(I)| \ge \gamma_2 K^{-\tau} - K\hat M\hat\rho \ge \frac{2}{3}\gamma_2 K^{-\tau} =: \alpha_2$. Also $\alpha_2 \le \alpha_1$ and (from $\hat\rho \le \gamma_i/(3\cdot K^{\tau+1}\cdot(\mathrm{M}\text{ or }\hat M))$)

$$\alpha_1 \ge 2MK\hat\rho, \qquad \alpha_2 \ge 2\hat M K\hat\rho. \tag{3.2}$$

Step S4.11: Smallness and $Ks \ge 6$. $Ks = 6\log_+(x^{-1}) \ge 6$. By (3.2) and $L \ge \hat M^{-1}$: $64e\,KE/(\alpha_2\hat\rho) \le 64e\,E/(2\hat M\hat\rho^2) \le 32e\,\hat E < 87\hat E < \hat c\hat E < 1$. So Lemma 4.8 applies with $r = \hat\rho$.

Step S4.12: Averaging. Lemma 4.8 gives Ψ, g, f_* on $D_{\hat\rho/2}\times\mathbb{T}_{\bar s+s/6}$. Since $e^{-Ks/6} = \min\{e^{-1}, x\} \le x$ ($Ks/6 = \log_+(1/x)$), $\|f_*\| \le xE = E^2LM^2/\gamma_1^2 = E_+$. Define Ψ_+ as the restriction of Ψ to $(D_+)_{\rho_+}\times\mathbb{T}_{\bar s+s_+}$ once D_+ is defined below ($(D_+)_{\rho_+} \subset D_{\hat\rho/8} \subset D_{\hat\rho/2}$ and $s_+ \le s/6$; the norm is monotone). Also $\sup_{D_{\hat\rho/2}}|g| \le \sup|g - f_0| + \sup|f_0| \le E/4 + E$. This proves the first part of (i).

Step S4.13: Displacements. By the last line of Lemma 4.8, (3.2), $Ks \ge 6$ and $M\hat M^{-1}$-bounds: $|I_1 - I_1'| \le \frac{9E}{\alpha_1 s} \le \frac{9EK}{6\alpha_1} \le \frac{9E}{12M\hat\rho} = \frac{3}{4}\frac{E}{M\hat\rho} \le \frac{3}{4}\frac{\hat M}{M}\frac{EL}{\hat\rho}$, using $\hat ML \ge 1$; and $\frac{EL}{\hat\rho} = \hat E\hat\rho$. Likewise $|I_2 - I_2'| \le \frac{9E}{\alpha_2 s} \le \frac{3}{4}\frac{E}{\hat M\hat\rho} \le \frac{3}{4}\hat E\hat\rho$, $2\pi|\theta - \theta'| \le \frac{9E}{\alpha_2\hat\rho} \le \frac{9E}{2\hat MK\hat\rho^2} \le \frac{9}{2}\frac{\hat E}{K} \le \frac{3}{4}\hat Es$ (as $K \ge 6/s$). This is (v).

Step S4.14: Derivatives of g. For $I \in D_{\hat\rho/4}$ the polydisc of radius $\hat\rho/4$ around I lies in $D_{\hat\rho/2}$. [K:bp:p1:cauchy-open] with $A = \frac{5}{4}E$, $\delta = \hat\rho/4$ gives $|\partial_i g(I)| \le 5E/\hat\rho$ and $|\partial_i\partial_j g(I)| \le 4A/\delta^2 = 80E/\hat\rho^2$, so

$$\sup_{D_{\hat\rho/4}}\|\partial^2 g\| \le \frac{80nE}{\hat\rho^2} = \frac{80n\hat E}{L}, \qquad \sup_{D_{\hat\rho/4}}|\partial g| \le \frac{5E}{\hat\rho}. \tag{3.3}$$

Since $L^{-1} \le M$, $L^{-1} \le \hat M$ and $\bar M \le L$: $\|\partial^2 g\| \le 80n\hat E\min\{M, \hat M\} \le \frac{1}{100}\min\{M, \hat M\}$ and $\bar M\|\partial^2 g\| \le 80n\hat E \le \frac{1}{100}$.

Step S4.15: Bounds on U_+. $U_+ = U + \partial^2 g$ on $D_{\hat\rho/4}$ and $\hat U_+ = \hat U + (\partial^2 g)_{\mathrm{rows}>\mathrm{n}_1}$ with $\|\mathrm{rows}\| \le \|\partial^2 g\|$. Hence $\|U_+\| \le M + \frac{M}{100} \le 2M$ and $\|\hat U_+\| \le \hat M + \frac{\hat M}{100} \le 2\hat M$. Further $U_+ = (1 + \partial^2 g\,U^{-1})U$, so $U_+^{-1} = U^{-1}(1 + \partial^2 g\,U^{-1})^{-1}$ ($\|\partial^2 g\,U^{-1}\| \le 1/100$, (X5)), hence $\|U_+^{-1}\| \le 2\bar M$ and, the rows of a product T_iB being T_i times B, $\|T_{i+}\| \le 2\|T_i\| \le 2\bar M_i$ (the left factor of $U^{-1}\cdot(\dots)$ keeps row blocks). Nonsingular. And $L_+ = 2L \ge \max\{2\bar M, (2M)^{-1}, (2\hat M)^{-1}\}$. This is (iii) on $D_{\hat\rho/4}$.

Step S4.16: Injectivity of ω_+ on $D_{\hat\rho/8}$. Let $I, I' \in D_{\hat\rho/8}$ with $\omega_+(I) = \omega_+(I')$ and put $w = \omega_0(I)$, $w' = \omega_0(I')$. Then $w - w' = \partial g(I') - \partial g(I)$ and $|w - w'| \le 10E/\hat\rho$ by (3.3). The ball $B(I, \hat\rho/8) \subset D_{\hat\rho/4} \subset D_\rho$, and ω_0 is injective on D_ρ with $\|U^{-1}\| \le \bar M$; Lemma 4.6 gives $\omega_0(B(I, \hat\rho/8)) \supset B(w, \hat\rho/(8\bar M))$. Since $10E/\hat\rho < \hat\rho/(8\bar M)$ (equivalent to $80E\bar M < \hat\rho^2$, implied by $80\hat E < 1$ and $\bar M \le L$), $w' = \omega_0(I'')$ with $I'' \in B(I, \hat\rho/8)$ and, by injectivity of ω_0 on D_ρ, $I'' = I'$. Thus the segment $[I, I'] \subset D_{\hat\rho/4}$ and, by Lemma 4.6 (second part) and the mean value inequality for ∂g along $[I, I']$, $|I - I'| \le \bar M|w - w'| \le \bar M\sup\|\partial^2 g\||I - I'| \le 80n\hat E|I - I'|$, whence $I = I'$. With (iii), ω_+ is a holomorphic injection with nonsingular derivative on $D_{\hat\rho/8}$; by (X1) an open map with holomorphic inverse, $\|D(\omega_+^{-1})\| \le 2\bar M$ (and $\|(D\omega_+^{-1})_i\| \le 2\bar M_i$ for the row blocks) on $\omega_+(D_{\hat\rho/8})$.

Step S4.17: The map $\hat\imath_+$ and D_+. For $I \in D_{\hat\rho/16}$, $B(I, \hat\rho/16) \subset D_{\hat\rho/8}$ and Lemma 4.6 for $F = \omega_+$, $m = 2\bar M$ gives $\omega_+(B(I, \hat\rho/16)) \supset B(\omega_+(I), \hat\rho/(32\bar M))$. But $\omega_0(I) = \omega_+(I) - \partial g(I)$

and $|\partial g(I)| \le 5E/\hat\rho < \hat\rho/(32\bar M)$ (as $160E\bar M < \hat\rho^2$, because $160\hat E < 1$). So $\hat\imath_+ := \omega_+^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, with values in $B(I,\hat\rho/16)$, $\omega_+\circ\hat\imath_+ = \omega_0$. By Lemma 4.6 (second part, with $w_1 = \omega_0(I)$, $w_2 = \omega_+(I)$): $|\hat\imath_+(I) - I| \le 2\bar M|\partial g(I)| \le 10\bar M E/\hat\rho$, and the same estimate holds for the first n_1 and last n_2 components with $\bar M_i$ instead of $\bar M$ (use the row blocks of $D\omega_+^{-1}$). By $\bar M \le L$: $10\bar M_i E/\hat\rho \le 10\frac{\bar M_i}{\bar M}\frac{EL}{\hat\rho} = 10\frac{\bar M_i}{\bar M}\hat E\hat\rho \le 10\hat E\hat\rho$. This is the sup bound in (iv). Then $\sup|\hat\imath_+ - \mathrm{id}| \le 10\hat E\hat\rho < \hat\rho/16$ since $\hat E < 2^{-13}$. Lipschitz bound: Lemma 4.7 with $S = D$, $r = \hat\rho/16$, $N = n$, $A = 10\hat E\hat\rho$ gives $L_D(\hat\imath_+ - \mathrm{id}) \le 4n\cdot 10\hat E\hat\rho\cdot 16/\hat\rho = 640n\hat E$. Since h is real analytic, ω_0 and ω_+ are real on $\mathbb{R}^n$; if $\omega_+(I'') = \omega_0(I)$ with $I\in D\subset\mathbb{R}^n$, then $\overline{I''}$ is a second preimage in the same ball, so $I'' = \overline{I''}\in\mathbb{R}^n$ by injectivity. Hence $\hat\imath_+(D)\subset\mathbb{R}^n$. Put $D_+ := \hat\imath_+(D)\subset D_{\hat\rho/16}\cap\mathbb{R}^n$. Since $\omega_+\circ\hat\imath_+ = \omega_0$ on D and ω_0 is injective, $\hat\imath_+$ is injective on D and $\omega_+(D_+) = \omega_0(D)$. Then $(D_+)_{\rho_+}\subset D_{\hat\rho/16+\hat\rho/16} = D_{\hat\rho/8}$. This is (ii) and the rest of (iv); in particular $\omega_+(D_+) = \omega_0(D)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Step S4.18: The new constants. Put $x = EM^2L/\gamma_1^2$; then $E_+ = Ex$ and $x_+ := E_+L_+M_+^2/\gamma_1^2 = Ex\cdot 2L\cdot 4M^2/\gamma_1^2 = 8x^2$. Hence $\log(1/x_+) = 2\log(1/x) - 3\log 2 \le 2\log_+(1/x)$ and $\log_+(1/x_+)\le 2\log_+(1/x)$ (as $1\le\log_+$), so

$$K_+ = \frac{6}{s_+}\log_+\frac{1}{x_+} \le \frac{72}{s}\cdot 2\log_+\frac{1}{x} = 24K.$$

Therefore $\gamma_1/(3M_+K_+^{\tau+1}) \ge \frac12 24^{-(\tau+1)}\gamma_1/(3MK^{\tau+1})$, likewise for $\gamma_2,\hat M$, and $\rho_+ = \hat\rho/16$; as $\hat\rho\le\min\{\gamma_1/(3MK^{\tau+1}),\gamma_2/(3\hat MK^{\tau+1})\}$, $\hat\rho_+\ge\hat\rho\min\{\frac{1}{2\cdot 24^{\tau+1}},\frac{1}{16}\} = \hat\rho/(2\cdot 24^{\tau+1})$.

Step S4.19: The quadratic bound. $E_+L_+ = 2E^2L^2M^2/\gamma_1^2$ and $\hat\rho_+^{-2}\le 4\cdot 24^{2(\tau+1)}\hat\rho^{-2}$, so $\hat E_+\le 8\cdot 24^{2(\tau+1)}\hat E^2\hat\rho^2M^2/\gamma_1^2$ (using $E^2L^2/\hat\rho^2 = \hat E^2\hat\rho^2$). From $\hat\rho\le\gamma_1/(3MK^{\tau+1})$ and $K\ge 6/s$: $\hat\rho^2M^2/\gamma_1^2\le\frac19(s/6)^{2(\tau+1)}$. Hence $\hat E_+\le\frac89(24s/6)^{2(\tau+1)}\hat E^2 = \frac89(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$ since $4s\le\bar s<1$. Finally $\hat c\hat E_+ < \hat c\hat E\cdot\hat E < 1$. This is (vi). □

Remark 4.10 (Differences from CP10 in the step). CP10 (74)–(75), (86)–(87) have $5n$, 2^6n and 2^7 from a mixture of 1-, ∞- and "sup" norms; the above constants are those of max norms. CP10 injectivity argument (p. 569–570) uses "$\sup\|\partial(\omega^{-1})\|\sup|\omega(I)-\omega(I')|$", a mean value inequality that silently needs $[\omega(I),\omega(I')]\subset\omega(D_\rho)$; Lemma 4.6 supplies it. The Lipschitz bound of $\hat\imath_+ - \mathrm{id}$ on the (non-convex) set D is repaired by Lemma 4.7. CP10 (91) prints $(4s)^{\tau+1}$ where the computation gives $(4s)^{2(\tau+1)}$ (a sharpening: the printed bound is true, being weaker).

4.7. **The iterative lemma.**

Definition 4.11 (The sequences)**.** Given the data and constants of Proposition 4.3, put $D_0 = D^{\mathrm{dio}} = \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau})\cap D$, $\rho_0 = \rho$, $s_0 = s$, $M_0 = M$, $\hat M_0 = \hat M$, $\bar M_0 = \bar M$, $\bar M_{i,0} = \bar M_i$, $L_0 = L$, $E_0 = E$, $h_0 = h$, $f_0 = f$ and, recursively for $j\ge 1$: $s_j = s_{j-1}/12$, $M_j = 2M_{j-1}$, $\hat M_j = 2\hat M_{j-1}$, $\bar M_j = 2\bar M_{j-1}$, $\bar M_{i,j} = 2\bar M_{i,j-1}$, $L_j = 2L_{j-1}$, $\rho_j = \hat\rho_{j-1}/16$, $E_j = E_{j-1}^2L_{j-1}M_{j-1}^2/\gamma_1^2$, and $K_j,\hat\rho_j,\hat E_j$ by Definition 4.2 with the data $(s_j,\rho_j,E_j,M_j,\hat M_j,\bar M_j)$ (so L_j there is $\max\{\bar M_j,M_j^{-1},\hat M_j^{-1}\}\le 2^jL$, and in the formulas the larger $L_j = 2^jL$ is used, as allowed by (S2) of Lemma 4.9).

Lemma 4.12 (Iterative lemma)**.** *Under the hypotheses of Proposition 4.3 (in particular $\hat c\hat E_0<1$), for every $j\ge 0$ there are a set $D_j\subset\mathbb{R}^n$, an analytic symplectic map $\Phi_j = \Psi_1\circ\cdots\circ\Psi_j$ on $W_j := (D_j)_{\rho_j}\times\mathbb{T}_{\bar s+s_j}$ (with $\Phi_0 = \mathrm{id}$, D_0 as above) and $H_j = H\circ\Phi_j = h_j + f_j$, such that*

(i) $D_j\subset(D_{j-1})_{\hat\rho_{j-1}/16}$, $\omega_j = \partial h_j$ is a diffeomorphism of $(D_j)_{\rho_j}$ onto an open set, $\omega_j(D_j) = \omega_0(D_0)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$; the map $\hat\imath_j := \omega_j^{-1}\circ\omega_{j-1} : D_{j-1}\to D_j$ (defined and holomorphic on $(D_{j-1})_{\hat\rho_{j-1}/16}$) satisfies $\sup|\hat\imath_{j,i} - \mathrm{id}_i| \le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ and $L_{D_{j-1}}(\hat\imath_j - \mathrm{id})\le 640n\hat E_{j-1}$;

(ii) $\|f_j\|_{\rho_j,\bar s+s_j}\le E_j$;

(iii) $\Psi_j : W_j \to W'_{j-1}$ *with* $W'_{j-1} := (D_{j-1})_{\hat\rho_{j-1}} \times \mathbb{T}_{\bar s+s_{j-1}} \subset W_{j-1}$ *is holomorphic (so* $\Psi_j(W_j) \subset W_{j-1}$*), real on reals,* $\Psi_j - \mathrm{id}$ *periodic, and on* W_j

$$|\Delta I_1| \le \tfrac{3}{4}\tfrac{\hat M}{M}\hat E_{j-1}\hat\rho_{j-1}, \quad |\Delta I_2| \le \tfrac{3}{4}\hat E_{j-1}\hat\rho_{j-1}, \quad 2\pi|\Delta\theta| \le \tfrac{3}{4}\hat E_{j-1}s_{j-1};$$

(iv) $\hat E_j < \hat E_{j-1}^2$ *for* $j \ge 1$*, whence* $\hat E_j \le \hat E_0^{2^j}$*, and* $\hat c\hat E_j < 1$*;*
(v) $\hat\rho_j \le \hat\rho_{j-1}/16$ *and* $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$*,* $s_j = s/12^j$*,* $K_j \le 24^j K$*.*

Proof. **Step S4.20: Induction hypothesis.** We prove by induction on $j \ge 0$ the statement $\mathcal{P}_j$: the data $(D_j, h_j, f_j, \rho_j, s_j, M_j, \hat M_j, \bar M_j, \bar M_{i,j}, L_j, E_j)$ satisfy the hypotheses of Lemma 4.9 (on the domain $(D_j)_{\rho_j} \times \mathbb{T}_{\bar s+s_j}$): $\hat c\hat E_j < 1$, $L_j \ge \max\{\bar M_j, M_j^{-1}, \hat M_j^{-1}\}$, $\omega_j(D_j) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, bounds $M_j, \hat M_j, \bar M_j, \bar M_{i,j}$ on $(D_j)_{\rho_j}$, ω_j injective on $(D_j)_{\rho_j}$ with nonsingular U_j, $\|f_j\| \le E_j$, $\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$, and $4s_j \le \bar s < 1$.

Step S4.21: Base. $\mathcal{P}_0$ is the hypothesis of Proposition 4.3 and $\omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by definition of D_0 (note $D_0 \subset D$, so the sup bounds on D_ρ give those on $(D_0)_\rho$; injectivity is inherited).

Step S4.22: Induction. Apply Lemma 4.9 to $\mathcal{P}_{j-1}$: it produces $D_j := D_+$, $\Psi_j := \Psi_+$, $h_j := h_+$, $f_j := f_+$, and (i)–(vi) of that lemma are (i)–(iii) here and the recursion of Definition 4.11 (the quantities named $\hat\rho_+, K_+, \hat E_+$ there are $\hat\rho_j, K_j, \hat E_j$). Items (iii) and (v) of the step give $\mathcal{P}_j$ ($\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$ propagate by doubling).

Step S4.23: Composition. $\Phi_j = \Phi_{j-1} \circ \Psi_j$ is defined on W_j because $\Psi_j(W_j) \subset W'_{j-1} \subset W_{j-1}$: $W_j \subset D_{\hat\rho_{j-1}/8} \times \mathbb{T}_{\bar s+s_j}$ (more precisely $(D_j)_{\rho_j}$ with $D_j \subset (D_{j-1})_{\hat\rho_{j-1}/16}$), the displacement is $\le \frac{3}{4}\hat E_{j-1}\hat\rho_{j-1} < \hat\rho_{j-1}/16$, so the action of $\Psi_j(y)$ lies in $(D_{j-1})_{\hat\rho_{j-1}/8+\hat\rho_{j-1}/16} \subset (D_{j-1})_{\hat\rho_{j-1}}$, and $\hat\rho_{j-1} \le \rho_{j-1}$ (definition of $\hat\rho$). The identity $H_j = H_{j-1}\circ\Psi_j = H\circ\Phi_j$ is then immediate, and Φ_j is symplectic, real, with $\Phi_j - \mathrm{id}$ periodic.

Step S4.24: (iv)–(v) of the Lemma. By Step (vi) of Lemma 4.9: $\hat E_j < \hat E_{j-1}^2$, $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$, $\hat\rho_j \le \rho_j = \hat\rho_{j-1}/16$, $K_j \le 24K_{j-1}$. □

5. Two-scale KAM theorem II: the limit, the tori and the measure

We continue with the notation of Definition 4.11 and Lemma 4.12; $c := 24^{\tau+1}/6$ and $q := 2\cdot 24^{\tau+1} = 12c$. Since $\hat c \ge 2c$, $\hat c \ge 2^{13}n$, the KAM condition gives

$$\hat E_0 < \frac{1}{8192\,n}, \qquad c\hat E_0 < \frac{1}{2}. \tag{3.4}$$

Lemma 5.1 (Sums)**.** *For* $m \ge 0$*:* $\hat E_k \le \hat E_m^{2^{k-m}}$ *for* $k \ge m$*,* $\sum_{k\ge m}\hat E_k \le 2\hat E_m$*, and* $\sum_{k\ge 0} c^k\hat E_k \le 2\hat E_0$*. Also* $\hat\rho_k \le \hat\rho_0\,16^{-k}$*,* $s_k = s\,12^{-k}$*.*

Proof. **Step S5.1: Doubling.** By Lemma 4.12(iv), $\hat E_k < \hat E_{k-1}^2$, so $\hat E_k \le \hat E_m^{2^{k-m}} \le \hat E_m^{k-m+1}$ (as $2^i \ge i+1$ and $\hat E_m < 1$). Hence $\sum_{k\ge m}\hat E_k \le \hat E_m/(1-\hat E_m) \le 2\hat E_m$.

Step S5.2: Weighted sum. $c^k\hat E_k \le c^k\hat E_0^{k+1} = \hat E_0(c\hat E_0)^k \le \hat E_0 2^{-k}$ by (3.4); sum $\le 2\hat E_0$.

Step S5.3: Radii. $\hat\rho_k \le \rho_k = \hat\rho_{k-1}/16$ (Definition of $\hat\rho_k$), iterate. □

5.1. **The limit actions (CP10 Step 1).**

Lemma 5.2 (Limit actions). *Put $\check\imath_0 = \mathrm{id}$, $\check\imath_j := \hat\imath_j \circ \cdots \circ \hat\imath_1 : D_0 \to D_j$. Then $\check\imath_j$ converges uniformly on D_0 to $\check\imath = (\check\imath_1, \check\imath_2)$ and*

(i) $\sup_{D_0}|\check\imath_i - \mathrm{id}_i| \le 20\frac{\bar M_i}{\bar M}\hat E_0\hat\rho_0 \le 20\hat E_0\hat\rho_0$; $\sup_{D_0}|\check\imath - \check\imath_j| \le 20\hat\rho_j\hat E_j$;
(ii) $D_* := \check\imath(D_0) \subset (D_j)_{\rho_j/2}$ *for all j, in particular* $D_* \subset \mathrm{Re}(D_0)_{20\hat E_0\hat\rho_0}$;
(iii) $\lambda := L_{D_0}(\check\imath - \mathrm{id}) \le 2560\, n\hat E_0 < \frac13$; *so $\check\imath$ is injective,* $(1-\lambda)|x-y| \le |\check\imath x - \check\imath y| \le (1+\lambda)|x-y|$;
(iv) $\omega_j(\check\imath_j(I_0)) = \omega_0(I_0)$ *for all* $I_0 \in D_0$, $j \ge 0$.

Source: CP10 Step 1, pp. 570–571, (97)–(104); constants recomputed (CP10: $10n$, $2^8 n\hat E$).

Proof. **Step S5.4: Well-posedness.** $\hat\imath_j : D_{j-1} \to D_j$ (Lemma 4.12(i)); so $\check\imath_j(D_0) \subset D_j$ and the composition is defined. (iv) follows from $\omega_j \circ \hat\imath_j = \omega_{j-1}$ by induction.

Step S5.5: Telescoping. $|\check\imath_{j,i}(x) - \check\imath_{j-1,i}(x)| = |\hat\imath_{j,i}(y) - y_i|$ with $y = \check\imath_{j-1}(x) \in D_{j-1}$, so $\le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ (the ratio $\bar M_{i,j}/\bar M_j = \bar M_i/\bar M$ is constant). Since $\hat\rho_{j-1} \le \hat\rho_0$, Lemma 5.1 gives the Cauchy property and (i): $\sum_{j\ge1} 10\hat E_{j-1}\hat\rho_{j-1} \le 20\hat E_0\hat\rho_0$; the tail is $\le 10\hat\rho_j\sum_{m\ge j}\hat E_m \le 20\hat\rho_j\hat E_j$.

Step S5.6: Location. $\check\imath_j(D_0) \subset D_j \subset \mathbb{R}^n$ and $20\hat\rho_j\hat E_j \le \rho_j/2$ ($\hat\rho_j \le \rho_j$, $\hat E_j < 1/40$), so $D_* \subset \mathrm{Re}(D_j)_{\rho_j/2}$; for $j = 0$ and $D_0 \subset D$ this gives the last claim of (ii).

Step S5.7: Lipschitz constant. Put $i_j = \check\imath_j - \mathrm{id}$. From $\check\imath_{j+1} = \hat\imath_{j+1}\circ\check\imath_j$: $i_{j+1} = (\hat\imath_{j+1} - \mathrm{id})\circ\check\imath_j + i_j$, so $1 + L(i_{j+1}) \le (1 + L(\hat\imath_{j+1} - \mathrm{id}))(1 + L(i_j))$ and, by Lemma 4.12(i), $1 + L(i_j) \le \prod_{k\le j}(1 + 640n\hat E_{k-1}) \le \exp(640n\sum_k \hat E_{k-1}) \le \exp(1280n\hat E_0)$. As $x := 1280n\hat E_0 < \frac16$, $e^x \le 1 + 2x$: $L(i_j) \le 2560n\hat E_0$ uniformly in j; the pointwise limit of Lipschitz maps keeps the constant. Injectivity and the two-sided bound follow from $|\check\imath x - \check\imath y| \ge |x-y| - \lambda|x-y|$. □

5.2. **The limit symplectic map (CP10 Step 2).** *Rescaling.* Let $S(I,\theta) := (I/\hat\rho_0,\ 2\pi\theta/s_0)$ and $|(I,\theta)|_* := |S(I,\theta)|$ (max norm); for a map G put $\check G = SGS^{-1}$ and L_* for Lipschitz constants in $|\cdot|_*$. Put $\mathcal{D}_j := \mathrm{Re}(D_j)_{\rho_j/2}$. All maps below are real on real points.

Lemma 5.3 (Domains and rescaled Lipschitz bounds for Ψ_k). *(i)* $\Psi_k(\mathcal{D}_k \times \mathbb{R}^n) \subset \mathcal{D}_{k-1} \times \mathbb{R}^n$ *($k \ge 1$); hence $\Phi_j = \Psi_1 \circ \cdots \circ \Psi_j$ is defined on $\mathcal{D}_j \times \mathbb{R}^n$ with values in $\mathcal{D}_0 \times \mathbb{R}^n$.*
(ii) $L_*(\check\Psi_k - \mathrm{id}; \mathcal{D}_k \times \mathbb{R}^n) \le 192\, n\, c^{k-1}\hat E_{k-1} =: \varsigma_k$, *and* $\sum_{k\ge1}\varsigma_k \le 384n\hat E_0 < \frac{1}{20}$.
(iii) $|\Psi_k(y) - y|_* \le \frac34\hat E_{k-1}12^{-(k-1)} \le \frac34\hat E_{k-1}$ *for $y \in \mathcal{D}_k \times \mathbb{R}^n$.*

Proof. **Step S5.8: Domains.** Let $y = (I,\theta)$, $I \in \mathcal{D}_k$: $\mathrm{dist}(I, D_k) < \rho_k/2 = \hat\rho_{k-1}/32$ and $D_k \subset \mathrm{Re}(D_{k-1})_{\hat\rho_{k-1}/16}$. By Lemma 4.12(iii) the real displacement is $\le \frac34\hat E_{k-1}\hat\rho_{k-1}$. Hence $\Psi_k(y)_I$ is within $\hat\rho_{k-1}(\frac{1}{32} + \frac{1}{16} + \frac34\hat E_{k-1}) < \hat\rho_{k-1}/8$ of D_{k-1}; and $\hat\rho_{k-1}/8 < \rho_{k-1}/2$ (for $k = 1$ because $\hat\rho_0 \le \rho$; for $k \ge 2$ because $\hat\rho_{k-1} \le \hat\rho_{k-2}/16$).

Step S5.9: Bound on displacement. Scaled: $|\Delta I|/\hat\rho_0 \le \frac34\hat E_{k-1}\hat\rho_{k-1}/\hat\rho_0 \le \frac34\hat E_{k-1}16^{-(k-1)}$ (Lemma 5.1; $\hat M \le M$) and $2\pi|\Delta\theta|/s_0 \le \frac34\hat E_{k-1}s_{k-1}/s_0 = \frac34\hat E_{k-1}12^{-(k-1)}$. Thus $A_k := \sup_{W_k}|\check\Psi_k - \mathrm{id}| \le \frac34\hat E_{k-1}12^{-(k-1)}$; this is (iii).

Step S5.10: Lipschitz. Apply Lemma 4.7 in the scaled coordinates, $N = 2n$, $S = S(\mathcal{D}_k \times \mathbb{R}^n)$, radius $\varepsilon_k = \min\{\rho_k/(2\hat\rho_0),\ s_k/s_0\}$: the scaled complex ε_k-neighbourhood of a real point of S lies in $(D_k)_{\rho_k} \times \mathbb{T}_{s_k} \subset W_k$ ($\varepsilon_k\hat\rho_0 \le \rho_k/2$ and $\varepsilon_k s_0 \le s_k \le \bar s + s_k$), where the bound A_k of (iii) holds (Lemma 4.12(iii) holds on the complex W_k). Now $\rho_k/(2\hat\rho_0) = \hat\rho_{k-1}/(32\hat\rho_0) \ge q^{-(k-1)}/32$ (Lemma 4.12(v)) and $s_k/s_0 = 12^{-k} \ge q^{-(k-1)}/32$, so $\varepsilon_k \ge q^{-(k-1)}/32$ and

$$L_* \le \frac{4\cdot 2n\, A_k}{\varepsilon_k} \le 8n\cdot\tfrac34\hat E_{k-1}12^{-(k-1)}\cdot 32q^{k-1} = 192n\,(q/12)^{k-1}\hat E_{k-1} = 192n\, c^{k-1}\hat E_{k-1}.$$

Step S5.11: Sum. Lemma 5.1: $\sum_k \varsigma_k = 192n\sum_{m\ge 0} c^m \hat{E}_m \le 384n\hat{E}_0 < 384/8192 < \frac{1}{20}$. □

Lemma 5.4 (Compactness of periodic holomorphic functions on a strip)**.** *Let $F_j : \{\vartheta \in \mathbb{C}^n : |\operatorname{Im}\vartheta| < w\} \to \mathbb{C}$ be holomorphic, $\mathbb{Z}^n$-periodic, $|F_j| \le B$, and suppose F_j converges uniformly on $\mathbb{R}^n$ to F. Then F_j converges uniformly, together with all derivatives, on $\{|\operatorname{Im}\vartheta| \le w/2\}$ to a holomorphic periodic function extending F, with $|F| \le B$ on the strip.*

Proof. **Step S5.12: Coefficient decay.** For $0 < w' < w$ and $k \ne 0$ shift the integration contour to $\operatorname{Im}\vartheta = -w'\operatorname{sign}(k)$ (componentwise; Cauchy's theorem for the periodic integrand, [K:bp:p1:ext-rectangle]): $|\hat{F}_{j,k}| \le Be^{-2\pi w'|k|_1}$, uniformly in j.

Step S5.13: Convergence of coefficients. Uniform convergence on $\mathbb{R}^n$ gives $\hat{F}_{j,k} \to \hat{F}_k$ for each k ((X2)), and $|\hat{F}_k| \le Be^{-2\pi w'|k|_1}$.

Step S5.14: Tail and finite part. Take $w' = 3w/4$. On $|\operatorname{Im}\vartheta| \le w/2$ the modes satisfy $|e^{2\pi ik\cdot\vartheta}| \le e^{\pi w|k|_1}$, so $\sum_{|k|_1>N}|\hat{F}_{j,k}e^{2\pi ik\vartheta}| \le B\sum_{|k|_1>N} e^{-\pi w|k|_1/2} =: \varepsilon_N \to 0$ uniformly in j (same for F); the finitely many modes $|k|_1 \le N$ converge uniformly. Hence $F_j \to F$ uniformly on the closed half strip; derivatives: multiply the terms by $(2\pi k_i)^m$, which keeps the tail summable. Holomorphy of the limit: (X3). □

Lemma 5.5 (The limit map Φ)**.** *On $W_* := D_* \times \mathbb{R}^n$ the maps Φ_j converge uniformly to a map Φ with $\Phi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ, real analytic in θ, and*

- *(i)* $\sup_{W_*}|\Pi_{I_1}\Phi - \mathrm{id}| \le \frac{3}{2}\frac{\hat{M}}{M}\hat{E}_0\hat{\rho}_0$, $\sup|\Pi_{I_2}\Phi - \mathrm{id}| \le \frac{3}{2}\hat{E}_0\hat{\rho}_0$, $\sup 2\pi|\Pi_\theta\Phi - \mathrm{id}| \le \frac{3}{2}\hat{E}_0 s_0$, *and the same bounds on complex θ with $|\operatorname{Im}\theta| < \bar{s}/2\pi$;*
- *(ii)* $L_*(\check{\Phi} - \mathrm{id}; W_*) \le 768\, n\hat{E}_0 < \frac{1}{10}$; *hence Φ is injective on W_* and $|x-y|_*\cdot\frac{9}{10} \le |\Phi x - \Phi y|_* \le \frac{11}{10}|x-y|_*$;*
- *(iii)* $\Phi(I,\theta+k) = \Phi(I,\theta) + (0,k)$ *for $k \in \mathbb{Z}^n$.*

Source: CP10 Step 2, pp. 571–572, (105)–(108).

Proof. **Step S5.15: Telescoping of the displacements.** Write $(I^{(k)},\theta^{(k)}) = \Psi_{k+1}\circ\cdots\circ\Psi_j(x)$ for $x \in W_* \subset W_j$ (Lemma 5.2(ii)); then $\Pi_I\Phi_j(x) - \Pi_I x = \sum_{k=1}^{j}(I^{(k-1)} - I^{(k)})$ and every summand is bounded by Lemma 4.12(iii). Summing with Lemma 5.1 ($\hat{\rho}_{k-1} \le \hat{\rho}_0$, $s_{k-1} \le s_0$, $\sum \hat{E}_{k-1} \le 2\hat{E}_0$) gives (i), uniformly in j and also for complex θ (Ψ_k is defined on the complex strip).

Step S5.16: Cauchy sequence. For $x \in W_*$: $\Phi_j(x) - \Phi_{j-1}(x) = \Phi_{j-1}(\Psi_j x) - \Phi_{j-1}(x)$, with $x \in \mathcal{D}_j \times \mathbb{R}^n$ and $\Psi_j x \in \mathcal{D}_{j-1}\times\mathbb{R}^n$ (Lemma 5.3(i)). By the composition rule $L_*(\check{\Phi}_{j-1} - \mathrm{id}) \le \prod_{k<j}(1+\varsigma_k) - 1 \le e^{\sum\varsigma_k} - 1 \le 2\sum\varsigma_k \le \frac{1}{10}$ (using $e^x \le 1+2x$, $x \le 1$; the composition formula is that of Lemma 5.2, proof step S5.7, valid because each Ψ_k maps its domain into the next one). Therefore $|\Phi_j(x) - \Phi_{j-1}(x)|_* \le \frac{11}{10}|\Psi_j x - x|_* \le \frac{11}{10}\cdot\frac{3}{4}\hat{E}_{j-1}$, which is summable: uniform convergence on W_*. The limit satisfies (ii) with $L_* \le 2\sum\varsigma_k \le 768n\hat{E}_0$ (the constant passes to the pointwise limit).

Step S5.17: Periodicity and injectivity. Each $\Psi_k - \mathrm{id}$ is $\mathbb{Z}^n$-periodic in θ (Lemma 4.12), so (iii) holds for Φ_j and in the limit. From (ii), $|\Phi x - \Phi y|_* \ge \frac{9}{10}|x-y|_*$.

Step S5.18: Analyticity in θ. Fix $I_* \in D_*$; $\vartheta \mapsto \Phi_j(I_*,\vartheta) - (0,\vartheta)$ is holomorphic on $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$ (as $I_* \in (D_j)_{\rho_j}$), $\mathbb{Z}^n$-periodic, bounded uniformly in j by (i), and converges uniformly on $\mathbb{R}^n$. By Lemma 5.4 (componentwise, $w = \bar{s}/2\pi$) it converges, with all derivatives, uniformly on $\{|\operatorname{Im}\vartheta| \le \bar{s}/4\pi\}$ to a holomorphic function whose real restriction is $\Phi(I_*,\cdot) - (0,\mathrm{id})$; the bounds (i) pass to the limit. (The statement (a) of Proposition 4.3 on the full strip $|\operatorname{Im}\vartheta| < \bar{s}/2\pi$ follows by applying the lemma with w arbitrarily close to $\bar{s}/2\pi$ on exhausting substrips; the Cauchy bound for $D_\vartheta u$ below only uses $\bar{s}/4\pi$.) □

5.3. **The tori (CP10 Steps 2–3 and the estimates (35)).** Define, for $\omega \in \Omega_* = \omega_0(D)\cap\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, $I_0(\omega) := \omega_0^{-1}(\omega) \in D_0$ (injectivity of ω_0; $\omega \in \omega_0(D)\cap\mathcal{D}$ means $I_0 \in D^{\mathrm{dio}} = D_0$), $I_*(\omega) := \check{\imath}(I_0(\omega)) \in D_*$ and

$$\varphi_\omega(\vartheta) := \Phi(I_*(\omega), \vartheta) = (v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)).$$

Lemma 5.6 (Estimates and location)**.** *For $\omega \in \Omega_*$: $|u| \le \frac{3}{4\pi}\hat{E}_0 s$ on $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$; $|v_1 - I_{0,1}| \le 20(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{\bar{M}})\hat{E}_0\hat{\rho}_0$, $|v_2 - I_{0,2}| \le 20(\frac{\bar{M}_2}{\bar{M}} + 1)\hat{E}_0\hat{\rho}_0$ on $\mathbb{R}^n$, and $v \in \operatorname{Re}(D_r)$ with $r = 32\hat{E}_0\hat{\rho}_0$, so $\varphi_\omega(\mathbb{R}^n) \subset \operatorname{Re}(D_r)\times\mathbb{R}^n$. This is Proposition 4.3(a), (d), and the location claim.*
Source: CP10 (35), (105)–(107), Step 2 end, p. 572. Misprint: *CP10 (35) bounds $|v_2 - I_2^0|$ by $10n(\bar{M}_2/\bar{M} + \hat{M}/M)\hat{E}\hat{\rho}$ although (107) only gives $2\hat{E}\hat{\rho}$ for I_2 without the factor $\hat{M}/M$.*

Proof. **Step S5.19: Sup bounds.** $|v_i - I_{0,i}| \le |\Pi_{I_i}\Phi(I_*,\vartheta) - I_{*,i}| + |I_{*,i} - I_{0,i}|$, the two terms bounded by Lemma 5.5(i) and Lemma 5.2(i): $\frac{3}{2}\frac{\hat{M}}{\bar{M}}\hat{E}\hat{\rho} + 20\frac{\bar{M}_1}{\bar{M}}\hat{E}\hat{\rho}$ for $i=1$ (and $\frac{3}{2} + 20\frac{\bar{M}_2}{\bar{M}}$ for $i=2$), both $\le$ the stated constants. $|u| \le \frac{3}{2}\hat{E}s/2\pi$ is the angle bound.

Step S5.20: Location. $I_* \in \operatorname{Re}(D_0)_{20\hat{E}\hat{\rho}}$ and $|v - I_*| \le \frac{3}{2}\hat{E}\hat{\rho}$, so $\operatorname{dist}(v, D) < 22\hat{E}\hat{\rho} < r$. □

Lemma 5.7 (Immersion and Lagrangian property)**.** *For $\omega \in \Omega_*$, $\|D_\vartheta u\| \le \frac{3}{4}n\hat{E}_0$ (so $D_\vartheta\varphi_\omega = \binom{D_\vartheta v}{1+D_\vartheta u}$ has injective lower block $1 + D_\vartheta u$, since $\frac{3}{4}n\hat{E}_0 < \frac{1}{2}$), and $\varphi_\omega^*(\sum_i dJ_i \wedge d\theta_i) = 0$.*

Proof. **Step S5.21: Cauchy in the strip.** u is holomorphic on $|\operatorname{Im}\vartheta| < \bar{s}/2\pi$ with $|u| \le 3\hat{E}s/4\pi$. At real ϑ the polydisc of radius $\bar{s}/4\pi$ lies in the strip, so $|\partial_a u_i| \le \frac{3\hat{E}s}{4\pi}\cdot\frac{4\pi}{\bar{s}} = 3\hat{E}\frac{s}{\bar{s}} \le \frac{3}{4}\hat{E}$ ($4s \le \bar{s}$), and the row sum is $\le \frac{3}{4}n\hat{E}$.

Step S5.22: Each Φ_j-torus is Lagrangian. Φ_j is symplectic and $\vartheta \mapsto (I_*, \vartheta)$ has pullback $\sum dI_i\wedge d\theta_i = 0$ (the I-coordinates are constant), so $\varphi^{(j)} := \Phi_j(I_*,\cdot)$ has $\Omega_j(a,b) := \sum_i(\partial_a v_i^{(j)}\partial_b\Theta_i^{(j)} - \partial_b v_i^{(j)}\partial_a\Theta_i^{(j)}) = 0$.

Step S5.23: Limit. By Lemma 5.5 (Step S5.18) the first derivatives of $\varphi^{(j)}$ converge to those of φ_ω (uniformly on $\mathbb{R}^n$, Lemma 5.4); Ω_j is bilinear in them, so $\Omega = \lim\Omega_j = 0$. □

Lemma 5.8 (Invariance)**.** *For $\omega \in \Omega_*$, $\vartheta \in \mathbb{R}^n$, $t \in \mathbb{R}$: $\frac{d}{dt}\varphi_\omega(\vartheta + t\omega) = X_H(\varphi_\omega(\vartheta + t\omega))$, $X_H = (-\partial_\theta H, \partial_J H)$.*
Source: CP10 Step 3, pp. 572–573.

Proof. Let $I_0 = I_0(\omega)$, $I_* = \check{\imath}(I_0)$, $\theta_t = \vartheta + t\omega$, $\gamma(t) = \Phi(I_*,\theta_t)$, $\gamma_j(t) = \Phi_j(I_*,\theta_t)$.

Step S5.24: Symplectic conjugation. Since Φ_j is a symplectic diffeomorphism onto its image and $H_j = H\circ\Phi_j$: $X_H\circ\Phi_j = D\Phi_j\, X_{H_j}$ ([N:bpN:p5:symplectic]). With $X_{H_j}(I_*,\theta) = (-\partial_\theta f_j,\ \omega_j(I_*) + \partial_I f_j)$ we get $\gamma_j'(t) - X_H(\gamma_j(t)) = D\Phi_j(I_*,\theta_t)\big(\partial_\theta f_j,\ \omega - \omega_j(I_*) - \partial_I f_j\big)$ (with $\gamma_j' = D\Phi_j(0,\omega)$).

Step S5.25: $D\Phi_j$ bounded. $\mathcal{D}_j\times\mathbb{R}^n$ is open and $L_*(\check{\Phi}_j) \le \frac{11}{10}$ (Lemma 5.5(ii) argument), so $|S\,D\Phi_j S^{-1}| \le \frac{11}{10}$: $\|D\Phi_j\| \le \kappa := \frac{11}{10}\max\{1, \frac{2\pi\hat{\rho}_0}{s_0}, \frac{s_0}{2\pi\hat{\rho}_0}\}$, independent of j.

Step S5.26: The three error terms. (1) $\omega = \omega_j(\check{\imath}_j(I_0))$ (Lemma 5.2(iv)); $|I_* - \check{\imath}_j(I_0)| \le 20\hat{\rho}_j\hat{E}_j$ and the segment lies in $(D_j)_{\rho_j}$ where $\|U_j\| \le M_j$, so $|\omega - \omega_j(I_*)| \le 20M_j\hat{\rho}_j\hat{E}_j \le 7\gamma_1\hat{E}_j$, using $M_j\hat{\rho}_j \le \gamma_1/(3K_j^{\tau+1}) \le \gamma_1/3$. (2) $I_* \in \operatorname{Re}(D_j)_{\rho_j/2}$, so the polydisc of radius $\rho_j/2$ about I_* is in $(D_j)_{\rho_j}$ and [K:bp:p1:cauchy-open] gives $|\partial_I f_j| \le 2E_j/\rho_j \le 2E_j/\hat{\rho}_j = 2\hat{E}_j\hat{\rho}_j/L_j \le 2\hat{E}_j M_j\hat{\rho}_j \le \gamma_1\hat{E}_j$ ($L_j \ge M_j^{-1}$). (3) For real θ the polydisc of radius $\bar{s}/4\pi$ in θ lies in $\mathbb{T}_{\bar{s}/2}$, where $|f_j| \le E_j$ (as $\bar{s}/2 \le \bar{s} + s_j$), so $|\partial_\theta f_j| \le 4\pi E_j/\bar{s}$ ([K:bp:p1:cauchy-open]); and $E_j = \hat{E}_j\hat{\rho}_j^2/L_j \le \hat{E}_j\hat{\rho}_j M_j\hat{\rho}_j \le \hat{E}_j\gamma_1\hat{\rho}_0$. All three tend to 0, uniformly in (ϑ,t), as $\hat{E}_j \le \hat{E}_0^{2^j}$. Hence $\sup_t|\gamma_j' - X_H(\gamma_j)| \to 0$.

Step S5.27: Passing to the limit. $\delta_j := \sup_t|\gamma_j(t)-\gamma(t)| \to 0$ (Lemma 5.5; unscaled distances are comparable to scaled ones with fixed constants). Only nearby pairs are needed (review P3-08). The action of $\gamma(t)$ lies in $\mathrm{Re}(D_r) \subset \mathrm{Re}(D_{\rho/4})$ ($r = 32\hat{E}\hat{\rho} \le \rho/4$ since $\hat{E} < 2^{-13}$, $\hat{\rho} \le \rho$) and its angle is real; for j so large that $\delta_j < \min\{\rho/4, \bar{s}/4\pi\}$, the real segment $[\gamma_j(t), \gamma(t)]$ lies in the convex polydisc of radius δ_j about $\gamma(t)$, hence in $D_{\rho/2} \times \mathbb{T}_{\bar{s}/2}$. There $\|DX_H\| \le B_0$, where DX_H consists of $U + \partial_J^2 f$, $\partial_\theta \partial_J f$, $\partial_\theta^2 f$ and $B_0 = M + C(E, \rho, \bar{s})$ by [K:bp:p1:cauchy-open] (applied to $\sup|f| \le E$ on $D_\rho \times \mathbb{T}_{\bar{s}}$). By the mean value inequality, $|X_H(\gamma_j(t)) - X_H(\gamma(t))| \le B_0\delta_j \to 0$ uniformly in t. A uniform limit of C^1 functions $\gamma_j \to \gamma$ with $\gamma_j' \to X_H(\gamma)$ uniformly is C^1 with $\gamma' = X_H(\gamma)$. Since $\gamma(t) = \varphi_\omega(\vartheta + t\omega)$, this is the claim. (Conjugation identity of Step S5.24: [N:bpN:p5:symplectic], [N:bpN:p5:canonical].) □

5.4. **Measure estimate (CP10 Step 4).** Let $K_\Omega = \bigcup_{\omega\in\Omega_*} \varphi_\omega(\mathbb{R}^n)$; by definition $K_\Omega = \Phi(D_* \times \mathbb{R}^n)$ ($\check{\imath}$ maps $D_0 = D^{\mathrm{dio}}$ onto D_* and $\omega \mapsto I_0(\omega)$ is a bijection $\Omega_* \to D_0$). Put $a := \mathrm{meas}(D \setminus D_0)$, $b := \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\lambda \le \frac{1}{3}$ as in Lemma 5.2, $\eta := \sup|\check{\imath} - \mathrm{id}| \le 20\hat{E}_0\hat{\rho}_0$.

Lemma 5.9 (Bi-Lipschitz extension and covering)**.** *Let $X \subset \mathbb{R}^N$ be arbitrary, $g : X \to \mathbb{R}^N$ with $L(g) \le \lambda < 1$ in the max norm and $\sup|g| \le \eta$. Then there is a bijection $F = \mathrm{id} + g_e : \mathbb{R}^N \to \mathbb{R}^N$, $F|_X = \mathrm{id} + g$, with $(1-\lambda)|x-y| \le |F(x)-F(y)| \le (1+\lambda)|x-y|$, and for $y \in X$: $F^{-1}(y) \in N_{\eta/(1-\lambda)}(X)$ (real max-norm neighbourhood).*

Proof. **Step S5.28: Extension.** g_e: McShane [K:bp:p1:mcshane], componentwise, with the same λ for the max norm.

Step S5.29: Bijectivity. $|F(x)-F(y)| \ge |x-y| - \lambda|x-y|$ gives injectivity and the lower bound; for $y \in \mathbb{R}^N$ the map $x \mapsto y - g_e(x)$ is a λ-contraction ((X4)), with fixed point $x = F^{-1}(y)$.

Step S5.30: Location of preimages. For $y \in X$ and $x = F^{-1}(y)$: $|x-y| = |g_e(x)| \le |g_e(y)| + \lambda|x-y| \le \eta + \lambda|x-y|$ so $|x-y| \le \eta/(1-\lambda)$. □

Lemma 5.10 (Measure of the complement of D_*)**.** $\mathrm{meas}(D_0 \setminus D_*) \le (4/3)^n(a+b)$ *and hence* $\mathrm{meas}(D \setminus D_*) \le a + (4/3)^n(a+b)$.

Proof. **Step S5.31: Extension and preimage.** Apply Lemma 5.9 with $X = D_0$, $g = \check{\imath} - \mathrm{id}$, $N = n$: $F = \check{\imath}_e$. Let $\rho_1 := \eta/(1-\lambda) \le 30\hat{E}_0\hat{\rho}_0 < r$. If $y \in D_0 \setminus D_*$, then $x = F^{-1}(y) \in N_{\rho_1}(D_0)$ and $x \notin D_0$ (else $y = F(x) = \check{\imath}(x) \in D_*$). So $D_0 \setminus D_* \subset F\big(N_{\rho_1}(D_0) \setminus D_0\big)$.

Step S5.32: Measure. By [K:bp:p1:lip-image] (F is $(1+\lambda)$-Lipschitz), $\mathrm{meas}(D_0 \setminus D_*) \le (1+\lambda)^n \,\mathrm{meas}(N_{\rho_1}(D_0) \setminus D_0)$, $(1+\lambda) \le \frac{4}{3}$. As $D_0 \subset D$: $N_{\rho_1}(D_0) \setminus D_0 \subset (N_{\rho_1}(D) \setminus D) \cup (D \setminus D_0)$, of measure $\le b + a$ ($\rho_1 < r$). The second claim: $D \setminus D_* \subset (D \setminus D_0) \cup (D_0 \setminus D_*)$. □

Lemma 5.11 (Measure of the complement of K_Ω over D)**.** $\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le \big[1 + (4/3)^n + 2^n\big(1 + (4/3)^n\big)\big](a+b)$.

Proof. Identify $\mathbb{T}^n$-measure with the unit cell; K_Ω is $\mathbb{Z}^n$-invariant in the angle.

Step S5.33: Splitting. $(D \times \mathbb{T}^n) \setminus K_\Omega \subset \big((D \setminus D_*) \times \mathbb{T}^n\big) \cup \big(((D_* \cap D) \times \mathbb{T}^n) \setminus K_\Omega\big)$. The first set has measure $\le a + (4/3)^n(a+b)$ by Lemma 5.10.

Step S5.34: Rescaled extension. Work in scaled coordinates $x \mapsto Sx$; S is linear, multiplies the Lebesgue measure of every subset of $\mathbb{R}^n_I \times \mathbb{R}^n_\theta$ by the constant $\hat{\rho}_0^{-n}(2\pi/s_0)^n$ and maps products to products, so all comparisons below between measures of subsets of $\mathbb{R}^n \times \mathbb{R}^n$ are unaffected. By Lemma 5.5, $g = \check{\Phi} - \mathrm{id}$ on $X = S(D_* \times \mathbb{R}^n) \subset \mathbb{R}^{2n}$ has $L_*(g) \le \lambda' := 768n\hat{E}_0 < \frac{1}{10}$ and $\sup|g| \le \eta' := \frac{3}{2}\hat{E}_0$ (Lemma 5.5(i), scaled). Lemma 5.9 ($N = 2n$) gives the bijection $\check{\Phi}_e = \mathrm{id} + g_e$ of $\mathbb{R}^{2n}$ extending $\check{\Phi}$ on X, $(1+\lambda')$-Lipschitz, with $|\check{\Phi}_e^{-1}(y) - y|_* \le \eta'/(1-\lambda') \le 2\hat{E}_0$ for $y \in X$.

Step S5.35: Preimage structure. Let $y = (I_y, \vartheta_y)$ with $I_y \in D_* \cap D$ and $\vartheta_y \in [0,1)^n$ be a point of the second set, i.e. $y \notin K_\Omega$ (mod $\mathbb{Z}^n$ in the angle), and let $x = (I_x, \vartheta_x) := \Phi_e^{-1}(y)$ (unscaled notation; $y \in D_* \times \mathbb{R}^n = S^{-1}X$). Then $|I_x - I_y| \le \rho_2 := 2\hat{E}_0\hat{\rho}_0$ and $|\vartheta_x - \vartheta_y| \le \hat{E}_0 s_0/\pi \le \hat{E}_0/(4\pi) =: \epsilon$ (as $s_0 \le \frac{1}{4}$). If $I_x \in D_*$, then $x \in D_* \times \mathbb{R}^n$, $\Phi_e(x) = \Phi(x) = y$ and $y \in K_\Omega$, a contradiction. Hence $I_x \in N_{\rho_2}(D_*) \setminus D_*$ and $\vartheta_x \in [-\epsilon, 1+\epsilon)^n$.

Step S5.36: Measure. Thus the second set is contained in $\Phi_e\big((N_{\rho_2}(D_*) \setminus D_*) \times [-\epsilon, 1+\epsilon)^n\big)$. Φ_e is $(1+\lambda')$-Lipschitz for $|\cdot|_*$ on $\mathbb{R}^{2n}$, so by [K:bp:p1:lip-image] in scaled coordinates (the scaling constants cancel on both sides) its measure is $\le (1+\lambda')^{2n}(1+2\epsilon)^n \operatorname{meas}(N_{\rho_2}(D_*) \setminus D_*) \le 2^n \operatorname{meas}(N_{\rho_2}(D_*) \setminus D_*)$, because $(1.1)^{2n}(1+\hat{E}_0/2\pi)^n \le (1.21 \cdot 1.01)^n < 2^n$.

Step S5.37: The shell. $D_* \subset N_\eta(D_0)$, so $N_{\rho_2}(D_*) \subset N_{\rho_2+\eta}(D_0) \subset N_r(D)$ ($\rho_2 + \eta \le 22\hat{E}_0\hat{\rho}_0 < r$), and $N_{\rho_2}(D_*) \setminus D_* \subset (N_r(D) \setminus D) \cup (D_0 \setminus D_*) \cup (D \setminus D_0)$: if $x \in N_{\rho_2}(D_*) \setminus D_*$ then either $x \notin D$ (first set), or $x \in D \setminus D_0$, or $x \in D_0 \setminus D_*$. Its measure is $\le b + a + (4/3)^n(a+b) = (1+(4/3)^n)(a+b)$ by Lemma 5.10.

Step S5.38: Sum. Total $\le [a + (4/3)^n(a+b)] + 2^n(1+(4/3)^n)(a+b) \le [1 + (4/3)^n + 2^n(1 + (4/3)^n)](a+b)$. □

5.5. Proof of Proposition 4.3.

Proof of Proposition 4.3. **Step S5.39: Construction.** The sequence of Lemma 4.12 exists (hypotheses (S1)–(S3) at $j = 0$ from the proposition's hypotheses and the definition of D_0); Lemma 5.2 defines $\check{\imath}$, Lemma 5.5 defines Φ, and φ_ω is defined in §5.3.

Step S5.40: (a), (d). Lemmas 5.5, 5.6.

Step S5.41: (b), (c). Lemmas 5.8, 5.7.

Step S5.42: Injectivity and Lipschitz structure. If $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') = (0, k)$, then $\Phi(I_*, \vartheta) = \Phi(I_*', \vartheta' + k)$ by periodicity (Lemma 5.5(iii)), so $(I_*, \vartheta) = (I_*', \vartheta' + k)$ by (ii); $I_* = I_*'$ gives $I_0 = I_0'$ ($\check{\imath}$ injective), so $\omega = \omega_0(I_0) = \omega'$. The factorization and the constants: Lemma 5.2(iii) and Lemma 5.5(ii). For ω_0^{-1}: if $\omega, \omega' \in \omega_0(D)$ with $|\omega - \omega'| < \rho/\bar{M}$, then Lemma 4.6 (with $F = \omega_0$ on D_ρ, $\delta = \rho$, $x_0 = \omega_0^{-1}(\omega)$) gives $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \bar{M}|\omega - \omega'|$; for far pairs $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \operatorname{diam} D \le \operatorname{diam} D \cdot \frac{\bar{M}}{\rho}|\omega - \omega'|$.

Step S5.43: Measure. $K_\Omega \subset \operatorname{Re}(D_r) \times \mathbb{T}^n$ (Lemma 5.6), so $(\operatorname{Re}(D_r) \times \mathbb{T}^n) \setminus K_\Omega \subset ((\operatorname{Re}(D_r) \setminus D) \times \mathbb{T}^n) \cup ((D \times \mathbb{T}^n) \setminus K_\Omega)$ (an inclusion; the shell may meet K_Ω, which only helps), and Lemma 5.11 gives (3.1) with $c_n = 1 + [1 + (4/3)^n + 2^n(1+(4/3)^n)] = 1 + (1+2^n)(1+(4/3)^n)$ ($a = \operatorname{meas}(D \setminus D^{\mathrm{dio}})$, $b = \operatorname{meas}(\operatorname{Re}(D_r) \setminus D)$, $\operatorname{meas} \mathbb{T}^n = 1$). □

5.6. What Part 5 consumes.

Part 5 ([P:bpP:p5:kam-application]) applies Proposition 4.3 to the Hamiltonian $\breve{H}$ of CP10 Step 6 (p. 560) after Part 2's averaging ([P:bpP:p2:averaged-remainder]). *Three radii must not be confused:* ϵ is the oscillator radius of CP10 (the quantity renamed r in the frozen cylinder $\mathcal{C}(V, r)$); $\hat{\rho}$ is the action radius of Definition 4.2; $r = 32\hat{E}\hat{\rho}$ is the enlargement radius of this proposition. The identification (CP10 p. 560, review P3-05): $n_1 = a$, $n_2 = b$, $h = H_0(J_1) + \mu\breve{N}(J)$, $f = \mu\,\epsilon^5\breve{P}$ (*not* $\mu^5\breve{P}$), $\rho = \breve{\rho}$, $s = \breve{s}/5$, $\bar{s} = 4\breve{s}/5$ (in coefficient-radius widths of §4.1), $D = D_{\mathrm{CP}}$ of CP10 (48).

(1) *Before* ϵ, μ (depend only on the fixed family): n, τ (hence $\hat{c}, c_n$), and the constant C with $M \le C$, $\hat{M} \le C\mu$, $\bar{M} \le C\mu^{-1}$, $\bar{M}_1 \le C$, $\bar{M}_2 \le C\mu^{-1}$, $E \le C\mu\epsilon^5$ (CP10 p. 560: "$M = C, \hat{M} = C\mu, \bar{M} = C\mu^{-1}, E = C\mu\epsilon^5, \bar{M}_1 = C, \bar{M}_2 = C\mu^{-1}$"), the N-norm of $\mu\epsilon^5\breve{P}$ being $\le C\mu\epsilon^5$.

(2) *Consequences, to be checked per* (ϵ, μ)*:* $L \le C\mu^{-1}$, hence $EL \le C\epsilon^5$ and $x = EM^2L/\gamma_1^2 \le C\epsilon^5/\gamma_1^2$; since $x \mapsto \log_+(1/x)$ is nonincreasing, $K \le \frac{6}{s}\log_+\big(\gamma_1^2/(C\epsilon^5)\big) \le C'\log(\epsilon^5/\gamma_1^2)^{-1}$ with $\log_+$ retained (CP10 p. 560, "$K \le C\log(\epsilon^5/\gamma_1^2)^{-1}$"). The KAM condition $\hat{c}\hat{E} < 1$ is CP10 (53)–(54); the normalizations $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$ hold for small μ.

(3) *Outputs used:* the tori φ_ω, $\omega \in \Omega_*$, with (a)–(d); the measure inequality (3.1) with the *absolute* constant c_n; $r = 32\hat{E}\hat{\rho} \le C\hat{\rho}$ so that $r \le C\bar{\gamma}_2$ as in CP10 p. 561; $|v - I^0|$, $|u|$ for the reconstruction (CP10 (51), (35)).

(4) *Conversions* ([P:`bpP:p1:period-convert`], [P:`bpP:p1:cyl-pipeline`]; Lemma 4.5): $H = \tilde{H}/2\pi$ in (I, θ); the frozen torus is $K = \Psi \circ K_3$ with $K_3 = Q \circ \varphi_\omega$ in the frozen order, and *the frozen frequency is* $w = \omega$ *(Part 3's frequency; CP's angular frequency is* $2\pi w$*)*, not $\omega/2\pi$; the output angle $\tilde{u} = 2\pi u$.

Not conclusions of Part 3 (obligations of Part 5): the complex neighbourhood and uniform bounds from the literal premises (H1)–(H6), the injectivity/nondegeneracy of ω_μ on a complex neighbourhood, the volume losses, the reconstruction of the oscillators and the seven clauses of [P:`bpP:p0:cyltorus`], measurability and whole-torus saturation of S, positivity and the strict inequalities.

Source gap: CP10 Prop. 3 claims *"a unique real-analytic embedding"*; translations $\vartheta \mapsto \vartheta + c$ preserve all stated properties, so uniqueness is neither proved nor claimed here. The Lipschitz continuity of $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$ passes through ω_0^{-1}, which is globally Lipschitz only for bounded D (locally, at scale $\rho/\bar{M}$, in general: separated action islands at j^2 with local frequency maps centred at j and Hessian 1 have inverse derivative 1 but no global Lipschitz inverse; review E9).

5.7. **Source gaps and suspected misprints (CP10, Prop. 3, Appendix B).** Dispositions follow the independent review (`reviews/p3-review-1.md`): V = verified misprint, R = proof repair or convention change (not an erratum), T = tentative.

(E1) **V.** p. 554: "$\hat{U}$ denote the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$": size and index range disagree; the proof needs $n_2 \times n$ (Remark 4.1).

(E2) **V.** p. 565 (Lemma B.1): "$M_0 := M$, $\hat{M}_0 := M$, $\bar{M}_0 := \bar{M}$"; $\hat{M}_0$ should be $\hat{M}$.

(E3) **V.** p. 566, B.1(i): "$\hat{\imath}_j : I \in D_{j-1} \mapsto \omega_{j-1} \circ \omega_{j-1}^{-1}(I) \in D_j$" is the identity; it should read $\omega_j^{-1} \circ \omega_{j-1}$ (cf. B.2, p. 568, $\hat{\imath}_+ = \omega_+^{-1} \circ \omega|_D$).

(E4) **V, with scope.** (74) uses $\bar{M}_i/\bar{M} \le 1$ without specifying compatible bounds (compatible bounds can be chosen: normalization, Remark 4.4); and (35), second line, carries a factor $\hat{M}/M$ for $|v_2 - I_2^0|$ that p. 572 (107) does not give (replacing it by 1 gives what is proved). No counterexample to a sharper estimate is claimed.

(E5) **T (not verified).** p. 551 assigns the 1-norm to the action, the angular sup metric, sup norms to p, q and the entrywise sup norm to matrices; the proof uses operator norms and mixed conversions. The blueprint's constants are a legitimate *convention change* (max norms, N-norm). No source constant is exhibited as false; the allegation that the printed constants "cannot all hold" is withdrawn.

(E6) **V (sharpening).** p. 569 (91) prints $\frac{8}{9}(4s)^{\tau+1}\hat{E}^2$; the computation gives exponent $2(\tau+1)$. The printed inequality is true (weaker).

(E7) **R.** pp. 569–570: the mean value step for ω^{-1} needs the frequency segment inside the domain of the inverse; and Lipschitz bounds on non-convex D, D_j need near/far pairs. Repaired by Lemmas 4.6, 4.7; composition domains stated (Lemma 5.3(i)).

(E8) **R (expository supplement).** Step 3 (pp. 572–573) does use (108) ("Using (108)"); the derivative bound on the open domain and analyticity on a fixed strip are not spelled out; supplied by Steps S5.25, S5.18 and Lemma 5.4. Not a missing estimate.

(E9) **V.** pp. 555, 573: "a unique real-analytic embedding" is not true as stated (translations); global Lipschitz dependence on ω fails for unbounded disconnected D (example above); bounded D suffices.

(E10) **R.** p. 573 footnote 32: Kirszbraun (Euclidean) is replaced by max-norm McShane and the cube-covering bound $\operatorname{meas} F(A) \le \Lambda^d \operatorname{meas} A$ ([K:`bp:p1:mcshane`], [K:`bp:p1:lip-image`]); c_n is recomputed, not reproduced; not an erratum.

(E11) **V (dependency).** Prop. 1 is proved in CP10 only through the unpinned [4, Lemma A.5] and a finite induction "as in [4]"; supplied by Part 2. The pointer to Arnold [3, p. 144] in Remark 3 is historical (Appendix B has its own induction) and is not a dependency.

Two further verified suspected defects found by the review (for the orchestrator's ledger, not used here): CP10 p. 565 (70) and the following split print "$k \in \Lambda$" where $k \notin \Lambda$ is meant (with $\Lambda = \{0\}$ the zero mode would be divided by zero); CP10 p. 564 (66) prints the target $W_{v,\sigma}$ where the angle width is s, i.e. $W_{v,s}$.

5.8. **References and imported facts.** Written parts cited: [P:`bpP:p1:period-convert`], [P:`bpP:p1:cyl-pipeline`], [P:`bpP:p2:avg`], [P:`bpP:p2:nfb`], [P:`bpP:p2:split`], [P:`bpP:p2:averaged-remainder`]; track N: [N:`bpN:p4:norm`], [N:`bpN:p5:symplectic`], [N:`bpN:p5:canonical`]. Forward reference to an unwritten part: [P:`bpP:p5:kam-application`]. Track K: [K:`bp:p1:cauchy-open`], [K:`bp:p1:uniform-limits`], [K:`bp:p1:mcshane`], [K:`bp:p1:lip-image`], [K:`bp:p1:ext-mvt`], [K:`bp:p1:ext-banach`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-rectangle`]. The only ingredient proved elsewhere is the averaging step (Lemma 4.8, proved in Part 2 from track N); the rest is proved here from the external facts (X1)–(X6). Formalization notes: the ball lemma needs a finite-dimensional path-continuation package; Lemma 5.4 is a bounded stand-alone task (no several-variable Vitali–Montel theorem is assumed).